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Quality-Driven Next-Best-View Planning for Target-Conditioned Egocentric 3D Reconstruction

Qualitätsgetriebene Next-Best-View-Planung für zielkonditionierte egozentrische 3D-Rekonstruktion

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Abstract

Active perception couples sensing and reconstruction: what can be reconstructed depends on what a sensing action makes visible. Under a limited acquisition budget, next-best-view planning must therefore choose which feasible observation should come next. The literature reviewed in Chapter 2 treats scene coverage, one-step reconstruction-quality ranking, target-aware information criteria, and sequential coverage separately, but leaves their conjunction for target-specific egocentric reconstruction unresolved. This thesis studies that conjunction as finite candidate selection under hard validity constraints in Aria Synthetic Environments (ASE). It adapts Relative Reconstruction Improvement (RRI) to a target-cropped point–mesh objective, separates privileged task construction and oracle evaluation from the deployable actor path, and marks GT-derived target or selected-depth inputs as non-deployable controls. The primary experiment first tests whether bounded oracle lookahead provides endpoint-quality headroom over one-step oracle greedy; a learned policy is evaluated only if that headroom passes a prespecified meaningful-effect and uncertainty rule. The current implementation supports target-specific oracle scoring and selected-action replay, with rendering memory limiting scale. Actor-visible target matching, metric repeatability, a validated held-out population, headroom, and paired policy outcomes remain unestablished. The present contribution is therefore an auditable method and experimental design, not evidence of policy superiority or deployment readiness.

Zusammenfassung

Aktive Wahrnehmung beruht auf einer geometrischen Prämisse: Was rekonstruiert werden kann, hängt davon ab, was eine Sensorhandlung sichtbar macht. Bei begrenztem Aufnahmebudget muss die Next-Best-View-Planung daher entscheiden, welche zulässige Beobachtung als Nächstes erfolgen soll. Die in Kapitel 2 ausgewertete Literatur behandelt Szenenabdeckung, einstufige Rekonstruktionsqualität, zielbezogene Informationsmaße und sequenzielle Abdeckung getrennt, lässt deren Verbindung für die zielspezifische egozentrische Rekonstruktion jedoch offen. Diese Arbeit untersucht diese Verbindung als Auswahl aus einer endlichen Kandidatenmenge unter harten Zulässigkeitsbedingungen in ASE. Sie überträgt die RRI auf ein zielbeschnittenes Punkt–Mesh-Maß, trennt privilegierte Aufgabenerzeugung und Orakelevaluation vom einsatzfähigen Akteurpfad und kennzeichnet GT-basierte Ziel- oder Tiefeneingaben als nicht einsetzbare Kontrollen. Das primäre Experiment prüft zunächst, ob eine beschränkte Orakelvorausschau gegenüber einer einstufigen gierigen Orakelstrategie einen Vorteil bei der Endpunktqualität bietet; eine gelernte Strategie wird nur bewertet, wenn dieser Vorteil ein vorab festgelegtes Relevanz- und Unsicherheitskriterium erfüllt. Die aktuelle Implementierung unterstützt zielspezifische Orakelbewertung und Replay ausgewählter Aktionen, wobei der Speicherbedarf des Renderers die Skalierung begrenzt. Nicht belegt sind bislang eine beobachtungsbasierte Zielzuordnung, die Wiederholbarkeit des Qualitätsmaßes, eine validierte zurückgehaltene Studienpopulation, ein Vorteil der Orakelvorausschau und gepaarte Strategieergebnisse. Der gegenwärtige Beitrag ist daher ein prüfbarer Methoden- und Versuchsaufbau, kein Nachweis für Strategieüberlegenheit oder Einsatzreife.

Contents

Glossary and Abbreviations

Dataset

ADT **– Aria Digital Twin**: Project Aria dataset with real-world captures and digital-twin scene annotations. ADT is adjacent to ARIA-NBV's sim-to-real context; the current experiments focus on ASE and EFM3D/ATEK exports, while ADT remains a relevant transfer surface.

AEO **– Aria Everyday Objects**: Small-scale real-world Project Aria object dataset used by EFM3D for egocentric 3D perception evaluation. AEO is relevant as a possible sim-to-real check for ARIA-NBV ideas that are first developed on ASE mesh-supervised snippets.

ASE **– Aria Synthetic Environments**: Large-scale synthetic indoor dataset with simulated Project Aria sensor characteristics, egocentric trajectories, and scene annotations. ARIA-NBV uses ASE snippets and the public mesh-supervised subset to generate oracle RRI labels from ground-truth meshes and semi-dense point clouds.

CPF **– Central Pupil Frame**: Coordinate frame placed at the midpoint between the left and right eye boxes of Project Aria glasses. CPF is used by Project Aria for gaze-related quantities and should remain distinct from rig, camera, world, and PyTorch3D frames in ARIA-NBV docs.

MPS **– Machine Perception Services**: Project Aria processing services that derive pose, mapping, gaze, hand, and related perception outputs from sensor recordings. ARIA-NBV mainly depends on MPS-style pose and semi-dense mapping products as the current reconstruction state for RRI computation.

MTD **– Motion Trajectory Data**: Device poses over time, usually represented as a sequence of 6-DoF transformations. MTD supplies the logged egocentric trajectory used to define snippet state, current reconstruction context, and candidate-view reference poses.

MFCD **– Multi-Frame Camera Data**: Synchronized camera streams from multiple Project Aria cameras over a temporal window. MFCD provides the egocentric image evidence consumed by EVL and aligned with pose, calibration, and semi-dense point streams.

***MSDPD* – Multi-Semi-Dense Point Data**: Semi-dense 3D point observations generated by SLAM-style processing across a snippet or trajectory window. MSDPD is the sparse observed geometry that ARIA-NBV fuses with candidate point clouds and compares against ground-truth meshes for RRI labels.

***Project Aria* – Project Aria**: Meta's egocentric research-device and tooling ecosystem for calibrated, time-aligned multimodal sensing. ARIA-NBV treats Project Aria and its MPS-style products as the actor-visible sensing contract, while ASE meshes remain offline supervision and evaluation assets.

***SSL* – SceneScript Language**: Structured language representation for indoor scene layout using primitives such as walls, doors, windows, and objects. SceneScript is relevant to ARIA-NBV as a possible semantic/global planning layer and as one source of ASE scene-structure context.

***snippet* – Snippet**: Short synchronized temporal window of Aria sensor data used as one EVL/VIN input sample. A snippet typically contains RGB or grayscale streams, poses, calibration, semi-dense points, and scene metadata that EVL lifts into a voxel grid.

***VRS* – Virtual Reality Standard**: File format used by Project Aria tooling to store multi-modal sensor recordings efficiently. VRS is part of the upstream Project Aria ecosystem, while ARIA-NBV primarily consumes ASE/ATEK-derived tensorized snippets for current experiments.

Entity

***target* – Target of Interest**: Selected entity, object crop, point, region, or surface-deficit hypothesis whose reconstruction quality should be improved. Target-conditioned ARIA-NBV variants use this target as an explicit input to candidate scoring and planning instead of optimizing only scene-level RRI.

Geometry

***DoF* – Degrees of Freedom**: Number of independent pose parameters available to a camera, object, or action representation. ARIA-NBV distinguishes full 6-DoF pose estimation from reduced 5-DoF candidate or action spaces used for practical view planning.

***frustum* – Frustum**: Truncated pyramidal camera-visible volume bounded by near and far clipping planes plus lateral field-of-view planes. Candidate-view frusta define which scene surfaces can project into a camera image and are therefore central to visibility, rendering, and RRI diagnostics.

***LUF* – Left-Up-Forward**: Camera coordinate convention whose x axis points left, y axis points up, and z axis points forward. LUF is one of the coordinate-frame conventions that must stay explicit when moving between Project Aria cameras, PyTorch3D cameras, and ARIA-NBV candidate poses.

***OBB* – Oriented Bounding Box**: 3D bounding box with arbitrary orientation, used to represent object extent more tightly than an axis-aligned box. OBBs are a natural target encoding for entity-aware ARIA-NBV because they provide center, extent, orientation, semantic class, and confidence signals.

***6DoF* – Six Degrees of Freedom**: Pose parameterization with three translational and three rotational degrees of freedom. Project Aria poses, candidate cameras, and world-to-device transforms are usually represented as 6-DoF rigid-body transforms.

Metrics

***AUC* – Area Under Curve**: Aggregate score computed by integrating a metric curve over an acquisition, threshold, or ranking axis. AUC can summarize how quickly reconstruction quality, coverage, or ranking quality improves as views are acquired or candidate thresholds change.

***CD* – Chamfer Distance**: Historical bidirectional distance family used to compare reconstructed points against reference geometry. Thesis-facing ARIA-

NBV notation uses point-mesh error D with directional components $D_{P \to M}$ and $D_{M \to P}$; older seminar material may still call this CD.

***CR* – Coverage Ratio**: Fraction of a target surface, scene, or region treated as observed under a chosen visibility or distance threshold. Coverage ratio is a useful diagnostic baseline for NBV, but ARIA-NBV treats reconstruction-quality improvement through RRI as the preferred optimization target.

***GT* – Ground Truth**: Reference data or annotations treated as the trusted target for training, validation, or evaluation. In ARIA-NBV, ground truth usually refers to ASE meshes, object annotations, poses, or labels used to compute oracle supervision and diagnostic metrics.

***RRI* – Relative Reconstruction Improvement**: Metric quantifying the relative reconstruction-quality improvement obtained by adding a candidate observation to the current reconstruction. ARIA-NBV computes RRI by comparing reconstruction error before and after fusing candidate-view geometry, usually against an ASE ground-truth mesh for oracle supervision and evaluation.

***reward* – Target Root-Gain Reward**: Quality-only immediate reward for thesis-core target-aware rollouts. It is the target-specific point-mesh error reduction normalized by the rollout-root target error; scalar motion, rule, diversity, and validity penalties are later ablations.

***target RRI* – Target-Specific RRI**: RRI computed only on the ground-truth and reconstructed geometry associated with a selected target of interest. Target-specific RRI lets the thesis compare views by how much they improve a selected object or region, even when scene-level RRI would prefer large background surfaces.

Model

***Q-CORAL* – CORAL Q-Value Decoder**: Ordinal Q-decoder ablation that discretizes continuous fitted-Q targets into ordered classes and decodes cumulative threshold outputs to a scalar conditional value through fixed return representatives.

***EFM3D* – Egocentric Foundation Model 3D**: Egocentric 3D foundation-model stack used as the frozen spatial backbone for ARIA-NBV candidate scoring. The current project uses EFM3D and its EVL architecture to expose voxel occupancy, centerness, semantic, and OBB evidence for VIN-style RRI prediction.

***EVL* – Egocentric Voxel Lifting**: EFM3D architecture that lifts synchronized egocentric observations into a gravity-aligned 3D voxel feature volume. ARIA-NBV uses EVL head outputs, pre-head features, and OBB predictions as local actor-visible evidence and target support, not as the complete frozen scene representation.

***Q__H* – Finite-Horizon Q Function**: Target-conditioned finite-candidate value family Q_theta(s,e,a,h) that scores each valid candidate for an explicit requested residual horizon h up to a configured maximum H. The thesis-core architecture is one shared horizon-conditioned scorer; dense Q_1 and exact Q_2 targets establish the recursion, horizon-indexed fitted Q supplies longer-horizon targets, and Double Q remains an optional correction for a noisy learned successor maximum.

***target-conditioned scorer* – Target-Conditioned Scorer**: VIN-style candidate scorer that receives scene state, a candidate view, and an encoding of the target of interest. The scorer predicts target-specific utility so view ranking can prioritize a selected entity or region instead of only optimizing scene-level RRI.

***VIN* – View Introspection Network**: Learned candidate-view scorer that predicts RRI or an ordinal RRI-derived utility without capturing the candidate view. ARIA-NBV adapts VIN-NBV by placing a lightweight RRI prediction head on top of frozen EVL features and candidate-pose evidence from ASE snippets.

Planning

***cost* – Acquisition Cost**: Budget consumed to acquire observations, measured by view count, path length, elapsed time, invalid-action rate, or a weighted combination. The thesis should first report cumulative root-normalized target gain, diagnostic target RRI, and acquisition cost separately, then use scalarized objectives only when the tradeoff is explicit.

***candidate* – Candidate View**: Proposed camera pose whose expected reconstruction utility is evaluated before selecting the next observation. ARIA-NBV samples candidate views around a reference pose or target, renders candidate depths from the mesh for oracle labels, and scores candidates with RRI or learned VIN predictions.

***transition* – Counterfactual Transition**: Deterministic or replayable rollout update that renders or retrieves only the selected candidate observation, backprojects it to candidate geometry, and fuses it into the accumulated reconstruction proxy.

***action set* – Finite Candidate Action Set**: Valid discrete action set for ARIA-NBV planning, obtained by masking a sampled candidate view table. The action is the candidate-table index, not a continuous pose; the selected pose is $q_t = q_{t,a_t}$. Invalid candidates are constraints, not low-quality actions.

***return* – Finite-Horizon Return**: Symbolic H-step return used by ARIA-NBV rollout evaluation and Q_H targets. The formula keeps γ explicit while the thesis-core comparison reports cumulative root-normalized target gain under equal acquisition budget.

***5DoF* – Five Degrees of Freedom**: Reduced camera-action parameterization commonly used when roll is fixed or otherwise constrained. ARIA-NBV uses 5-DoF language for candidate and planning abstractions where position and viewing direction matter while roll is not an independently optimized control dimension.

***CF+ state* – Geometry-Rich Counterfactual State**: Ablation actor state that extends the minimal counterfactual state with selected prior synthetic observations only. The executable CF-GT carrier stores rendered depth, valid mask, calibration, selected-camera pose, source identity, and causal support; implemented S1 derives an identity-start, fixed-width current-camera surface-point residual from it.

***historic state* – Historic Snippet State**: Raw actor-visible state available along the logged Project Aria/ASE trajectory before counterfactual rollout synthesis. It includes captured camera streams, calibration, timestamps, trajectory/gravity, semidense points with support or uncertainty, EVL/EFM evidence, and detected or predicted OBBs; GT mesh and GT OBBs are excluded as actor inputs.

***CF0 state* – Minimal Counterfactual Actor State**: Main thesis-core actor-visible counterfactual rollout state. It keeps the fused point proxy $\mathcal{P}_t$ as broad scene state, optional lifted image-foundation features attached to points, local root EVL evidence for target support, selected-view history, target descriptor, budget, candidate table, validity masks and reason codes, and current candidate-query features, without exposing all-candidate GT renders.

***NBV* – Next-Best-View**: Problem of selecting the next sensor viewpoint to improve an active reconstruction or inspection objective under a limited acquisition budget. In ARIA-NBV, the preferred objective is reconstruction quality improvement rather than coverage alone, with candidate views ranked by oracle or learned RRI scores.

***oracle state* – Oracle Rollout State**: Privileged state used only for ASE mesh-supervised label generation, upper-bound planning, and evaluation. It includes GT mesh and target crops, all-candidate renders, points, normals, mesh-face visibility, RRI and point-mesh error metrics, oracle scores, and GT OBBs.

***offline state* – Persisted Offline Sample State**: Compact VIN offline-store state persisted for training and diagnostics. It contains the VinSnippetView payload, candidate table and cameras, candidate count, validity or count metadata, oracle metrics as labels, optional rendered depths, compact OBBs, trajectory metadata, and selected EVL numeric fields.

***state* – Rollout State**: Family of rollout state variants used by ARIA-NBV. The actor-visible variants contain logged or replayed observations, reconstruction proxies, candidates, validity masks, target descriptors, history, and budget; the oracle variant adds privileged GT geometry and all-candidate labels for supervision and evaluation only.

***NBV MDP* – Target-Conditioned NBV MDP**: Finite-horizon Markov decision process used to define ARIA-NBV's target-conditioned rollout and fitted Q_H training contract. The actor observes only current reconstruction state, sampled candidate views, validity masks, target descriptors, history, and budget state; oracle meshes and GT crops are supervision and evaluation signals.

***mask* – Validity Mask**: Hard candidate-level feasibility mask used during candidate ranking, rollout generation, and Q_H training. The scalar mask $m_{t,i}$ gates selectable rows, while the reason code $\rho_{t,i}$ records why a row was rejected. Invalidity is represented separately from reconstruction quality and should not be encoded as the lowest RRI class.

Protocol

***GT-EVAL* – Ground-Truth Target Evaluation**: Main thesis protocol component restricting ground-truth OBBs and target mesh crops to oracle labels, target-RRI computation, matching checks, and evaluation. GT target crops are not actor-visible inputs in the main result.

***OBS-SEL* – Observed Target Selection**: Main thesis protocol component requiring target selection to use only actor-visible evidence such as predicted or tracked OBBs, class probabilities, confidence, projected area, and semidense or EVL support. Ground-truth target annotations are not visible to the selector in this protocol.

***PRED-Q* – Predicted-Target Q**: Main thesis protocol component requiring the scorer or fitted Q_H model to condition on predicted or observed target descriptors rather than ground-truth target inputs. It covers target-conditioned one-step scoring and finite-candidate Q_H selection.

Reconstruction

***MVS* – Multi-view Stereo**: Reconstruction family that estimates dense or semi-dense scene geometry from multiple calibrated views. MVS is part of the broader reconstruction context for NBV, although ARIA-NBV's current oracle labels are built from ASE meshes and Aria-style semi-dense point observations.

***PC* – Point Cloud**: Set of 3D points representing observed scene geometry. ARIA-NBV compares current and candidate-fused point clouds against ground-truth meshes to compute oracle RRI and surface-distance diagnostics.

***SLAM* – Simultaneous Localization and Mapping**: Method for estimating sensor motion while building a map of the surrounding scene. In ARIA-NBV, logged SLAM poses and semi-dense points form the current reconstruction state against which candidate-view RRI is evaluated.

***track* – Track**: Temporal sequence of corresponding image-feature detections across frames, usually carrying per-frame image coordinates, timestamps, and camera IDs. Tracks can be triangulated or optimized into 3D points, making them a bridge between image evidence and the semi-dense point clouds used by ARIA-NBV.

***VIO* – Visual-Inertial Odometry**: Pose-estimation method that combines visual measurements from cameras with inertial measurements from IMUs. Project Aria pose and mapping products build on visual-inertial estimation, and ARIA-NBV depends on those pose streams for snippet state and candidate frame contracts.

Representation

***Occupancy Grid* – Occupancy Grid**: Spatial grid whose cells encode whether space is occupied, free, unknown, or represented by a related occupancy probability. Occupancy-style voxel evidence appears in EVL outputs and VIN feature construction, where it helps summarize local 3D scene state for candidate scoring.

***3DGS* – 3D Gaussian Splatting**: Explicit radiance-field representation using optimized 3D Gaussian primitives for real-time novel-view synthesis. ARIA-NBV treats active 3DGS work as proposal, uncertainty, and simulator-bridge literature, not as a replacement for ASE mesh-supervised RRI.

Supervision

***oracle RRI* – Oracle RRI**: RRI label computed with privileged ground-truth geometry, used for supervised training and evaluation. The current oracle renders candidate depth maps from ASE ground-truth meshes, backprojects candidate point clouds, fuses them with the current semi-dense reconstruction, and scores the resulting surface-distance improvement.

List of Symbols

Symbol	Meaning	Key
	Oracle and Reconstruction	
RRI	Relative Reconstruction Improvement score for a candidate view.	`oracle.rri`
$\mathcal{P}$	Actor-visible point set accumulated from observed or replayed views.	`oracle.points`
$\mathcal{P}_q$	Candidate-view point contribution used in oracle or counterfactual updates.	`oracle.points_q`
$\mathcal{Q}$	Finite candidate-view set considered by the planner.	`oracle.candidates`
$\mathcal{Q}_t$	Candidate-view set available at rollout step t.	`oracle.candidates_t`
$q_{t,i}$	Candidate pose i at rollout step t.	`oracle.candidate_qti`
$\boldsymbol{c}$	World-space camera-center vector; subscripts identify a rollout root or candidate pose.	`oracle.center`
$\boldsymbol{D}_q$	Oracle-rendered or candidate-specific depth map for a proposed view.	`oracle.depth_q`
$D_{P\to M}$	Point-to-mesh directional error component.	`oracle.acc`
$D_{M\to P}$	Mesh-to-point directional error component.	`oracle.comp`
D	Aggregate point-mesh reconstruction error used by RRI definitions.	`oracle.err`
	ASE Assets	
$\mathcal{M}^{\mathrm{GT}}$	Ground-truth scene mesh used for oracle labels and evaluation.	`ase.mesh`
$\mathcal{M}_e^{\mathrm{GT}}$	Ground-truth mesh crop for target e.	`ase.mesh_target`
$\mathcal{F}^{\mathrm{GT}}$	Ground-truth mesh faces used in point-mesh distance calculations.	`ase.faces`
$\boldsymbol{T}_{\mathrm{rig}}^{\mathrm{w}}(t)$	Project Aria rig pose in the world frame at time t.	`ase.traj`
$\boldsymbol{T}_{\mathrm{rig}}^{\mathrm{w}}(T)$	Final rig pose used as an anchor for gravity-aligned local fields.	`ase.traj_final`
	VIN Scorer	
r	Target or scene RRI value used as a model target.	`vin.rri`
$\hat{r}$	Predicted RRI value emitted by the learned scorer.	`vin.rri_hat`
$\mathcal{L}$	Training loss for scorer or value-model objectives.	`vin.loss`
m	Candidate validity indicator used by scorer diagnostics.	`vin.cand_valid`

Symbol	Meaning	Key
	Entity and Target	
RRI_e	Target-specific Relative Reconstruction Improvement for entity e.	`entity.rri_e`
$\mathrm{RRI}^e_{t,i}$	State-relative marginal target RRI for candidate i at rollout step t.	`entity.target_rri_marginal`
$C_t^{\mathrm{RRI},e}$	Running sum of selected state-relative target RRIs.	`entity.target_rri_cumulative`
J_t^e	Running cumulative target gain normalized by rollout-root error.	`entity.target_root_gain_cumulative`
$\boldsymbol{\varphi}_e$	Actor-visible target/entity descriptor used to condition target-specific value prediction.	`entity.target_desc`
$\boldsymbol{p}_e^w$	World-space center of the selected target entity e.	`entity.center`
	Planning and Rollout	
s	Abstract planning state.	`rl.s`
a	Action index selecting a candidate row.	`rl.a`
r	Scalar reward or immediate gain.	`rl.r`
G	Finite-horizon return accumulated across rollout steps.	`rl.G`
j	Factual rollout-chain index used to preserve common trajectory heritage across selected steps.	`rl.rollout_index`
$\mathcal{M}_{\mathrm{NBV}}$	Target-conditioned finite-candidate NBV decision process.	`rl.mdp_nbv`
$\mathcal{A}(s_t)$	Valid action set available in state s_t.	`rl.action_set`
T	State-transition operator for selected candidate actions.	`rl.transition`
s_t^{hist}	Logged historic snippet state at rollout step t.	`rl.s_hist`
s_t^{off}	Persisted offline sample state used by training readers.	`rl.s_off`
s_t^{cf0}	Minimal actor-visible counterfactual state.	`rl.s_cf0`
$s_t^{\mathrm{S0\text{-}pose}}$	Implemented fixed-root pose-history state used by qh_cf0_v1.	`rl.s_pose`
$s_t^{\mathrm{cf+}}$	Geometry-enriched counterfactual successor state.	`rl.s_cf_geom`
$s_t^{\mathrm{CF\text{-}GT\text{-}carrier}}$	Implemented privileged selected-depth carrier used by qh_cfplus_gt_depth_v1.	`rl.s_cf_gt_carrier`
s_t^{oracle}	Oracle-only state used for labels, upper bounds, and evaluation.	`rl.s_oracle`
r_t^e	Target-specific reward at rollout step t.	`rl.reward_target`
	Entity and Target	

Symbol	Meaning	Key
r_t^e	Canonical target-specific reward at rollout step t.	`entity.target_reward`
	Planning and Rollout	
$G_t^{(H)}$	H-step finite-horizon return from rollout step t.	`rl.return_h`
Q_H	Finite-horizon candidate-value function.	`rl.qh`
$Q_{h,\theta,e,i}^{\text{cond}}$	Action-mask-independent conditional candidate value emitted by the scorer.	`rl.conditional_q`
$\ell_{t,i}^{\text{feas}}$	Physical or observed feasibility logit emitted by the scorer's feasibility head.	`rl.feasibility_logits`
e_k^Q	Fixed continuous fitted-Q boundary between adjacent CORAL classes.	`rl.coral_q_edge`
u_k^Q	Fixed continuous-Q representative used to decode one CORAL class.	`rl.coral_q_value`
c_n^Q	Ordinal class assigned to one continuous fitted-Q target.	`rl.coral_q_label`
γ	Return discount factor.	`rl.gamma`
H	Planning or rollout horizon length.	`rl.H`
$H_{\max}$	Maximum residual horizon admitted by a scorer/data contract.	`rl.H_max`
h	Scalar residual horizon requested for one scorer query; it must satisfy 1 <= h <= b_t <= H_max.	`rl.requested_horizon`
$m_{t,i}$	Hard validity mask for candidate i at rollout step t.	`rl.validity_mask`
$m_{t,i}^{\text{cand}}$	Materialized candidate row versus structural padding.	`rl.candidate_row_mask`
$m_{t,i}^{\text{act}}$	Authoritative physically valid action support.	`rl.action_mask`
$m_{t,i}^{Q,h}$	Availability of a finite factual value target for candidate i at requested horizon h.	`rl.q_label_mask`
$m_{t,i}^{F}$	Availability of a trustworthy feasibility label for candidate i.	`rl.feasibility_label_mask`
m_t^{succ}	Availability of a factual successor backup for transition t.	`rl.successor_mask`
$\ell_{t,i}^{\text{src}}$	Actor-oracle source-provenance role for candidate evidence.	`rl.source_role`
$\rho_{t,i}$	Invalid-action reason code for candidate i at rollout step t.	`rl.invalid_reason`
e_t	Selected target entity at rollout step t.	`rl.target`
b_t	Remaining acquisition budget at rollout step t.	`rl.budget`
$C(\tau)$	Acquisition cost of a selected trajectory.	`rl.acquisition_cost`

Symbol	Meaning	Key
	Shape and Size	
N_q	Number of candidate views in a candidate table.	`shape.Nq`
K	Number of ordinal bins or discrete levels when used in scorer losses.	`shape.K`
	scene	
$\boldsymbol{M}_t^{\mathrm{ray}}$	Sparse actor-visible ray-aware memory separating occupied, free, and unknown support at step t.	`scene.ray_memory_t`
	Observation and Actor State	
$\boldsymbol{F}_t^{\mathrm{DINO@pt}}$	Visibility-gated logged DINO feature bank attached to semidense or fused world points.	`obs.dino_point_bank_t`
$\boldsymbol{X}_t^{\mathrm{pt}}$	Point-token set combining geometry, support, uncertainty, history, and optional compressed logged descriptors.	`obs.point_tokens_t`
	scene	
$\boldsymbol{\Phi}_t^{\mathrm{scene}}$	Composite actor-visible scene-memory descriptor queried by the finite-candidate value model.	`scene.scene_memory_t`
$\boldsymbol{E}_0^{\mathrm{EVL\text{-}local}}$	Root local EVL evidence field used for target/candidate support reads.	`scene.evl_local`
$\omega_{t,i}^{\mathrm{EVL}}$	Candidate or target-query fraction supported by the root EVL voxel extent.	`scene.evl_support_frac`
$\boldsymbol{g}_{t,i}^{\mathrm{EVL}}$	Pooled local EVL evidence token for a target, candidate frustum, or target-candidate intersection query.	`scene.evl_support_token`
$\boldsymbol{g}_e^{\mathrm{tgt}}$	Pooled target-support descriptor for the selected target hypothesis.	`scene.target_support_pool`
$\boldsymbol{g}_{t,i}^{\mathrm{fr}}$	Candidate-frustum support descriptor pooled from actor-visible scene memory.	`scene.frustum_support_pool`
$\boldsymbol{g}_{t,e,i}^{\mathrm{cap}}$	Pooled descriptor over the intersection between target support and candidate frustum support.	`scene.target_frustum_pool`
$\boldsymbol{g}_{t,i}^{\mathrm{ray}}$	Candidate-conditioned ray query over occupied, free, unknown, hit, target-support, and uncertainty summaries.	`scene.ray_query_ti`
RenderQuery	Abstract query operator that summarizes scene memory in a candidate camera/frustum without introducing fresh counterfactual RGB features.	`scene.render_query`
	spatial	
r_t	Reference pose for candidate-relative descriptor construction at decision step t.	`spatial.ref_pose`

Symbol	Meaning	Key
$\boldsymbol{T}^{\mathrm{rel}}_{r_t,i}$	Relative transform from the reference pose r_t to candidate camera i.	`spatial.ref_candidate_transform`
$\boldsymbol{h}^{\mathrm{pose}}_{t,i}$	Relative/local candidate pose descriptor derived from a reference pose rather than raw world coordinates.	`spatial.candidate_pose_feat`
$\boldsymbol{h}^{\mathrm{rel}}_{t,e\|i}$	Candidate-target relation descriptor in the candidate/ query local frame.	`spatial.candidate_target_rel_feat`
$\boldsymbol{e}^{\mathrm{rel}}_{a\|i}$	Query-local relative positional embedding for target, history, support, or candidate relations.	`spatial.relation_rpe`
r_e	Geometric-mean target-proxy scale derived from the target OBB semi-axis lengths.	`spatial.target_obb_scale`
$\hat{\boldsymbol{\delta}}^{e}_{j,t}$	Unit factual selected-camera displacement expressed in target-object coordinates for rollout j and step t.	`spatial.target_frame_motion_direction`
$\hat{\boldsymbol{v}}^{e}_{j,t}$	Unit selected-camera forward optical axis expressed in target-object coordinates.	`spatial.target_frame_view_direction`
$\mathcal{F}^{e}_{j,t}$	Front-facing target-proxy surface directions that project inside the calibrated selected-camera image.	`spatial.target_frame_frustum`
$\kappa^{e}_{j,t}$	Fraction of the target-centred proxy sphere geometrically supported by one selected calibrated frustum.	`spatial.target_frame_frustum_fraction`
	model	
$\boldsymbol{h}^{\mathrm{tgt}}_{e}$	Learned selected-target token built from the actor-visible target descriptor and scene support.	`model.target_token`
	spatial	
$\Omega^{\mathrm{FOV}}_{j,t}$	Intrinsic solid angle of one calibrated selected-camera pinhole field of view.	`spatial.frustum_solid_angle`
	model	
$\boldsymbol{x}_{t,i}$	Per-candidate row feature assembled from pose, relation, support, validity, provenance, and history descriptors.	`model.candidate_row`
$\boldsymbol{p}^{\mathrm{hist}}_{t,j}$	Previously selected pose j encoded from the current camera at decision state t.	`model.history_pose_feature`
$a^{\mathrm{hist}}_{t,j}$	Normalized relative age of selected pose j at decision state t; the immediate predecessor has age zero.	`model.history_relative_age`
$\boldsymbol{h}^{\mathrm{hist}}_{t}$	Fixed-width causal selected-pose history token supplied to scorer state fusion.	`model.history_token`
	Entity and Target	

Symbol	Meaning	Key
$J_e^{(H)}$	Endpoint reconstruction gain for target e over horizon H.	`entity.endpoint_gain`
$J_{e,\log}^{(H)}$	Log-scale endpoint reconstruction gain for target e.	`entity.log_gain`
Δ_{look}	Gain available to an oracle lookahead policy beyond one-step selection.	`entity.lookahead_headroom`
η_Q	Fraction of the learned-myopic-to-oracle-lookahead endpoint gap closed by the Q policy.	`entity.q_recovery`
$G_t^{(H)}$	Target-conditioned finite-horizon return.	`entity.return_h`
Observation and Actor State		
$\boldsymbol{D}$	Depth observation sequence.	`obs.depth`
$\boldsymbol{n}$	Per-point or per-face normal observation.	`obs.face_normal`
$\boldsymbol{I}^{\mathrm{rgb}}$	RGB image observation.	`obs.img_rgb`
$\mathcal{P}^{\mathrm{cf}}$	Counterfactual point observation.	`obs.points_cf`
$\mathcal{P}^{\mathrm{semi}}$	Semi-dense observed point set.	`obs.points_semi`
$\mathcal{P}_t$	Point set available at rollout step t.	`obs.points_t`
Planning and Rollout		
$\mathcal{Q}_t$	Finite candidate-view table at rollout step t.	`rl.candidate_table`
$y_t^{(2,\mathrm{exact})}$	Exact two-step fitted-Q control using factual dense successor one-step rewards.	`rl.exact_q2_target`
$\varepsilon_t^{(2)}$	Absolute learned-recursion target error against the exact two-step control.	`rl.q2_recursion_error`
VIN Scorer		
$\boldsymbol{F}_v$	Learned value field evaluated at v.	`vin.field_v`

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1 Introduction

Active perception begins from a geometric premise: what can be reconstructed depends on what a sensing action makes visible [AWB88, Baj88]. A new view changes both the available surface evidence and the state from which later views are chosen. Next-best-view (NBV) planning turns this coupling between action, observation, and reconstruction into the repeated choice of where to look next [Jia+25, SRR03].

With a limited acquisition budget, a view is “best” only relative to an objective. Coverage rewards newly observed surface, information criteria reward reduced uncertainty, and quality-driven methods evaluate the reconstruction itself [Fra+25, GML22, JLD24]. These objectives can disagree. A view may reveal more of a room without exposing the occluded surface of a requested object, while a small side-step may improve that object but add little scene-wide coverage. Target conditioning therefore specifies **whose geometry** should improve; it is not generic semantic awareness.

Existing methods supply the necessary ingredients under different assumptions. VIN-NBV ranks sampled candidates by direct reconstruction improvement, object-aware 3D Gaussian Splatting conditions an information criterion on a requested object, and GenNBV and Hestia learn sequential coverage policies [Che+24, Fra+25, Jeo+26, Lu+26]. What remains unresolved is their conjunction: whether target-specific reconstruction quality contains useful multi-step structure and whether that structure can be recovered from a bounded egocentric information state.

The actor-state question includes a representation-transfer hypothesis. Prior NBV pipelines already use generic 2D pretraining or foundation-model components for view prediction, depth estimation, or scene construction [AKC20, Gué+23, Str+24a]. These uses do not establish whether an egocentric 3D representation that combines a frozen 2D foundation feature extractor with learned upsampling and 3D processing improves target-conditioned endpoint-value learning. ARIA-NBV therefore treats EFM3D/EVL evidence as a hypothesis to be tested against matched actor-visible geometric controls, not as an assumed source of gain [Str+24b].

Project Aria supplies calibrated egocentric observations, EFM3D lifts them into local 3D evidence, and ARIA Synthetic Environments (ASE) provides the privileged geometry needed for controlled target tasks and counterfactual evaluation [Eng+23, Met25a, Str+24b]. This combination makes the information boundary testable. Task construction, action-set construction, hard admission, supervision, and evaluation may use privileged geometry, while the scorer consumes only the fields admitted by its declared actor-facing protocol. Scientific interpretation must account for both paths. Chapter 3 therefore treats actor visibility as a property of the complete decision process, not merely of the scorer tensor.

1.1 Aim and bounded problem

The evaluation proceeds in two stages. First, it measures whether bounded oracle lookahead yields better fixed-budget endpoint reconstruction of a requested object than one-step oracle-greedy selection. Conditional on such headroom, it then measures how much of the endpoint gap from an actor-visible learned-myopic control to oracle lookahead an offline finite-horizon model closes. This order separates the existence of a planning opportunity from the ability of a learned model to exploit it; the gap-closure ratio and oracle headroom have different baselines.

At every step, each policy selects from the same finite generated candidates; hard-invalid actions are excluded, and the target, horizon, candidate support, and endpoint evaluation are matched. The current oracle tasks use geometry-valid ground-truth boxes, while actor-visible target discovery and matching remain separate research requirements. The objective adapts VIN-NBV rather than copying it: VIN-NBV defines RRI from point-cloud Chamfer distance [Fra+25], whereas ARIA-NBV evaluates a target-cropped point–mesh error after an equal acquisition budget. The resulting claims concern this bounded finite-candidate setting, not a deployable target-selection system or an objective invariant to other reconstruction protocols.

1.2 Contributions and current evidence

The thesis operationalizes this evaluation through four scientific functions:

1. It separates one-step RRI, finite-horizon return, and fixed-budget endpoint gain within target-specific reconstruction quality.
2. It defines an end-to-end information boundary across task and action construction, state updates, model inputs, supervision, and evaluation.
3. It defines candidate value relationally over target, causal state, admissible action, budget, horizon, and continuation rule.
4. It evaluates measurement, support, oracle headroom, immediate-value learning, recursion, and learned-control endpoint gap closure in that order.

Before model capacity becomes the main question, the evidence needed to define the learning problem must itself be identifiable. The outcome must be repeatable, the study population and candidate support must contain the relevant opportunity, bounded lookahead must exhibit headroom, and factual replay must identify the required targets. Only after those conditions hold can a failure to recover non-myopic value be attributed primarily to representation or learning. The order makes a negative result informative: it distinguishes a measurement or support failure from a failure of immediate-value learning or finite-horizon recursion.

The present evidence supports a methodological contribution rather than a policy result. The implementation executes target-specific oracle scoring and selected-action replay, while rendering memory limits scale. Actor-visible target matching, metric repeatability, a validated held-out population, oracle-lookahead headroom, and paired policy outcomes remain unestablished. Accordingly, the thesis does not yet claim policy superiority or deployment readiness.

1.3 Research Questions

1.3.1 Research aim and evaluation logic

The study evaluates a sequence of linked questions. RQ1 defines the endpoint outcome. RQ2 then asks whether bounded oracle lookahead improves that outcome relative to one-step oracle-greedy selection and, conditionally, how much of the separate actor-visible myopic-to-oracle-lookahead gap an offline model closes. RQ3 defines the admissible information, and RQ4 establishes the target, action, and replay support over which both comparisons are interpretable. RQ5 and RQ6 extend the study only if the offline finite-candidate evidence warrants online or continuous-control claims.

1.3.2 RQ1 — Target-specific objective and endpoint outcome

Can target-conditioned finite-candidate NBV be evaluated through a stable, target-specific reconstruction objective under a fixed acquisition budget?

For a target e, the oracle measures reconstruction error on a frozen target crop. The primary estimand is the quality change from the root state to the endpoint after H acquisitions:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

State-relative RRI is a one-step diagnostic, root-normalized gain supplies the training reward, and endpoint gain remains the policy outcome. A positive answer requires a frozen, repeatable metric and equal acquisition horizons; runtime, invalid actions, and oracle calls are reported separately.

1.3.3 RQ2 — Bounded lookahead and learned gap closure

Does bounded oracle lookahead improve fixed-budget endpoint target gain over one-step oracle-greedy selection and, if so, how much of the actor-visible myopic-to-oracle-lookahead endpoint gap does an offline finite-horizon value model close?

The paired endpoint difference under the same scene, target, candidates, validity rules, horizon, and budget defines oracle-lookahead headroom:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

The learned comparison proceeds only if this headroom passes a predeclared meaningful-effect and uncertainty rule. Its ratio uses a different baseline:

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Here the numerator is the gain of the finite-horizon learned policy over the actor-visible learned-myopic control, and the denominator is the full gap from that control to oracle lookahead. The ratio is therefore conditional learned gap closure, not a fraction of the oracle-lookahead headroom above oracle-greedy. It is reported only when the paired denominator $J_e^H(\pi_{\text{oracle-look}}) - J_e^H(\pi_{\text{learned-1}})$ exceeds a predeclared positive tolerance $\tau_{\text{gap}} > 0$; otherwise the analysis reports the raw paired policy effects and classifies gap closure as not estimable. Because the headroom rule and required gap-closure fraction are not yet frozen, RQ2 remains prospective. If headroom is absent, the evaluated support does not expose a non-myopic advantage; this does not establish its universal absence.

1.3.4 Conditions for interpretation

1.3.4.1 RQ3 — Actor-visible target and information state

Which end-to-end target, action-support, and history protocol supports one-step and finite-horizon candidate scoring without privileged target geometry, oracle labels, or unselected candidate renders at decision time?

Ground-truth target tasks may define supervision, evaluation, and a separately reported privileged `V0/GT` task-construction sanity control; they do not satisfy the core RQ3 gate. The core learned comparison requires an observation-derived target instruction and matching path (`V1/observed`) in addition to actor-visible candidate support, hard validity, selected-state updates, and scorer inputs. Until that route exists, the core actor-visible comparison remains unavailable. The scorer then receives only the declared target descriptor, causal egocentric evidence, remaining budget, and requested horizon; evaluation separately audits matching failures, ranking and calibration, and actor/oracle leakage.

1.3.4.2 RQ4 — Candidate, replay, and population support

Do the candidate generator, validity rules, rollout recipes, and replay population provide adequate and diverse support for the RQ1–RQ3 estimands?

The candidate table combines exploration and target-bearing views, records generator provenance, and excludes geometric infeasibility through a hard mask. RQ4 measures candidate-family survival, invalidity, scene and target coverage, replay and horizon support, and resource cost. Scene-disjoint splits and scene-level aggregation prevent dense sampling from masquerading as population coverage.

1.3.5 Scope extensions

1.3.5.1 RQ5 — Online discrete bridge

If offline headroom, replay support, and actor-visible scoring are established, does online interaction over the unchanged discrete candidate set and validity rules improve endpoint target gain or calibration over the offline policy?

RQ5 keeps the target, information, candidate, validity, and endpoint assumptions of RQ1–RQ4 unchanged. It is not required for the offline finite-candidate evaluation.

1.3.5.2 RQ6 — Continuous or simulator-backed control

If the finite-candidate evidence is stable, does a continuous or hierarchical target-then-pose policy provide measurable headroom over the best discrete policy under the same target-specific objective and a comparable acquisition or motion budget?

RQ6 changes the action space and feasibility assumptions and therefore requires separate evidence for support, safety, simulator realism, and comparable cost. It remains deferred; the current implementation implies no continuous, simulator, or real-device result.

1.3.6 Shared evidence constraints

Action validity, actor support, target validity, oracle-label validity, and training eligibility remain distinct; missing evidence is not encoded as low utility. Confirmatory comparisons are paired under the same scene, target, candidates, validity regime, and budget. The scene is the experimental unit, and the analysis plan freezes exclusions, aggregation, uncertainty, and decision rules before policy outcomes are inspected.

1.3.7 Research-to-evidence map

RQ	Question role	Primary evidence	Interpretation gate
RQ1	outcome measurement	target reconstruction endpoint gain	frozen repeatable metric; fixed horizon and budget
RQ2	lookahead and learned gap closure	paired greedy, lookahead, myopic-control, and learned-policy outcomes	meaningful oracle headroom before learned gap closure
RQ3	information boundary	matching, ranking, calibration, and leakage audits	end-to-end actor-visible protocol
RQ4	population and support	candidate, replay, validity, and coverage diagnostics	scene-disjoint aggregation
RQ5	conditional extension	matched online discrete-policy evaluation	offline gates satisfied first
RQ6	deferred extension	continuous or simulator-backed evaluation	separate action space and cost comparison

Table 1: Research-question-to-evidence map.

1.4 Thesis structure

The remaining chapters follow that dependency chain. Chapter 2 derives the objective, support, temporal, and representation concepts needed to state the gap precisely. Chapter 3 constructs the experimental world by defining information access, admissible actions, state transitions, and measured outcomes. Chapter 4 then specifies the one finite-horizon value interface evaluated within that world. Chapter 5 states which evidence admits or blocks each claim. Chapter 6 reports the admitted answers in gate order; Chapter 7 attributes their implications and alternatives; and Chapter 8 synthesizes what those bounded answers establish.

2 Foundations and Related Work

The Introduction located this thesis within active perception: sensing actions change the evidence on which later inference depends [AWB88, Baj88]. Three-dimensional view planning turns that principle into a repeated choice of where to observe next [SRR03]. The unresolved dependency is not another planner implementation but a precise account of what a view is valuable **for**, which actions are available, and what information a sequential score may condition on. By the end of the chapter, the thesis question is reduced to three coupled requirements—utility alignment, temporal dependence, and support/state adequacy—before the experimental world is constructed.

The argument follows the dependencies of the decision problem. It first separates the next-best-view mechanism from the objective used to rank views, then distinguishes target-conditioned reconstruction quality from coverage and uncertainty proxies. It next separates candidate-view support from endpoint, transition, and human-motion feasibility before explaining why partial observability and delayed consequences require a finite-horizon information state. The chapter then derives the geometric properties a candidate scorer should preserve and positions the thesis question against the resulting literature dimensions.

2.1 Active Perception and the Next-Best-View Problem

In active perception, an observation is also an intervention on the future information state. A next-best-view (NBV) method operationalizes this idea by generating possible viewpoints, estimating the benefit of observing from each one, and selecting an action. PB-NBV makes this generate–score–select decomposition explicit for a finite candidate set [Jia+25]. The decomposition identifies three different scientific objects: the proposal mechanism determines which views can be considered, the utility determines how they are ordered, and the selection rule turns that ordering into an action.

How those objects are represented varies across NBV systems. Finite-candidate methods evaluate a bounded set of poses, whereas learned continuous policies predict camera position and orientation directly. GenNBV formulates the latter as a five-degree-of-freedom reinforcement-learning problem, while Hestia factorizes the continuous action into a look-at point followed by a camera position [Che+24, Lu+26]. A finite set makes the evaluated choices explicit and comparable; a continuous or hierarchical policy can search a broader space but couples view proposal and control more tightly.

These action formulations do not determine what makes a view useful. PB-NBV, for example, accelerates candidate evaluation through projected frontier and occupied evidence, so its proposal and scoring mechanism is tied to a coverage objective [Jia+25]. The same finite candidate table could instead be ranked by uncertainty reduction or by reconstruction error. Consequently, changing the utility changes the scientific task even when candidate generation and selection remain unchanged.

The NBV mechanism therefore answers only **how** a sensing choice is made. To interpret the choice, the utility must state **which reconstruction consequence** is valued and **for which spatial support**. This distinction leads from the general active-perception loop to the objective families compared next; only after fixing that objective can candidate support and feasibility be separated from view value [Che+24, Jia+25].

2.2 View Utility and Target-Conditioned Reconstruction

Coverage-based utilities value evidence about previously unobserved surface. SCONE estimates surface-coverage gain from occupancy and visibility fields, MACARONS anticipates coverage gain while reconstructing online, and Hestia rewards newly visible voxel faces [GML22, Gué+23, Lu+26]. These objectives connect a view to geometric completeness, but they weight every counted surface element through the coverage definition; they do not directly ask whether the reconstructed geometry became more accurate.

Information-based utilities instead value a view through its expected reduction of model uncertainty. FisherRF derives an information-gain criterion from the Fisher information of a radiance field [JLD24]. Such a criterion can prefer a view that constrains uncertain model parameters even when its immediate surface coverage is small. The benefit is model-aware exploration; the limitation is that lower parameter uncertainty and lower geometric reconstruction error are related only through assumptions about the model and the evaluated scene.

Direct quality utilities close this proxy gap by evaluating the reconstruction itself. VIN-NBV defines Relative Reconstruction Improvement (RRI) as the reduction in Chamfer distance between reconstructed and ground-truth point clouds after adding a candidate observation, and learns to rank sampled query cameras by this quantity [Fra+25]. The present work follows this direct reconstruction-improvement principle but not VIN-NBV's metric unchanged: it evaluates a target-cropped point–mesh error instead of scene-level point-cloud Chamfer distance. This project-specific definition makes the requested target, rather than all reconstructed geometry, determine which errors count. It does not remove the myopia of one-step scoring, because each candidate is still valued by its immediate improvement rather than by the sequence it may enable.

Target conditioning changes which reconstruction errors contribute to that quality. Object-centric 3DGS planning associates features and confidence with individual Gaussian primitives and gates its information score by the requested object [Jeo+26]. This mechanism demonstrates that a view can be informative for one object without being equally useful for the scene as a whole. It also exposes a separate assumption: supplying an object identifier or region for conditioning is not the same problem as discovering that object and maintaining its identity from observations.

The thesis therefore treats coverage and uncertainty as non-equivalent diagnostics and adopts target-specific reconstruction quality as the endpoint of interest [Fra+25, Jeo+26]. These works motivate direct reconstruction improvement and object conditioning separately; their combination in a target-cropped endpoint is the thesis-specific objective. This choice determines what the decision should improve, but not which viewpoints may be proposed or reached. Candidate support and motion feasibility therefore remain separate from utility before delayed consequences are introduced through the finite-horizon decision problem [Jia+25].

2.3 Candidate-View Support and Motion Feasibility

A finite-candidate policy selects only from the pose rows in its proposed candidate table $\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_t}$. PB-NBV uses a reachability- and camera-conditioned partial hemisphere of target-facing candidates [Jia+25]. Proposal geometry therefore bounds what even a perfect scorer can select.

Target-relative coordinates provide a useful proposal prior without defining the objective. PB-NBV points candidates toward the object centre, while Hestia factorizes a viewpoint into a predicted look-at point and a camera position [Jia+25, Lu+26]. Such parameterizations concentrate support on a requested target, but an orbit or target-facing arc remains a coverage geometry, not evidence that a wearer normally moves that way. Project Aria explicitly contrasts carefully curated scanning motion with natural, unconstrained egocentric activity, so wearable plausibility must be checked against trajectory evidence rather than inferred from a target-relative construction [Eng+23].

Geometric admissibility has at least two levels. Hestia shifts a proposed camera position to its nearest collision-free endpoint, whereas Next Best Sense obtains a list of feasible candidate views but still falls back to the next-ranked view when inverse kinematics or trajectory planning fails [Lu+26, Str+24a]. A collision-free endpoint therefore does not by itself establish that the transition from the current state is executable.

Physical feasibility and wearable plausibility are likewise different requirements. Robot reachability constrains PB-NBV's hemisphere, while Project Aria supplies calibrated device trajectories recorded during natural activities [Eng+23, Jia+25]. Robot-specific reachability can define a hard admissibility test for that platform; it cannot establish a distribution of realistic human head and body motion. For egocentric data generation, that distribution is an empirical prior to be measured, reported, and validated.

These distinctions map the proposed poses $q_{t,i} \in \mathcal{Q}_t$ to the canonical admissible row-index set $\mathcal{A}(s_t) = \{i : m_{t,i} = 1\}$, where $m_{t,i}$ is the hard-valid indicator after endpoint and transition checks. The utility or value function ranks only those admitted rows; a rejected candidate is unavailable rather than merely low-value [Jia+25, Str+24a]. Separating proposal, admission, and ranking makes the state dependence of available actions explicit [SB18, Secs. 3.1 and 3.5, pp. 47–53 and 58–62] and prepares the sequential decision model introduced next.

2.4 Sequential Decision-Making under Partial Observability

An egocentric reconstruction system never observes the complete physical scene. Its decision can depend on the sequence of earlier actions and observations, not only on the latest frame. In a partially observable Markov decision process, a belief state is a sufficient statistic of that action–observation history for policy choice [LHP23]. ARIA-NBV does not attempt general belief inference; the relevant consequence is narrower: any finite representation used for view selection is a hypothesis about which parts of the history are sufficient for predicting future target improvement.

The preceding section defined the admissible choices as the state-dependent row-index set $\mathcal{A}(s_t)$ after proposal and feasibility checks. That set can change as the target-conditioned reconstruction and current pose change, while a hierarchical controller can also condition camera position on an earlier look-at decision [Jia+25, Lu+26]. Sequential value must therefore be conditioned on the currently available action rather than treating every camera pose as permanently selectable.

Sequentiality matters when the best immediate view is not the best first step of a sequence. For target e, candidate i, and requested horizon h, the finite return and corresponding conditional value are

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \le h \le b_t \le H_{\max}$$

$$Q_{h,e}^{\star}(s_t, i) = \sup_{\pi \in \Pi^{\mathrm{act}}} \mathbb{E}_\pi \left[G_{t,e}^{(h)} \mid s_t, a_t = i \right], \quad i \in \mathcal{A}_t, \quad 1 \le h \le b_t \le H_{\max}, \quad Q_{0,e}^{\star}(s, i) = 0$$

where continuation policies choose only from the admissible candidates at each step. These equations specialize the standard discounted-return and action-value constructions to a requested target and finite horizon [SB18, Secs. 3.3–3.6, pp. 54–67]. Fixed-Horizon TD defines horizon-indexed predictions with the boundary $Q_0 = 0$ and relates horizon h to the shorter horizon $h - 1$ [De +20]. If an episode ends before all h rewards are collected, later rewards are zero. The value thus states the consequence of taking a candidate now and then following a named continuation rule; it is not an intrinsic property of the camera pose alone.

The horizon is distinct from two related bounds. The requested horizon h states how many future acquisition rewards the query represents; the remaining budget b_t limits how many acquisitions are still possible; and $H_{\max}$ marks the largest horizon supported by the chosen model and evidence. Universal Value Function Approximators establish the general possibility of conditioning one approximator on an additional task variable, while Fixed-Horizon TD shares representations across horizon-indexed predictions [De +20, Sch+15]. Neither result licenses extrapolation beyond the horizons actually supported and evaluated.

Learning such values from a fixed data collection introduces an epistemic constraint in addition to the mathematical horizon. Batch-constrained and conservative offline reinforcement-learning methods are motivated by the error that arises when a learned policy assigns high value to actions outside the data distribution [FMP19, Kum+20]. These methods differ in how they control that error, but they share the warning relevant here: a longer horizon does not create evidence for transitions that the data never resolves.

Together, partial observability, state-dependent choices, and finite-horizon return make candidate value explicitly relational [De +20, LHP23]. A camera pose has no context-free value. Its value depends on the causal information state, requested target, currently admissible support, horizon, state-update rule, and continuation policy. The next dependency is therefore representational. The actor state must preserve the distinctions on which those future consequences depend.

2.5 Egocentric and Geometric Representations

The preceding section established that future view value depends on a causal information state rather than on a camera pose alone. Project Aria provides calibrated, time-aligned egocentric streams, and EFM3D lifts posed image features and semidense geometry into a local gravity-aligned representation [Eng+23, Str+24b]. The representation question is therefore not how faithfully to reproduce the entire world, but which distinctions must survive so that target-specific counterfactual return remains predictable.

Pretraining provenance and decision-time role must be distinguished. 3D-NVS uses ImageNet initialization in its NBV classifier, whereas MACARONS reuses pretrained image features within an online mapping pipeline [AKC20, Gué+23]. Next Best Sense instead uses SAM2 and monocular-depth priors to improve 3DGS construction while retaining an analytic Fisher-information selector [Str+24a]. EVL freezes its DINOv2.5 foundation feature extractor, then learns the feature upsampler, 3D U-Net, and task heads that form a local 3D field from posed egocentric streams and semidense geometry [Str+24b]. This structure may preserve semantic and geometric distinctions relevant to target visibility, but whether those distinctions improve candidate value is an empirical property of the downstream task rather than a consequence of pretraining alone.

Four requirements follow. First, **causal sufficiency** requires the actor state to retain the observation history relevant to future return without importing future or unselected evidence. EFM3D supplies strong local evidence but has a finite voxel extent, while Hestia demonstrates that accumulated directional history can alter later choices [Lu+26, Str+24b]. A spatial state is therefore not sufficient merely because it is geometric; sufficiency is a hypothesis to be tested against the task.

Second, **spatial relationality** requires the target, observer, candidates, and observed support to be comparable in a common local geometry. Query-centric models express scene elements relative to the active query, reducing dependence on an arbitrary global origin [Zho+23]. For target-conditioned NBV, the relevant quantity is therefore not candidate position in isolation but candidate geometry relative to the current observer, target, and accumulated evidence. Coordinate choice also determines which changes should be treated as nuisances. Translating the world origin should not change a score, whereas gravity, metric scale, camera direction, target orientation, occlusion, and temporal order can remain informative [Bro+21]. The useful prior is thus relative geometry that removes arbitrary coordinates without erasing task-relevant physical structure.

Third, **candidate-order equivariance** follows from the action set itself. A candidate table denotes physical viewpoints rather than a ranked sequence, so permuting its rows must induce the same permutation of their scores; an invariant output would instead discard which score belongs to which action. Deep Sets provides invariant aggregation for shared set context, while Set Transformer provides permutation-equivariant candidate interaction [Lee+19, Zah+17]. This is a behavioral contract rather than an architecture prescription: independent row scoring, pooled context, and attention can all satisfy it.

Fourth, **physical observability** requires the representation to preserve what the causal sensing process actually distinguishes. EFM3D derives a surface mask from observed semi-dense points and a free-space mask from camera-to-surface rays [Str+24b]. Voxels supported by neither mask remain unobserved; they must not be treated as observed free space or negative surface evidence. This surface/free/unobserved distinction is the observation contract. The choice of carrier is a separate architecture tradeoff: point models retain irregular samples and local relations, sparse-voxel models regularize space at a chosen resolution and extent, and equivariant message passing commits to a chosen symmetry family [CGS19, SHW21, Zha+21]. These families trade spatial fidelity, computational structure, and strength of geometric prior, but none by itself guarantees the observation contract or improves the endpoint utility.

These four requirements—causal sufficiency, spatial relationality, candidate-order equivariance, and physical observability—complete the conceptual dependency chain. The literature synthesis can now compare methods by the scientific distinctions they preserve, while the Method chapter remains responsible for one concrete realization and its tests [Bro+21, Str+24b].

2.6 Related-Work Synthesis and Research Gap

The reviewed approaches differ along six coupled dimensions: utility, target scope, represented evidence, representation provenance and decision-time role, candidate support and admission, and decision horizon [Fra+25, GML22, Jia+25, JLD24]. Comparing only one dimension can obscure the scientific task: a sequential coverage controller does not answer the same question as a greedy reconstruction-quality scorer, and neither comparison is meaningful if the relevant view lies outside candidate support.

Family	Utility and target scope	State evidence and representation role	Support, feasibility, and decision
PB-NBV [Jia+25]	Projected frontier and occupied coverage; reconstruction object	Analytic classified voxel clusters and compact projection proxies	Reachability- and camera-conditioned partial hemisphere; greedy score
3D-NVS [AKC20]	Pairwise voxel reconstruction; object	ImageNet-pretrained VGG16 directly scores fixed next-view classes	Eleven fixed view classes; one-step classifier
SCONE / MACARONS [GML22, Gué+23]	Expected surface coverage; scene-wide	Task-trained occupancy and visibility; pretrained ResNet features support MACARONS depth	Iterative candidate selection
GenNBV / Hestia [Che+24, Lu+26]	Coverage gain; reconstruction object	Task-trained geometric, history, and shallow-image encoders; Hestia uses no external pretraining	Continuous or hierarchical policy; collision handling
FisherRF [JLD24]	Information gain; radiance-field scene model	Per-scene model uncertainty enters an analytic score	Candidate views; greedy score
VIN-NBV [Fra+25]	Direct reconstruction improvement; reconstructed object	Task-trained CNN over candidate-conditioned projections of current reconstruction	Sampled candidates; greedy score
Next Best Sense [Str+24a]	Color–depth information gain; radiance-field scene model	SAM2 and depth priors support 3DGS; Fisher selection remains analytic	Feasible candidates; fallback after kinematic or planning failure
Object-centric 3DGS [Jeo+26]	Object-conditioned information gain; requested object	Foundation-model masks and CLIP support object state and selection; target score remains analytic	Candidate views; greedy score

Table 2: Concept-centred comparison of the literature families that bound the thesis question. Rows are distinguished by utility, target scope, state evidence and representation role, candidate support and feasibility, and decision structure; they are not ranked by reported performance across incompatible settings.

Pretraining alone does not determine the scientific role of a representation. 3D-NVS uses ImageNet initialization directly in a fixed-view classifier, while MACARONS uses pretrained image features inside an online mapping pipeline [AKC20, Gué+23]. Next Best Sense places foundation-assisted masks and depth upstream of an analytic Fisher selector [Str+24a]. By contrast, VIN-NBV, GenNBV, and Hestia learn their decision state from task-specific reconstruction cues or shallow encoders [Che+24, Fra+25, Lu+26]. These approaches establish different uses for transferred visual features, but they do not isolate the value of an egocentric 3D representation that combines frozen 2D foundation features with learned 3D processing for target-conditioned endpoint prediction.

The comparison yields two requirements for the value definition. First, **utility alignment** requires the score to reflect the endpoint of interest: coverage and information criteria reward proxies for reconstruction, whereas VIN-NBV evaluates realized reconstruction error [Fra+25, GML22, JLD24]. Second, **temporal dependence** requires the score to represent delayed consequences. Sequential policies retain history and optimize delayed reward, but the reviewed examples optimize coverage rather than the reconstruction quality of a requested target [Che+24, Lu+26]. Existing work therefore establishes direct quality, target conditioning, and sequentiality separately under different assumptions.

A third requirement is **support/state adequacy**: valuable actions must be available, and the causal state must retain the consequences needed to value later actions. PB-NBV constrains proposal support; Hestia carries directional history while separately repairing infeasible endpoints; and Next Best Sense distinguishes feasible candidates from execution fallback [Jia+25, Lu+26, Str+24a]. Even a perfect scorer cannot exploit a view outside its support, and a longer horizon cannot recover a distinction erased from the state.

Within the actor-state question, representation transfer is a downstream diagnostic. A pretrained carrier can help only if it preserves distinctions aligned with target-specific return; semantic strength in 2D does not by itself provide causal scene memory, metric support, or candidate-relative geometry. EFM3D/

EVL supplies a foundation-derived local field in which a frozen 2D feature extractor feeds learned upsampling, 3D processing, and task heads alongside observed surface and free-space evidence [Str+24b]. Whether that field improves target-conditioned candidate value must therefore be measured against matched actor-visible geometric controls after the core evidence gates, rather than inferred from pretraining.

This thesis studies the core requirements jointly within a bounded setting: whether a finite-candidate policy can improve endpoint reconstruction quality for a specified target by valuing more than the next observation. Actor-visible foundation-derived egocentric evidence is then a conditional representation diagnostic for the corresponding value model [De +20, Fra+25, Str+24b]. Among the reviewed reconstruction-NBV methods, none establishes this conjunction: direct-quality prediction, target conditioning, bounded lookahead, and a matched test of a foundation-derived egocentric representation with frozen 2D features and learned 3D processing. The literature therefore cannot determine whether non-myopic headroom exists or whether EFM3D/EVL-derived evidence improves its recovery; the latter remains a conditional diagnostic rather than another principal question.

The resulting principle is relational rather than architectural: view value is defined by a target, a causal state, admissible support, a horizon, a state update, and a continuation rule. The thesis does not generalize this bounded relation to exhaustive novelty priority, continuous control, actor-visible target discovery, closed-loop motion execution, or natural human movement [Che+24, Eng+23, Lu+26, Str+24a]. Chapter 3 makes the bounded relation operational by separating observable state from the privileged geometry used to construct tasks and evaluate outcomes.

3 Oracle and Data Generation

Chapter 2 established that view value is relational: it depends on the target, causal state, admissible support, horizon, update rule, and continuation policy. This chapter turns that relation into a controlled experimental world. It first separates logged, selected-causal, and privileged information; then constructs target tasks and finite actions; and finally defines the intervention outcome and the immutable evidence needed to audit it. By the end, every value query has an explicit information boundary, action support, transition meaning, and outcome definition for the Method chapter to inherit.

3.1 Information Boundary

The learned component predicts finite-horizon values for a hard-masked candidate table, but its inputs come from only one layer of a larger information lattice. The experiment distinguishes logged observations, evidence caused by selected actions, and privileged counterfactual quantities. ASE contributes both sensor-like streams and ground truth; EFM3D transforms only the logged subset into local 3D evidence [Met25a, Str+24b]. Co-location in one adapted sample or replay row never makes those layers equally observable.

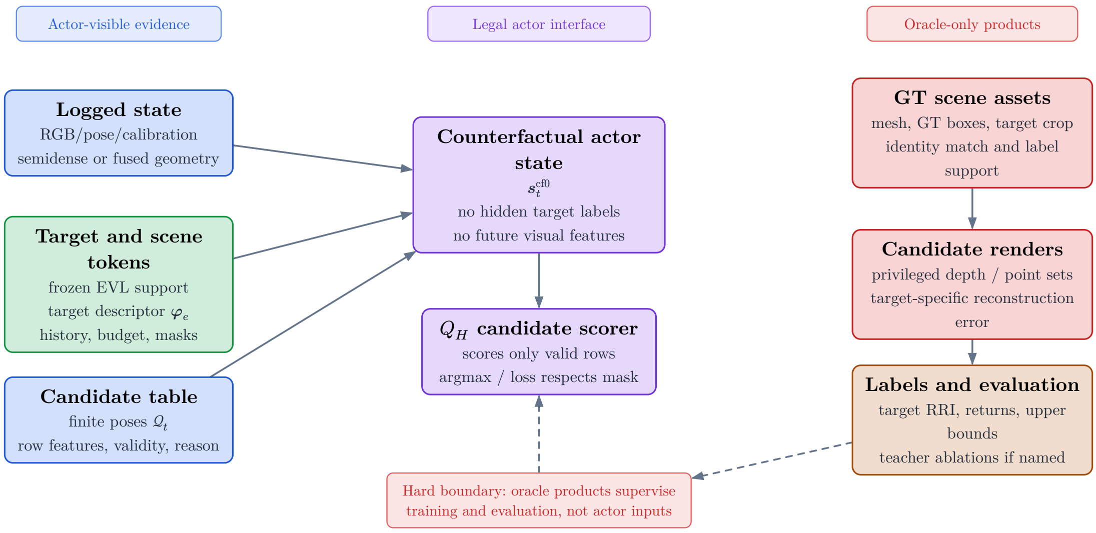


Figure 1: Actor and oracle boundary. Legal Q_H inputs are calibrated logged image evidence, poses, semi-dense geometry with uncertainty and observation support, frozen EVL features or predictions, an explicitly sourced target instruction, selected-view history, remaining budget, candidates, and masks. Reason codes remain audit evidence rather than scorer inputs. GT depth, segmentation, boxes, meshes, target crops, counterfactual renders, labels, and endpoint evaluation remain privileged.

The visibility boundary is protocol-relative and temporal. A dense render for an unselected candidate at the current decision step is oracle evidence. A render from an already selected action may enter a later state only under an explicitly named source protocol: privileged mesh-rendered depth, a declared sensor simulation, or an actor-visible observation. The same array shape can therefore denote different information, and source role must remain explicit.

3.1.1 Logged Observations

Following the conditional action-value definition in Section 2.4, the target is treated as external task context, so the value is conceptually $Q(s_t, e, a)$ rather than a target-independent state value. The logged historic state contains the recorded trajectory evidence,

$$s_t^{\mathrm{hist}} = \left(\boldsymbol{I}^{\mathrm{rgb}}, \boldsymbol{X}, \mathcal{P}^{\mathrm{semi}}, \boldsymbol{F}_v, e_t, b_t\right) \in \mathcal{S}^{\mathrm{hist}}$$

Calibrated image streams, trajectory, camera models, and gravity establish the spatial, metric, and temporal frame in which evidence is compared. EFM3D lifts posed image features into a finite gravity-aligned voxel volume rather than a complete world model [Str+24b]. An out-of-extent target or candidate therefore has missing representation support, not a measured zero and not automatically an invalid pose.

Semi-dense points add sparse surface evidence with uncertainty, timestamps, and observation lineage. EFM3D uses their observing camera centers to distinguish surface support from sampled free-space rays [Met25a, Str+24b]. That distinction is causal: a missing point is not evidence that a voxel is free, and free-space evidence is justified only when its camera lineage is retained.

The target and remaining budget are decision context rather than sensing modalities. The current oracle task derives its target from a selected GT box; an actor-visible target must instead come from a declared observation-derived association path. The information role is fixed by provenance, not by tensor shape.

3.1.2 Selected Causal Evidence

The conceptual counterfactual actor state retains the immutable logged substrate and adds only information causally available after selected actions,

$$s_t^{\mathrm{cf0}} = \left(\boldsymbol{F}_v^{\mathrm{root}}, \mathcal{P}_t, \mathcal{Q}_t, m_{t,i}, \rho_{t,i}, e_t, b_t\right) \in \mathcal{S}^{\mathrm{cf0}}$$

For the canonical model, this compact tuple is read together with the explicit ordered selected-view history:

$$\boldsymbol{h}_{t,e,i} = \mathrm{Read}\left(\boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{h}_e^{\mathrm{tgt}}, q_{t,i}, \boldsymbol{H}_t, t, H\right)$$

In s_t^{cf0}, the root field remains fixed while the accumulated point set may grow only through selected observations. Candidate rows contain poses and proposal provenance, not observations from those poses. The action mask defines admission, the ordered history retains the factual approach sequence, and invalid-reason codes remain audit evidence rather than model input.

The current baseline materializes only root evidence and selected-pose history (the implementation anchor is `qh_cf0_v1`), a deliberately weaker `S0-pose` state:

$$s_t^{\text{S0-pose}} = \left(\mathcal{S}_{\text{root}}^{\text{VIN}}, \boldsymbol{F}_v^{\text{root}}, \left(\boldsymbol{T}_{r,e}, \boldsymbol{l}_e\right), \mathcal{Q}_t, m_{t,i}, \boldsymbol{H}_t^{\text{pose}}, b_t\right)$$

It carries root evidence, the admitted target descriptor, current candidates and hard mask, factual selected-pose prefix, and remaining budget. A richer state would update causal geometry after each selected action:

$$s_t^{\text{cf+}} = \left(s_t^{\text{cf0}}, \left(\boldsymbol{D}, \boldsymbol{V}, \mathcal{P}^{\text{cf}}, \boldsymbol{n}\right)_{1:t}\right)$$

The implemented richer control stops earlier: it persists selected mesh-rendered depth with calibration under the privileged `qh_cfplus_gt_depth_v1` source protocol. The carrier is causal with respect to the selected action but remains privileged and is not yet fused into surface, free-space, uncertainty, or a refreshed EFM3D field. It therefore cannot be interpreted as the full geometry-updated actor state.

The transition from t to $t + 1$ may use only the observation produced by the selected row a_t. It may fuse its surface and ray evidence and update support, uncertainty, direction, and recency. It may not attach image features, detections, or EVL descriptors unless the protocol also provides the calibrated camera streams from which EFM3D derives them. Counterfactual geometry must therefore retain a source role and explicit absence masks for unavailable image-derived channels.

The counterfactual state is thus an **information state** for decision making, not automatically a complete Markov state of the physical scene. It becomes task-sufficient only if its retained root context, causal dynamic evidence, explicit history, target context, action support, and budget preserve every distinction needed to predict future target-specific return. `qh_cf0_v1` is the deliberately weaker `S0-pose` baseline; `qh_cfplus_gt_depth_v1` is a privileged depth carrier, not yet a realization of the geometry-updated state.

3.1.3 Privileged Counterfactual Evidence

The oracle state adds complete or counterfactual quantities outside the actor input graph:

$$s_t^{\text{oracle}} = \left(s_t^{\text{cf+}}, \mathcal{M}^{\text{GT}}, \mathcal{M}_e^{\text{GT}}, \boldsymbol{D}_q, \mathcal{P}_q, \text{RRI}\right) \in \mathcal{S}^{\text{oracle}}$$

$$\mathcal{M}_{\text{NBV}} = \left(\mathcal{S}^{\text{hist}}, \mathcal{S}^{\text{cf0}}, \mathcal{S}^{\text{oracle}}, \{\mathcal{A}_t\}, T, r_t^e, \gamma, H\right)$$

ASE provides per-frame metric GT depth aligned with RGB, per-pixel instance identifiers, class mappings, and the scene trajectory; the EFM3D release adds ASE OBB metadata and validation meshes for object-detection and surface-reconstruction supervision [Met25a, Str+24b]. EFM3D uses the depth channel to supervise occupancy at sampled free, surface, and behind-surface points, while visible OBBs supervise centerness, class, and box geometry [Str+24b]. These uses establish label provenance, not actor observability.

For ARIA-NBV, $\mathcal{M}^{\mathrm{GT}}$ is the privileged reference surface used for candidate rendering and reconstruction evaluation, while $\mathcal{M}_e^{\mathrm{GT}}$ fixes the selected entity's evaluation support. All-candidate depth $\boldsymbol{D}_q$ and backprojected point sets $\mathcal{P}_q$ answer counterfactual queries for unselected rows. RRI then compares target reconstruction error before and after adding that candidate evidence. Depth, points, crops, and RRI are legal for label generation, oracle search, diagnostics, and named upper bounds; none is a contemporaneous input to the student value model.

A GT OBB may define the current oracle identity, pose, extent, crop, and target-bearing instruction. It does not show that the actor detected or associated that object. A deployable protocol must obtain the target hypothesis from EVL predictions or another observation-derived path, retaining ASE identity, segmentation, OBBs, and meshes only for matching and evaluation.

The information lattice therefore separates **what was logged**, **what a selected history causally adds**, and **what only the oracle can know**. Its layers are connected by controlled projections, not by freely copying arrays between stores. Persisting an oracle tensor beside actor evidence for efficient replay never changes its information role.

3.1.4 End-to-End Actor Visibility

Visibility answers whether a modality contains evidence for a spatial element from a particular view; feasibility answers whether an action is admissible; utility answers how beneficial an admissible action is for the target. These concepts must not be collapsed. A target can be weakly visible from a geometrically valid candidate, a candidate can lie outside EVL support while remaining physically reachable, and an oracle render can fail even though the candidate pose itself is feasible.

Invalidity is a constraint rather than a reward value. Geometry-invalid candidates receive a hard action mask and a persisted candidate reason code before policy selection, stochastic normalization, loss construction, and bootstrap maximization. A geometrically feasible candidate may still have low or negative target gain. Failure of the separate oracle evaluation does not create a candidate reason code: depending on the configured recipe, the affected row or table is skipped, or its oracle-label validity and Q-training eligibility are cleared and the oracle failure is reported separately. Oracle-derived feasibility must be distinguished from a deployable validity estimate when a policy moves from privileged target tasks to an actor-visible target protocol.

Actor visibility is consequently an end-to-end property of the decision protocol. It includes how the target instruction is obtained, how candidate support is proposed, how the hard mask is computed, which selected observation updates the next state, and which fields reach the scorer. A scorer can be free of privileged tensors while still choosing from support oriented or pruned with privileged geometry. Such a protocol is a valid bounded oracle or control experiment, but its result cannot silently become a claim about independently actor-constructed actions.

3.2 Task and Action Construction

Validation TODO: Validate repeatability, failure handling, construct interpretation, root normalization, and the endpoint estimand before describing RRI as a reliable study objective.

Source: docs/literature/tex-src/arXiv-VIN-NBV; metric-validation artifacts

Gate: frozen metric protocol, repeatability table, and claim-level source locators

An oracle target task fixes entity identity, evaluation support, and the target instruction used to construct target-bearing actions. An actor-visible task would instead obtain its descriptor from observations. The present descriptor therefore defines a controlled requested object and its geometry; it does not establish actor-visible target discovery.

The general descriptor notation remains

$$\boldsymbol{\varphi}_e = \mathrm{Enc}_{\mathrm{tgt}}\Big(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{\pi}_e, A_e^{\mathrm{proj}}, n_e^{\mathrm{semi}}, n_e^{\mathrm{EVL}}, \omega_e^{\mathrm{EVL}}, \ell_e^{\mathrm{src}}, \boldsymbol{T}_{r_t,e}, \boldsymbol{T}_{c_t,e}\Big)$$

The finite action interface consists of a candidate table $\mathcal{Q}_t$, hard validity mask $\boldsymbol{m}_t$, and invalid-reason vector $\boldsymbol{\rho}_t$. The admissible set is $\mathcal{A}_t = \big\{i : m_{t,i} = 1\big\}$. Invalid rows remain logged for coverage and failure analysis but lie outside policy argmax, sampling, loss targets, and bootstrap maximization. Low RRI is a valid low-utility outcome; it is never an encoding for infeasibility.

For target e, the oracle computes the target-cropped point–mesh error defined by the frozen outcome protocol. Candidate selection and Q_H supervision use root-normalized target gain; state-relative RRI is retained as a diagnostic. Fixed-budget endpoint gain is the intended policy estimand, but the current replay store records cumulative selected-chain gains rather than an independent post-horizon endpoint reconstruction for every policy. Confirmatory policy comparisons must therefore add matched oracle endpoint re-evaluation or an explicitly persisted endpoint record.

3.2.1 Target Selection

The implemented sampler does not match actor proposals to GT objects. It enumerates the non-padding GT OBB rows in a snippet, accepts rows with finite positive geometry, and applies seeded uniform sampling without replacement up to the configured per-snippet cap. Thus the stored `matched` status currently means geometry-valid GT task row, not successful proposal-to-identity association. IoU, ambiguity-gap, visibility, and support thresholds are absent from this admission rule.

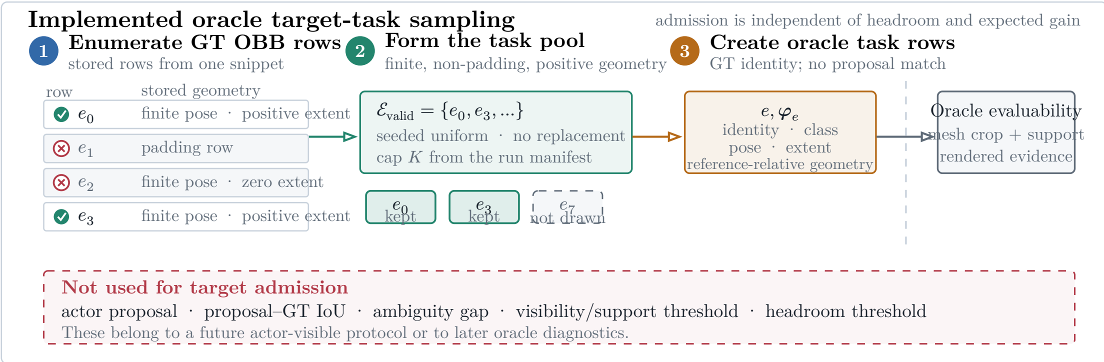


Figure 2: Implemented oracle target-task sampler. Geometry-valid GT OBB rows form the task pool, and seeded uniform sampling without replacement applies the manifest-defined cap. Rollout scoring later decides whether the selected crop is evaluable. No actor proposal, IoU match, visibility gate, or support gate is used.

The sampler bounds rollout cost without pre-filtering on headroom. Candidate scoring may subsequently invalidate the task when its mesh crop, current support, or rendered evidence is unusable; otherwise near-solved and negative-gain targets remain scientifically informative. A later actor-visible protocol must introduce proposal identity and observation-quality diagnostics as a separate selection stage rather than retroactively interpreting the present oracle fields as measurements.

3.2.2 Observed-Target Admission

The observed-target audit (implementation anchor `V1`) keeps the oracle task sampler separate from actor-visible target selection. For each observed or predicted target record $\hat{e}$, it compares only same-class GT OBB rows using oriented 3D IoU:

$$\mu_{\text{IoU}}(\hat{e}, e) = \text{IoU}_{3\text{D}}\left(\hat{\boldsymbol{B}}_{\hat{e}}, \boldsymbol{B}_e\right)$$

$$\tau_{\text{IoU}} = 0.20$$

A GT row qualifies only when its IoU is strictly greater than 0.20. The matcher admits the observed target iff exactly one GT row qualifies:

$$n_{\text{qual}}(\hat{e}) = \text{card}(\{e \in \mathcal{E} : \kappa(\hat{y}_{\hat{e}}, y_e) = 1 \wedge \mu_{\text{IoU}}(\hat{e}, e) > \tau_{\text{IoU}}\})$$

$$a_{\text{id}}(\hat{e}) = 1 \text{ iff } n_{\text{qual}}(\hat{e}) = 1$$

Equality at 0.20 is rejected, as are zero or multiple qualifying rows. The matching rule has no runner-up-gap or composite score. The current rollout sampler remains a distinct oracle-task path: geometry-valid GT-row admission is not evidence of actor-visible matching.

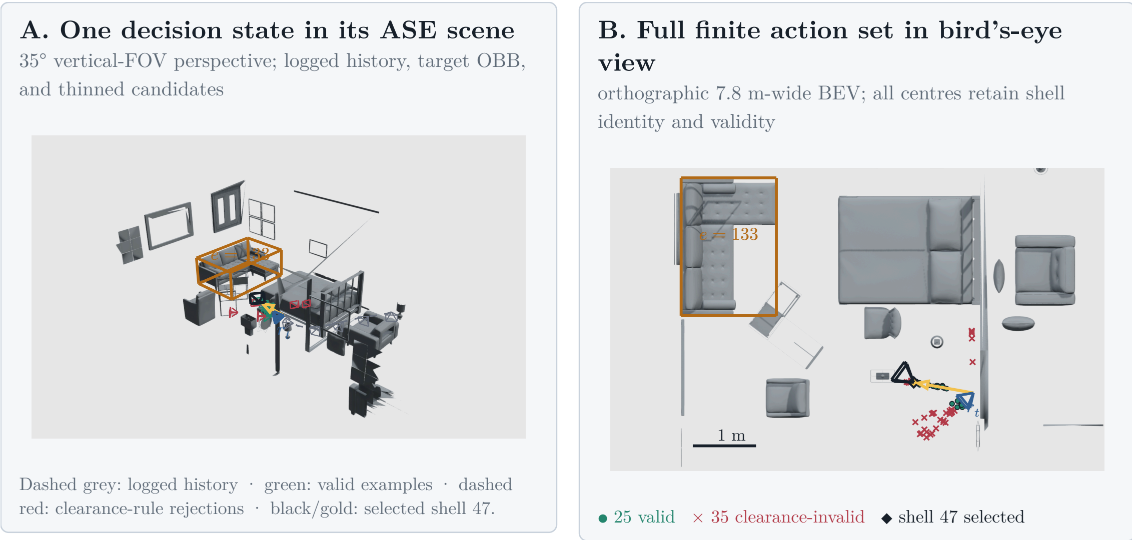
3.2.3 Candidate View Generation

Candidate generation turns one oracle instruction into a finite action table. Every quantitative choice—source population, target cap, shell size, family weights, motion limits, pruning thresholds, rollout recipes, renderer settings, and retention policy—belongs to the resolved run manifest and report bundle. The equations in this subsection instantiate the frozen `realistic_core_60` intervention profile: $N_q = 60$ rows split 24/24/12 across the forward-local, target-bearing-local, and lateral-target-bypass families; $\kappa = 8$; family-conditioned radius intervals of $[0.25, 1.1]$ metres for forward-local and lateral-target-bypass rows and $[0.4, 1.1]$ metres for target-bearing-local rows; the displayed family coefficients and motion limits; and the $N_{\text{valid}} \geq \max\big(12, \text{ceil}\big(0.25 N_q\big)\big)$ root-support gate. These values are neither universal candidate semantics nor library defaults. A run's resolved manifest remains authoritative, and another profile defines a different intervention that must be reported and analysed separately.

At rollout step t, candidate generation constructs a full shell

$$\mathcal{Q}_t = \left\{q_{t,i}\right\}_{i=1}^{N_q}, \quad N_q = 60$$

with a fixed provenance component $k(i)$ per row. The core mixture contains forward-local, target-bearing, and lateral-bypass motion; the diversity challenger may additionally allocate mass to local refinement and revisit/backtrack components. The resolved manifest, rather than prose, determines which components and counts apply to a run.



Scene 81286, sample `ASE_81286_Atek_000035`, rollout row 73, step row 121 24 forward-local + 24 target-bearing

1. 12 lateral-bypass candidates. Dense mesh layers are

z-buffered raster; OBBs, paths, centres, and wire frusta remain vector geometry.

Figure 3: One pinned finite-candidate decision state from ASE scene 81286, sample `ASE_81286_Atek_000035`, rollout row 73, and step row 121. Panel A uses a 35-degree vertical-FOV perspective view to place logged camera history, root state, target OBB, selected path, and a deterministically thinned set of wire frusta in the real scene. Panel B uses a 7.8-metre-wide orthographic bird's-eye view and retains all 60 candidate centres: 25 rows are admissible, 35 are hard-rejected by the clearance rule, and oracle-greedy selects shell 47. Dense scene geometry is z-buffered; OBBs, paths, centres, and camera glyphs remain vector overlays. Full eye, look-at, up, clipping, and resolution parameters are recorded in the figure JSON. The example fixes action support and validity for inspection; it is not a policy-performance result.

For each row, the raw direction is sampled in the reference rig frame. The frozen `realistic_core_60` profile uses the forward-biased Power Spherical distribution from the current implementation [DAT20]:

$$\boldsymbol{u}_i \sim \mathrm{PS}(\boldsymbol{e}_z, \kappa), \quad p(\boldsymbol{u}) = c_\kappa (1 + \boldsymbol{e}_z^T \boldsymbol{u})^\kappa, \quad \kappa = 8$$

Sampler construction or sampling failure aborts generation with the resolved strategy, device, and concentration in the error; the implementation does not silently substitute another distribution. This preserves the intervention identity asserted by the manifest. A successful draw is mapped into configured azimuth and elevation caps without rejection. With $\psi = \mathrm{atan2}(u_x, u_z)$ and $u_y = \sin\theta$, the cap transform is

$$\psi' = \psi \frac{\Delta_\psi}{2\pi}, \quad y' = \sin\theta_{\min} + \frac{u_y + 1}{2} \cdot (\sin\theta_{\max} - \sin\theta_{\min})$$

$$\boldsymbol{d}_i^0 = \left(\sqrt{1 - y'^2}\sin\psi', y', \sqrt{1 - y'^2}\cos\psi'\right), \quad \left\|\boldsymbol{d}_i^0\right\|_2 = 1$$

The sampler draws a capped direction in the reference rig frame and reinterprets it as egocentric forward motion, target-bearing motion, lateral bypass, local refinement, or backtracking according to component provenance. Target-looking families orient their optical axis toward the oracle instruction. In this chapter that point is privileged; calling the generator target-conditioned does not make the point actor-visible.

Let $\boldsymbol{f} = \boldsymbol{e}_z$ be the rig-forward unit vector, $\boldsymbol{b}_e$ the supplied target bearing in the reference frame, $\boldsymbol{l}_e = \frac{\boldsymbol{e}_y \times \boldsymbol{b}_e}{\|\boldsymbol{e}_y \times \boldsymbol{b}_e\|_2}$ the horizontal lateral direction, and $\boldsymbol{e}_y$ the world-up direction expressed in the sampling frame. The three core position families reinterpret the capped direction through raw family vectors $\boldsymbol{g}_i^k$ and their explicit Euclidean normalization:

$$\boldsymbol{g}_i^{\text{forward}} = \boldsymbol{f} + \alpha_f\big(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{f})\boldsymbol{f}\big), \quad \boldsymbol{d}_i^{\text{forward}} = \frac{\boldsymbol{g}_i^{\text{forward}}}{\left\|\boldsymbol{g}_i^{\text{forward}}\right\|_2}, \quad \alpha_f = 0.45$$

$$\boldsymbol{g}_i^{\text{target}} = \boldsymbol{b}_e + \alpha_t\big(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{b}_e)\boldsymbol{b}_e\big), \quad \boldsymbol{d}_i^{\text{target}} = \frac{\boldsymbol{g}_i^{\text{target}}}{\left\|\boldsymbol{g}_i^{\text{target}}\right\|_2}, \quad \alpha_t = 0.4$$

$$\boldsymbol{g}_i^{\text{bypass}} = 0.55\boldsymbol{b}_e + 0.85\,\text{sign}\big(d_{i,x}^0\big)\boldsymbol{l}_e + \text{clip}\Big(d_{i,y}^0, -0.35, 0.35\Big)\boldsymbol{e}_y, \quad \boldsymbol{d}_i^{\text{bypass}} = \frac{\boldsymbol{g}_i^{\text{bypass}}}{\left\|\boldsymbol{g}_i^{\text{bypass}}\right\|_2}$$

Finally, the sampler draws a radius and transforms the reference-frame offset into world coordinates:

$$r_i \sim \mathcal{U}\Big(r_{\min}^{k(i)}, r_{\max}^{k(i)}\Big), \quad \boldsymbol{c}_i^w = \boldsymbol{T}_r^w\Big(r_i \boldsymbol{d}_i^{k(i)}\Big)$$

The forward-local family keeps the reference rig orientation. Target-looking families orient the camera toward the supplied target center $\boldsymbol{p}_e$:

$$\boldsymbol{z}_i^w = \frac{\boldsymbol{p}_e - \boldsymbol{c}_i^w}{\|\boldsymbol{p}_e - \boldsymbol{c}_i^w\|_2}, \quad \boldsymbol{g}_i^{\text{up}} = \boldsymbol{e}_y - \big(\boldsymbol{e}_y^T \boldsymbol{z}_i^w\big)\boldsymbol{z}_i^w, \quad \boldsymbol{y}_i^w = \frac{\boldsymbol{g}_i^{\text{up}}}{\|\boldsymbol{g}_i^{\text{up}}\|_2}, \quad \boldsymbol{x}_i^w = \boldsymbol{y}_i^w \times \boldsymbol{z}_i^w$$

These equations describe the sampler, not an optimal proposal distribution. The forward-local family preserves egocentric continuity, the target-bearing-local family moves along the target ray, and the lateral-target-bypass family introduces side-step views; optional challenger families test smaller corrections and reversals. Candidate-profile utility must be judged after pruning. A store in which target-aware families rarely survive cannot support a target-conditioned planning claim.

The optional `target_orbit` family is a bilateral partial-orbit endpoint proposal prior: it samples paired signed angular offsets around the supplied target while preserving the current horizontal target standoff. Feasibility pruning may remove either side, so bilateral attempt does not imply bilateral surviving support. This family is not part of the production mixture and is not a learned human-motion prior; it is an opt-in challenger for testing whether local orbit coverage improves candidate support.

Pruning converts the full shell into a compact valid-action table. A row remains valid only if it lies in the snippet occupancy support, stays clear of the GT mesh, avoids straight-line path collision, and satisfies local egocentric motion limits:

$$\|\boldsymbol{o}_i\|_2 \leq 1.0 \text{ m}, \quad |\Delta h_i| \leq 0.25 \text{ m}, \quad \max\big(0, -o_{i,z}\big) \leq 0.25 \text{ m}, \quad \Delta\psi_i \leq 70 \text{ deg}$$

The full shell is retained with position, strategy, mixture, sampling probability, rule masks, diagnostics, and invalid-reason bitsets. Panel B of Figure 3 makes the distinction concrete for one stored table: invalid rows remain inspectable but sit outside the admissible set. Invalid candidates cannot enter Q_H selection, stochastic normalization, or loss targets. Conversely, a feasible row with weak target support or low expected gain remains a valid low-utility example rather than receiving an invalid-reason code. A manifest-defined root-support threshold may reject an entire rollout task when too few actions remain:

$$N_{\text{valid}} \geq \max\big(12, \text{ceil}\big(0.25 N_q\big)\big)$$

This threshold is a data-support guard, not a reward threshold. Preflight reporting must retain rejected-root counts and failure reasons by candidate family.

3.2.4 Candidate-Support and Geometric-Coverage Diagnostics

Candidate quality is audited before reward interpretation. Let $\mathcal{I}_s$ be all attempted, non-padding rows in state s, $\mathcal{V}_s \subset \mathcal{I}_s$ the rows that pass actor-valid geometry and hard-action checks, $\mathcal{L}_s \subset \mathcal{V}_s$ the rows with a finite oracle label. The horizon-specific mask $m^{Q,h}_{s,i}$ selects rows in $\mathcal{L}_s$ for Q-training; it does not redefine the canonical candidate table $\mathcal{Q}_t$. The first transition is represented by the canonical action mask $m^{\mathrm{act}}_{t,i}$; finite factual value-target availability is represented separately by $m^{Q,h}_{t,i}$. These are distinct populations: a row can be actor-valid without being renderable, oracle-labelled, or retained for training. The candidate actor-valid fraction is therefore

$$f_{\mathrm{actor}(s)} = \frac{\sum_{i \in \mathcal{I}_s} m^{\mathrm{act}}_{s,i}}{|\mathcal{I}_s|}$$

and hard valid support is

$$n_{\mathrm{valid}(s)} = |\{i \in \mathcal{I}_s : m^{\mathrm{act}}_i = 1\}|$$

Both quantities are computed per decision state before pooling. The configured family zero rate retains every configured family/state pair in its denominator, including pairs with zero valid rows:

$$z_{\mathrm{family}(s)} = \left(\frac{1}{|\mathcal{F}_s|}\right) \sum_{f \in \mathcal{F}_s} \mathbb{1}\left[n_{\mathrm{valid}(s,f)} = 0\right]$$

Every state statistic $q(s)$ is then averaged within its scene and finally macro-averaged across scenes:

$$\mathcal{S}_{c,q} = \{s \in \mathcal{S}_c : q(s) \text{ is defined}\}, \quad |(q)_c = \left(\frac{1}{|\mathcal{S}_{c,q}|}\right) \sum_{s \in \mathcal{S}_{c,q}} q(s), \quad \mathcal{C}_q = \left\{c \in \mathcal{C} : |\mathcal{S}_{c,q}| > 0\right\}, \quad |(q) = \left(\frac{1}{|\mathcal{C}_q|}\right) \sum_{c \in \mathcal{C}_q} |(q)_c$$

Here $\mathcal{S}_{c,q}$ is the metric-specific eligible-state set and $\mathcal{C}_q$ contains scenes with at least one eligible state. A scene with no defined state for q is absent from the descriptive scalar and counted in the accompanying undefined-scene table. A paired profile contrast uses the intersection of the two profiles' eligible-scene sets and reports every excluded scene. Thus the available-case denominator is explicit rather than an implicit row-level complete-case analysis.

The analysis manifest freezes the missing-state rule before result inspection. For support counts and fractions, a failed generation root contributes zero support; for the family-zero rate it contributes one. Target-relative balance and span are not numerically imputed when no target-conditioned direction is defined: the failed state remains in the failure table, and a paired scene is ineligible for a geometric profile contrast. Thus no missing state disappears through an implicit complete-case denominator.

For geometric comparison, candidate centres and target centres use the existing proposal-support normalization

$$d_{t,e}^{\text{current}} = \left\| \boldsymbol{p}_e^w - \boldsymbol{c}_{r_t}^w \right\|_2, \quad \tilde{\boldsymbol{c}}_{t,i}^{\text{support}} = \left(\boldsymbol{B}_{r,t}^{\text{Z-up}}\right)^\top \frac{\boldsymbol{c}_{t,i}^w - \boldsymbol{c}_{r_t}^w}{d_{t,e}^{\text{current}}}, \quad \tilde{\boldsymbol{p}}_{t,e}^{\text{support}} = \left(\boldsymbol{B}_{r,t}^{\text{Z-up}}\right)^\top \frac{\boldsymbol{p}_e^w - \boldsymbol{c}_{r_t}^w}{d_{t,e}^{\text{current}}}, \quad \left\| \tilde{\boldsymbol{p}}_{t,e}^{\text{support}} \right\|_2 = 1$$

: the origin is the factual expansion/root pose, the target has unit norm, the vertical axis is world-up, and distances are divided by the current root-to-target distance $d_{t,e}^{\text{current}}$. This is a target-relative support frame, not a camera image frame and not a claim that the root or target is visible. Side balance is computed from non-neutral attempted target-conditioned rows in the `target_bearing_local` and `target_orbit` families, using their target-relative azimuth signs. The neutral bin is $|y_{s,i}^{\text{target}}| \le \varepsilon$ with $\varepsilon = 10^{-9}$ in normalized support coordinates; its count is reported but does not enter the two-sided balance denominator,

$$> \varepsilon\Big], \quad n_-(s) = \sum_{i \in \mathcal{J}_s^{\text{target}}} \mathbb{1}\Big[y_{s,i}^{\text{target}} < -\varepsilon\Big], \quad n_0(s) = \sum_{i \in \mathcal{J}_s^{\text{target}}} \mathbb{1}\Big[\!\!\Big[y_{s,i}^{\text{target}}| \le \varepsilon\Big], \quad b_{\text{side}(s)} = 1 - |n_+(s) - n_-(s) \frac{|}{n_+(s) + n_-(s}$$

and orbit coverage uses the circular minimum-covering span rather than a Cartesian maximum-minus-minimum range:

$$\alpha_{s,1} \le \ldots \le \alpha_{s,n_s}, \quad \alpha_{s,n_s+1} = \alpha_{s,1} + 2\pi, \quad s_{\text{orbit}(s)} = 2\pi - \max_{1 \le j \le n_s} \left(\alpha_{s,j+1} - \alpha_{s,j}\right)$$

The projection fraction records whether the calibrated target centre lies inside the calibrated image domain for evaluated rows in one state,

$$f_{\text{proj}(s)} = \frac{\sum_{i \in \mathcal{J}_s^{\text{evaluated}}} \mathbb{1}\left[i \in \mathcal{J}_s^{\text{projected}}\right]}{|} \mathcal{J}_s^{\text{evaluated}}|, \quad |\mathcal{J}_s^{\text{evaluated}}| > 0$$

and is then reduced through the same state–scene–cohort macro operator. A state with no evaluated projection rows is undefined and counted separately; it is not assigned a zero fraction or silently removed from a paired profile contrast. This remains an evaluation/framing diagnostic, not visibility or actor support. The finite-support oracle opportunity is the per-state maximum target-root gain,

$$o_{\mathrm{oracle}(s)} = \max_{i \in \mathcal{L}_s} g_{s,i}^{\text{target-root}}, \quad |\mathcal{L}_s| > 0$$

which is defined only when the actor-valid, finite-label set $\mathcal{L}_s$ is nonempty. Its undefined-state count is reported, and eligible values are reduced through the same state–scene–cohort macro operator. The quantity measures available candidate headroom and is not a policy score. View jitter is likewise computed per state and macro-reduced. States without jitter rows, or without bounded rows for the cap-compliance diagnostic, are counted as undefined. The nonzero seminar-profile check remains separate from legacy zero-cap spherical support:

$$f_{\mathrm{jitter}(s)} = \frac{\sum_{i \in \mathcal{J}_s^{\text{jitter}}} \mathbb{1}[b_i = 1] \cdot \mathbb{1}[|\Delta\psi_i| \le |(\psi)_i] \cdot \mathbb{1}[|\Delta\theta_i| \le |(\theta)_i]}{\sum_{i \in \mathcal{J}_s^{\text{jitter}}} \mathbb{1}[b_i = 1]}, \quad \sum_{i \in \mathcal{J}_s^{\text{jitter}}} \mathbb{1}[b_i = 1] > 0$$

The production mixture retains the seminar model's nonzero jitter invariant. Legacy `uniform_sphere` and `forward_powerspherical` rows with a zero cap are annotated as uncapped spherical support and are not represented by a bounded box.

Quantity	Population	Unit	Interpretation boundary
Actor-valid fraction	Attempted rows per state	fraction	Hard feasibility; not label or training availability.
Valid support	Actor-valid rows per state	rows	Report the lower tail and failed roots, not only a pooled mean.
Configured-family zero rate	Configured families per state	fraction	Retains absent and zero-support families; state values are scene-macro reduced.
Side balance / circular span	Non-neutral attempted target-conditioned rows	fraction / angle	Neutral rows and undefined states are reported separately; proposal coverage precedes pruning.
Target-centre projection	Evaluated candidate views per state	fraction	State–scene–cohort macro; zero-evaluated states are reported as undefined. Calibrated framing, not visibility.
Oracle opportunity	Finite actor-valid rewards per eligible state	gain	Undefined when no finite actor-valid label exists; available one-step headroom, not achieved policy return.
Jitter compliance	Bounded-jitter rows per state	fraction	State–scene–cohort macro; report undefined, nonzero, and uncapped support separately.

Table 3: Candidate-generation diagnostics and their aggregation populations. Every state-level quantity is reduced before scene and cohort aggregation.

3.2.5 Rollout Branch Sampling and Dataset Impact

Rollout recipes select and retain finite chains from the valid action table. The implemented families are uniform valid sampling, one-step oracle greedy selection, bounded oracle lookahead, and temperature-softmax sampling. Their horizon, branch factor, beam width, temperature, and seed are resolved parameters. These recipes generate replay diversity and bounded references; they do not constitute a learned policy.

Bounded oracle lookahead can select a different first action from one-step greedy because it ranks retained finite-horizon chains rather than immediate gain alone (Figure 4). In reinforcement-learning terms, this is decision-time planning: computation is focused on the current choice by expanding possible consequences before acting [SB18, Sec. 8.8, pp. 180–181]. The persisted artifact contains these selected or beam-retained chains and their full per-step candidate shells; it is not an exhaustive materialization of the counterfactual action tree.

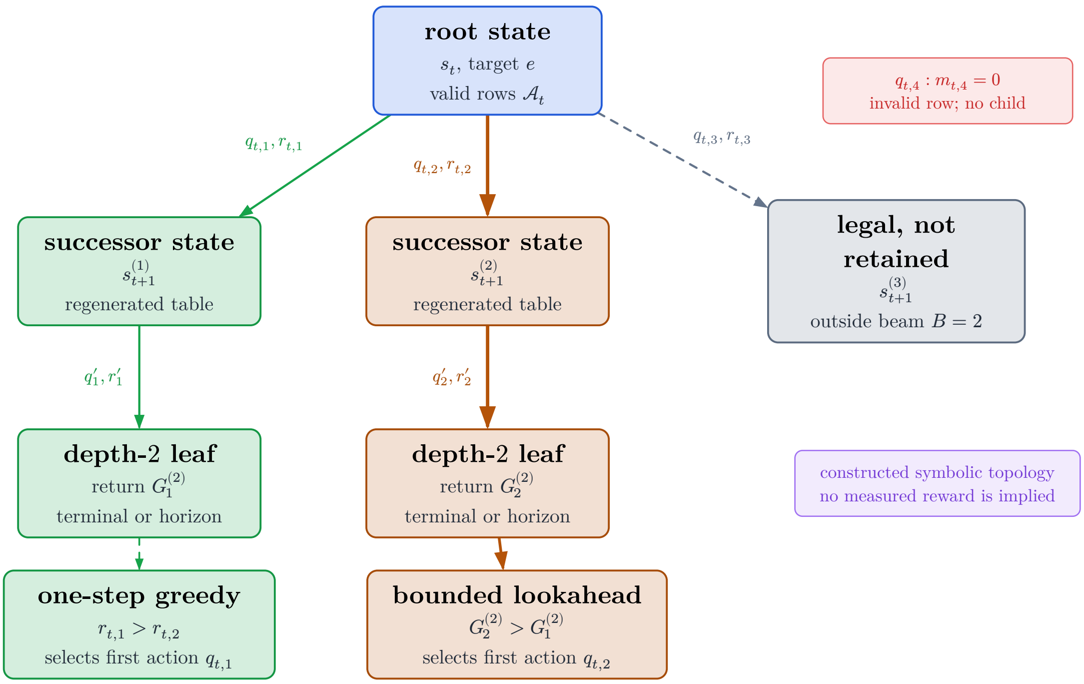


Figure 4: Constructed symbolic topology of the bounded oracle-lookahead reference. Nodes are counterfactual states and edge labels are candidate actions with immediate rewards. Only valid first-action rows produce children; the beam retains a bounded prefix set, and invalid rows receive no branch. The inequalities illustrate how the selected first action can differ from one-step greedy without inventing measured reward values.

For stochastic branches, let s_i be the finite oracle score of valid row i. The robust logit used for temperature-softmax is

$$\ell_i = \frac{s_i - \operatorname{median}(s)}{\operatorname{IQR}(s)\tau}, \quad P(i \mid m_i = 1) = \frac{\exp(\ell_i)}{\sum_{j:m_j=1}} \exp(\ell_j)$$

The target source defines the supervised task, the candidate mixture defines learnable action support, validity defines admissibility, and the recipe defines which chains enter replay. Reporting must therefore cover target-task coverage, valid fanout and invalid reasons by family, selected-family diversity, gain distributions, and retention cost before attributing policy behavior to non-myopic planning. The paired train-only pilots are bandwidth and candidate-profile probes; they are non-confirmatory and provide no held-out policy-performance evidence.

3.3 Measurement and Outcomes

Task construction and action admission specify the intervention; measurement specifies what consequence is compared. The oracle renders a selected candidate from the GT mesh, backprojects its depth, updates the privileged evaluation cloud through the frozen fusion rule, crops points and mesh to the selected target, and evaluates point–mesh error [Fra+25]. The crop, rendering, fusion, and scoring path are oracle-only.

3.3.1 Operational Reconstruction Error

Let $C_e(\mathcal{P}_t)$ denote the oracle-only crop of accumulated evaluation points to the selected target region. In the thesis protocol, its root points are reconstructed from the observed prefix of ASE GT depth; a candidate contributes renderer-derived depth that is backprojected and fused with the same root cloud. MPS semi-dense points remain actor-visible representation input and a legacy diagnostic source, not the current oracle evaluation cloud.

Writing $P = C_e(\mathcal{P}_t)$ and $M = \mathcal{M}_e^{\mathrm{GT}}$ for the target-cropped point set and mesh, the selected directional accuracy is the mean squared distance from each reconstructed point to its closest GT triangle,

$$D_{P\to M}\left(\mathcal{P}, \mathcal{M}^{\mathrm{GT}}\right) = \frac{1}{\|\mathcal{P}\|} \sum_{\boldsymbol{p}\in\mathcal{P}} \min_{\boldsymbol{f}\in\mathcal{F}^{\mathrm{GT}}} d(\boldsymbol{p}, \boldsymbol{f})^2$$

It penalizes reconstructed points that lie away from the target surface and therefore exposes extra, noisy, or misregistered geometry. The reverse term is the mean squared distance from each GT face to its closest reconstructed point,

$$D_{M\to P}\left(\mathcal{P}, \mathcal{M}^{\mathrm{GT}}\right) = \frac{1}{\|\mathcal{F}^{\mathrm{GT}}\|} \sum_{\boldsymbol{f}\in\mathcal{F}^{\mathrm{GT}}} \min_{\boldsymbol{p}\in\mathcal{P}} d(\boldsymbol{p}, \boldsymbol{f})^2$$

It penalizes target faces without nearby reconstructed evidence and therefore exposes missing support. Their sum is the bidirectional point–mesh error,

$$\Delta_t^e = d\big(C_e(\mathcal{P}_t), \mathcal{M}_e^{\mathrm{GT}}\big) = D^e_{P\to M,t} + D^e_{M\to P,t}$$

whose directional terms and total have units of square metres. The current and candidate-augmented clouds pass through the same deterministic fusion and point-cap configuration before scoring. This fusion step makes the before/after comparison operationally reproducible; it does not turn the point set into a continuous surface measure.

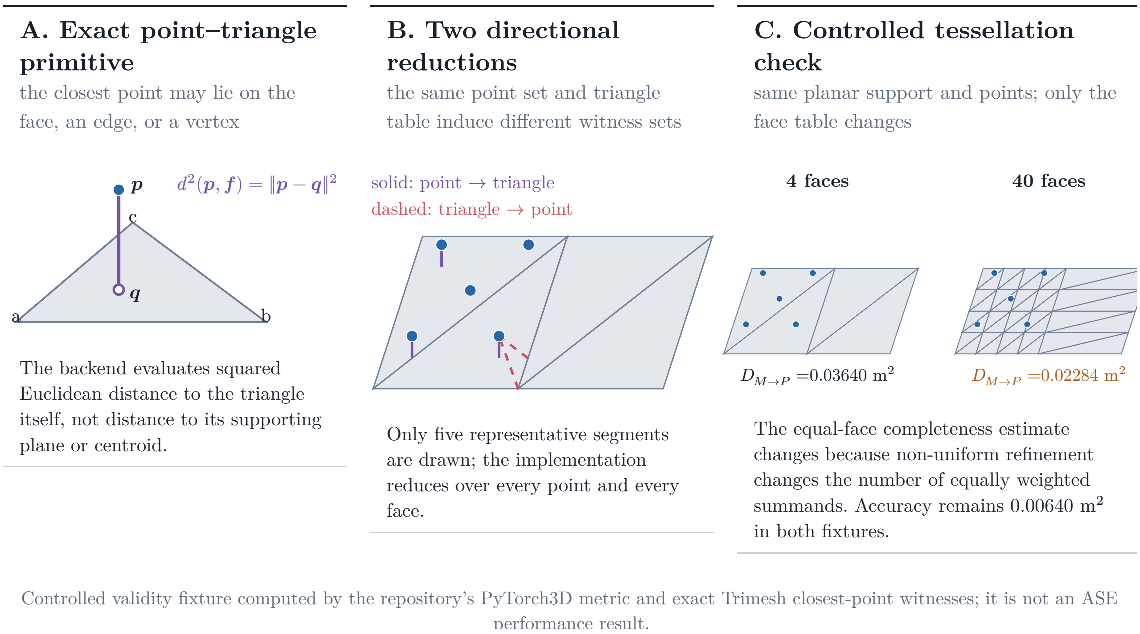


Figure 5: Point–mesh metric definition and controlled validity fixture. Panel A isolates the exact point-to-triangle primitive. Panel B distinguishes the point-to-face accuracy and face-to-point completeness reductions using computed closest-point witnesses. Panel C holds the planar support and point set fixed while changing only the triangle table; the measured equal-face completeness term changes from 0.03640 to 0.02284 m^2. These values are generated by the repository's PyTorch3D metric on a synthetic fixture and demonstrate a tessellation-sensitivity mechanism; they are not an ASE performance result.

The completeness reduction gives every mesh face equal weight. It is deliberately neither an area-weighted surface integral nor invariant to retessellation: subdividing part of an unchanged surface changes how much that region contributes. The controlled fixture in Figure 5 demonstrates this property. It bounds the interpretation of the operational estimand but does not invalidate comparisons that share one fixed mesh and scoring protocol.

3.3.2 Reliability under the Frozen Protocol

Reliability asks whether repeated execution of the same declared intervention produces stable errors and candidate rankings. It therefore covers rendering, backprojection, crop construction, point fusion and capping, numerical failure handling, and any stochastic sampling used by the metric. Deterministic code paths and a fixed mesh make the protocol reproducible in principle, but they do not establish repeatability across the intended study population. That evidence remains a prerequisite for RQ1 rather than an inferred property of the formula.

Metric family	Question answered	Dependency and thesis role
Equal-face bidirectional point–mesh error	Are reconstructed points close to GT faces, and does every GT face have nearby point evidence?	Depends on point fusion and mesh tessellation; selected operational RRI error.
Point-sampled Chamfer / accuracy–completeness	Are two sampled surface point sets mutually close?	Depends on sampling density, randomness, norm, and squaring convention; important alternative, not the current metric.
Thresholded precision, recall, and F-score	What fraction of predicted and reference samples fall within tolerance τ, and how do the two fractions balance?	Interpretable at a physical tolerance but threshold-dependent; useful diagnostic rather than the scalar RRI owner.
TSDF, occupancy, coverage, or information gain	How much volumetric surface, free space, or previously unknown space is reconstructed or observed?	Depends on grid, truncation, and visibility definitions; answers a different question from target surface quality.

Table 4: Reconstruction-metric alternatives and their role in this thesis. No alternative is promoted to a co-primary metric.

For comparison, a common squared point-sampled Chamfer convention draws a reference set Q_e from the mesh and averages nearest-neighbour distances in both directions,

$$D_{\text{PS-Chamfer}(\mathcal{P},\mathcal{Q})} = \frac{1}{\|\mathcal{P}\|} \sum_{\boldsymbol{p}\in\mathcal{P}} \min_{\boldsymbol{q}\in\mathcal{Q}} \|\boldsymbol{p}-\boldsymbol{q}\|_2^2 + \frac{1}{\|\mathcal{Q}\|} \sum_{\boldsymbol{q}\in\mathcal{Q}} \min_{\boldsymbol{p}\in\mathcal{P}} \|\boldsymbol{q}-\boldsymbol{p}\|_2^2$$

Its apparent symmetry does not remove sampling-density dependence: changing the number or distribution of samples changes the finite approximation. Some benchmarks also use unsquared distances, so the norm and squaring convention are part of the metric definition rather than an interchangeable naming detail.

At tolerance τ, precision counts reconstructed samples with a reference neighbour closer than τ, recall counts reference samples with a reconstructed neighbour closer than τ, and $F_{1,\tau} = 2\mathrm{Prec}_\tau \frac{\mathrm{Rec}_\tau}{\mathrm{Prec}_\tau + \mathrm{Rec}_\tau}$. These scores are physically interpretable but can change with the selected threshold and discard error magnitude on either side of it. Volumetric coverage and information gain are valuable for exploration and mapping, yet they measure discovered space or uncertainty reduction rather than the target reconstruction-quality change defined here.

3.3.3 Construct Validity

Construct validity asks what the operational error represents. In this thesis it measures idealized target-surface evidence under ASE depth, a selected target crop, equal-face mesh reduction, and the declared reconstruction update. It is not a representation-independent measure of object quality, perceived quality, or scene completeness. The rendering and fusion operator are part of the estimand: two systems using the same point–mesh formula but different state updates do not measure the same intervention effect.

The implementation crops mesh faces when any vertex lies inside the oriented target Oriented Bounding Box (OBB) and evaluates the selected point–mesh scorer. Geometry-invalid candidates are removed by the hard action mask and retain persisted candidate reason codes. Empty mesh crops, insufficient current support, or unusable renders instead invalidate the separate oracle evaluation: the recipe either skips the affected row or table, or clears oracle-label validity and Q-training eligibility, while reporting the oracle failure independently rather than assigning a candidate reason code.

Every compared policy must therefore share the same target mesh and crop, ASE-depth source, renderer and backprojection, fusion and point cap, and metric configuration. Replacing ASE depth with MPS semi-dense evidence would define a different intervention and would require separate analysis of texture, gradient, uncertainty, and sampling effects.

3.3.4 RRI, Training Reward, and Endpoint Gain

The thesis objective is not the absolute point–mesh error alone, but its normalized reduction after a fixed acquisition budget. It distinguishes state-relative RRI, the additive target-root reward, and endpoint gain: all are dimensionless reductions of the same target-cropped error, but their denominators and scientific roles differ.

The state-relative marginal target RRI remains the VIN-compatible candidate diagnostic:

$$\mathrm{RRI}^{e}_{t,i} = \frac{\Delta^{e}_{t} - \Delta^{e}_{t|i}}{\max(\Delta^{e}_{t}, \varepsilon)}$$

The implementation also reports its selected-chain running sum:

$$C^{\mathrm{RRI},e}_{t} = \sum_{k=0}^{t-1} \mathrm{RRI}^{e}_{k,a_k}$$

Because the marginal RRI denominator changes with state, this cumulative diagnostic does not telescope to endpoint improvement. The immediate training reward instead adapts the same error reduction to a target crop and normalizes by the rollout-root target error [Fra+25]:

$$r^{e}_{t} = \frac{\Delta^{e}_{t} - \Delta^{e}_{t+1}}{\max(\Delta^{e}_{0}, \varepsilon)}$$

With matched sequential states and unit discount, its selected-chain sum is the root-normalized endpoint difference under that same clamped root denominator:

$$J^{e}_{t} = \sum_{k=0}^{t-1} r^{e}_{k} = \frac{\Delta^{e}_{0} - \Delta^{e}_{t}}{\max(\Delta^{e}_{0}, \varepsilon)}$$

Using the finite-horizon return defined in Section 2.4, the discounted sum of those target rewards along a selected counterfactual branch is a training target for Q_H, not a claim that the deployed system has an online continuous-control policy:

$$G^{(h)}_{t,e} = \sum_{k=0}^{h-1} \gamma^{k} r^{e}_{t+k}, \quad 1 \le h \le b_t \le H_{\max}$$

Endpoint gain is the primary fixed-budget comparison metric because it measures the target quality after the same number of acquisitions for each policy:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

The log-gain variant is retained only as an internal scale-sensitivity ablation:

$$J_{e,\log}^{(H)} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon)$$

$J_e^{(H)}$ is the fixed-budget evaluation metric, $G_t^{(H)}$ is the learning return, and $J_{e,\log}^{(H)}$ is a sensitivity diagnostic. The endpoint equation uses $\Delta_0^e + \varepsilon$, whereas the additive target-root reward uses $\max(\Delta_0^e, \varepsilon)$; the two are therefore not canonically interchangeable, even when geometry, horizon, and acquisition budget match. State-relative one-step RRI remains a VIN-compatible diagnostic. The resolved manifest must freeze the discount, clipping, target cap, crop policy, and all evaluation-geometry parameters for each reported experiment.

3.4 Evidence and Reproducibility

The experiment preserves two evidence roles. An immutable source store owns the logged actor substrate and one-step oracle products; a replay store references those rows and owns target tasks, retained chains, per-step candidate shells, hard masks, reason codes, and selected-action successor evidence. This boundary prevents counterfactual experiments from mutating their source and prevents dense one-step labels from being mistaken for factual multi-step transitions.

Lineage follows the scientific units. One source may define several target tasks; one task may produce several recipe-specific retained chains; one chain contains ordered selected steps; and each step owns a full finite candidate shell. The store therefore represents selected or beam-retained evidence rather than an exhaustive counterfactual tree. Derived padded training arrays remain caches over these factual relations. Invalid rows stay auditable while action, training, and bootstrap masks exclude them from their respective decisions.

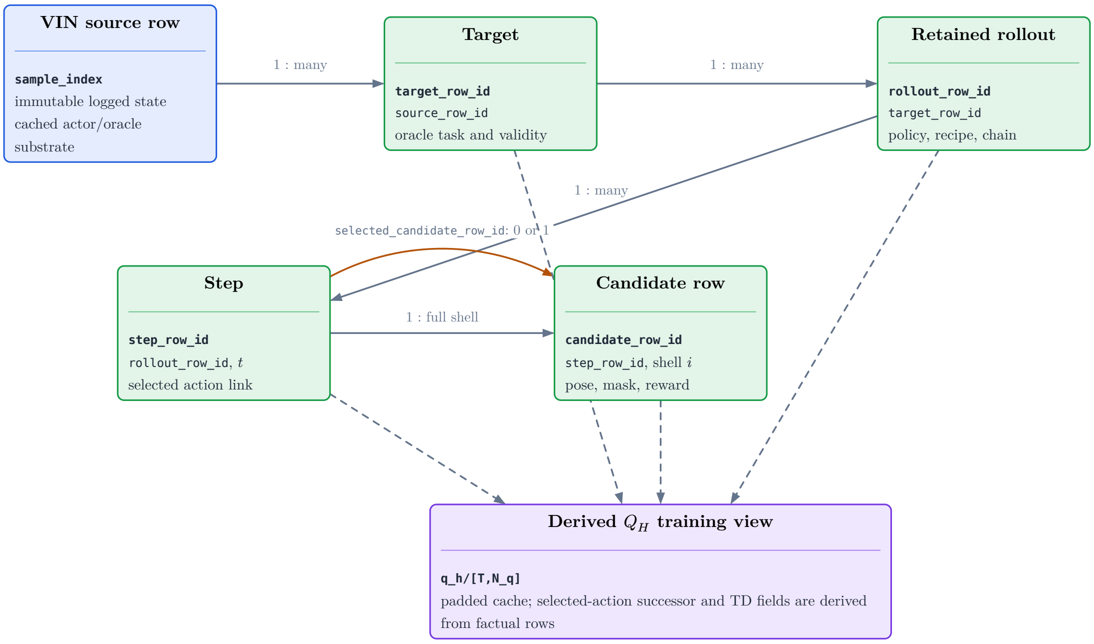


Figure 6: Normalized lineage of the replay evidence. An immutable source row may define several target tasks; each target may produce several retained policy chains; each chain contains ordered steps; and each step owns one full candidate shell. Selected-action successor and finite-horizon training fields are derived from those factual relations rather than treated as an independent counterfactual transition table.

Evidence family	Scientific interpretation
Sources and targets	Manifest-backed task coverage; not proof of actor-visible target discovery.
Candidates and invalidity	Full-shell support with separate hard-action, training, padding, and future deployable-feasibility roles.
Retained chains and steps	Recipe-selected evidence; not a persisted exhaustive search tree.
Selected depth	Chosen-action successor observation with calibration and source role; actor input only under an explicitly admitted later-state protocol.
Finite-horizon training view	Derived training cache whose rewards and masks must agree with factual rows; not a scene-memory representation.
Candidate-support metrics	Attempted-row actor-valid fraction, per-state valid support, configured-family zero rate, attempted target-conditioned side balance and circular span, calibrated target-centre projection fraction, finite-support oracle opportunity, and bounded-jitter compliance.
Metric populations	State metrics aggregate state then scene then cohort; failed roots and zero-valid configured family/state pairs remain in denominators. Projection is framing, oracle opportunity is headroom, and jitter is QC—not visibility or policy performance.

Table 5: Scientific interpretation of replay evidence. Numeric values are rendered from the resolved report bundle in the experiment and reproducibility sections.

Selected-depth persistence stores only the depth raster and calibration for the chosen action at each retained step. It can reconstruct the selected-observation prefix without duplicating dense all-candidate renders, but persistence does not make the source actor-visible. A privileged reader may consume selected mesh-rendered depth; an actor-visible reader must consume an admitted sensor-like or observed source. Unselected candidate renders remain oracle-only, and selected depth is not an independently scored endpoint artifact.

Likewise, rollout rows summarize final cumulative selected-chain metrics; they do not preserve every rejected branch or a policy-neutral endpoint reconstruction. These limitations must be resolved by matched endpoint re-evaluation before confirmatory policy comparison.

Missingness is part of the evidence rather than an ordinary zero. Reporting therefore retains target-task coverage, candidate validity and failure reasons, family survival and selection, selected-history sanity, gain distributions, source-role counts, and runtime or storage exclusions. Exact schema columns, joins, compression, chunking, hashes, and execution commands belong to the reproducibility record and resolved manifests. Development bandwidth pilots may size later jobs but cannot support held-out reconstruction or policy claims.

4 Method

Remove or rewrite before submission: Rewrite this chapter around one frozen, implemented, and evaluated method. Move rejected objectives, alternative carriers, architecture ladders, and unresolved scorer interfaces to development-only notes or a compact limitations/future-work account.

Source: this chapter and its source-owner decision gates

Gate: production scorer, frozen objective, matched controls, and end-to-end acceptance evidence

ARIA-NBV is formulated as target-conditioned selection from a finite candidate table. The implemented substrate generates and evaluates masked multi-step pose rollouts, records selected transitions, and exposes a persisted supervision profile that proves when every hard-valid realized action has a one-step Q label. It trains an actor-only Q_H scorer through fitted Double-Q optimization. The modular `A0/A1--S0-pose--root-moments` controls expose identical scene, target, causal-history, budget, and horizon inputs to an independent-row MLP or candidate-to-state attention, respectively; A1 remains the default. They are executable architecture controls, not yet a policy result or evidence that the compact state is task-sufficient. One-epoch CUDA contract tests cover regression and CORAL for deployable CF0 bundle reconstruction and privileged CF+ H0/S1 optimization; these bounded fixtures establish executability only, while the learned exact-Q2 gate remains closed.

The frozen interface distinguishes remaining budget b_t, scalar requested residual horizon h, and configured support bound $H_{\max}$. The `bounded_scalar_v1` scorer admits exactly one query per state over $1 \leq h \leq b_t \leq H_{\max}$: omission selects the full-budget diagonal $h = b_t$, while $h < b_t$ is a shorter return from the same factual state and padding alone uses zero. This syntactic family is wider than empirical capability: dense hard-valid labels train $h = 1$, factual selected transitions train supported $h > 1$, and a bundle rejects any requested horizon absent from its manifest-bound training support. A conditional value fixes the represented state, target, candidate selected first, scalar horizon, and named continuation rule; it remains independent of both the action mask and feasibility. The scorer returns those action-mask-independent conditional values and feasibility logits, while Lightning and online inference retain hard-mask ownership. Direct continuous Huber regression is the canonical value objective. A modular CORAL decoder over the same continuous fitted-Q targets is an ordinal ablation with closed train-fitted or physically predeclared support provenance and continuous representatives; exact horizon-two certification, recursive lower-horizon backups, Double-Q selection/evaluation, and behavior-policy returns remain separately named controls or ablations.

A retained rollout is not one recurrent neural-network sample. The normalized replay view expands it into a padded sequence of realized decision states: the root is one state, and every factual selected-action successor is another. For each state, the scorer receives only the causal prefix available before the current action and returns one value per materialized candidate row for one scalar requested horizon. Several states from one chain may occupy the batch/state axes and be evaluated in parallel; that colocation is a compute layout, not an information path from a later state into an earlier prediction. The current scorer contains no temporal interaction across those realized states.

Accordingly, **recursion** in this chapter means Bellman target recursion across factual successors, as formalized in Section 4.5.3. A longer-horizon target combines the current selected reward with a detached, frozen, or delayed shorter-horizon value at the next causal state. It is neither RNN-style hidden-state recurrence nor DETR-style iterative refinement of one prediction. The current Lightning transaction privately duplicates the actor batch so one scorer call evaluates two query domains: dense $h = 1$ for every realized state and the factual residual horizon $h = b_t$ for the same states. Dense one-step loss uses every hard-valid labelled candidate; recursive $h > 1$ loss uses only factual selected-transition targets, with nonterminal targets additionally requiring a supported successor backup. Off-diagonal queries $h < b_t$ are valid members of the public value family, but the present objective does not automatically enumerate every shorter horizon from each retained chain.

The chapter first fixes the actor-visible state and the coverage limitation of local EFM3D evidence. It then separates the implemented compact scorer from richer scene carriers, interaction ablations, and bounded Q_H learning targets.

Descriptive status blocks distinguish executable maturity from scientific evidence throughout.

4.1 Actor State and Representation Boundary

4.1.1 Implemented carrier and information boundary

Implementation: partial **Evidence: pending**

The replay carrier, actor/supervision separation, scalar requested-horizon scorer, fitted-Q adapter, and privileged `S1-points` selected-surface control are implemented. The deployable `S0-pose` root-moments state and privileged S1 state are deliberately incomplete; task-sufficient ray memory and frozen scientific validation remain pending.

Literature: [Str+24b]

Promotion gate: retain actor/oracle provenance checks and name the admitted state protocol in every run

Development source: aria_nbv/aria_nbv/data_handling/qh_data/views.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/tests/lightning/test_qh_module.py

The implemented one-step scorer consumes an actor-visible snippet view, a row-aligned candidate table, a reference rig pose, calibrated candidate cameras, and optionally cached EVL output. Oracle RRI labels enter the loss after prediction and are not scorer inputs. The current oracle target descriptor contains semantic identity, pose, positive metric extents, and reference-relative pose in an actor-safe shape, but its values are derived from privileged GT target tasks. A deployable claim therefore still requires observed or predicted target fields with explicit provenance.

The implemented `QhActorTensors` contract carries root semidense evidence, target geometry admitted by the declared target-input protocol, root-relative candidate poses, selected-pose history, and remaining budget to the finite-horizon scorer. The non-deployable `v0_gt_input` protocol derives that geometry from Oracle GT tasks; deployable `v1_observed` instead requires an actor-visible descriptor whose source, construction provenance, and descriptor hash are bound in training and held-out receipts. The current model reduces root evidence to global moments and selected history to a causal pose summary. Candidate regeneration changes pose, history, budget, and the finite action table; richer models must still define how causal selected observations update scene memory without widening this actor/oracle boundary.

Three boundaries are invariant. Invalidity is a hard mask with versioned reason codes, not a small value. GT meshes, target crops, current all-candidate renders, associations, and returns remain oracle or audit fields. Previously selected GT-mesh depth may enter a later state only under an explicitly privileged `CF-GT` protocol; the corresponding sensor-like or observed variants must use distinct provenance. Candidate rows remain tied to stable shell identities and documented coordinate frames.

4.1.2 Task-sufficient actor state

Implementation: partial **Evidence: pending**

The canonical state separates immutable root context from causal dynamic memory. A bounded, privileged identity-start S1 selected-surface residual is executable behind the scene-carrier seam. Its first five-chain comparison is development evidence because it informed subsequent initialization design and has no immutable run receipt; S1 remains unpromoted and S2 free/unknown memory remains planned.

Literature: [Bro+21, Zah+17]

Promotion gate: selected-observation reader, deterministic fusion, source-dropout tests, and held-out target-RRI ranking

Development source: aria_nbv/aria_nbv/vin/modules/qh_scene_encoders.py; aria_nbv/aria_nbv/rendering/unproject.py; aria_nbv/tests/vin/test_qh_scene_encoders.py; docs/contents/theory/efm3d_scene_embeddings.qmd; docs/contents/theory/candidate_view_dependence.qmd

ARIA-NBV does not require one universal reconstruction format. It requires an actor-visible state from which a model can compare a target e and candidate $q_{t,i}$ at decision step t. The frozen scorer interface includes the scalar requested residual horizon h in addition to the factual remaining budget b_t:

$$\boldsymbol{h}_{t,e,i} = \mathrm{Read}\big(\boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{h}_e^{\mathrm{tgt}}, q_{t,i}, \boldsymbol{H}_t, t, H\big)$$

For architectural and DTO purposes, the resulting scene memory is represented as the state consumed by the value model,

$$\boldsymbol{\Phi}_t^{\mathrm{scene}} = \Big(\boldsymbol{M}_t^{\mathrm{ray}}, \boldsymbol{X}_t^{\mathrm{pt}}, \boldsymbol{F}_t^{\mathrm{DINO@pt}}, \boldsymbol{E}_0^{\mathrm{EVL\text{-}local}}, \boldsymbol{O}_t^{\mathrm{pred}}, \boldsymbol{M}^{\mathrm{dir}}\Big)$$

The immutable root evidence and causally updated dynamic memory remain separate conceptually: root semidense geometry and supported local EVL features are not silently replaced by selected-observation evidence. The target is external task context in $Q(s_t, e, a)$ and is represented by a separate target state; it is not duplicated inside target-independent scene memory.

The selected-pose history $\boldsymbol{H}_t$ remains explicit unless a promoted memory is demonstrated to be a sufficient statistic for it. Raw selected depth is an observation consumed by the memory update; it need not remain a direct scorer input once its surface, free-space, support, source, and recency information have been fused.

State protocol	Dynamic evidence	Interpretation
`S0-pose`	selected poses only	Implemented feature-matched A0/A1 root-moments controls over replay tensors; not a complete reconstruction state.
`S1-points`	causally fused selected surface points	Implemented privileged identity-start, fixed-width residual over current-camera point sets; confirmatory evidence remains pending and observed free is not distinguished from unknown space.
`S2-ray`	surface, free, unknown, support, recency	Canonical planned dynamic state for candidate-frustum and target-support queries.
Privileged / sensor-like / actor-visible	source tag on selected observations	Orthogonal information protocol: privileged mesh depth, declared sensor-like simulation, or actor-visible observation.

Table 6: Counterfactual state and source protocols. Scene carrier, information source, interaction architecture, and learning objective are orthogonal experimental choices.

A useful representation must preserve distinctions that can change target-specific return, distinguish missing evidence from predicted free space, and admit a causal update after selection. The implemented S1 control deliberately stops earlier: it backprojects only selected $j < t$ depth, expresses points from the factual current camera, applies one shared point map, and mean/max pools the union into a root-width residual. It is point-order invariant but density weighted—each valid strided pixel contributes one set element—and retains only explicit point, pixel-support, and view-support summaries. Candidate queries do not yet read candidate-relative points. Zeroing only S1's final residual projection makes fresh S1 predictions exactly equal to matched H0, lets that projection move on the first backward pass, and delays task gradients into the point map until the residual path opens. The initial five-chain comparison was inspected during architecture development, so it is not an untouched test and supports neither an identity-start improvement nor a geometry-attribution claim. Candidate-local point relations remain unpromoted; S2 must additionally preserve ray/free/unknown evidence. A common world translation or yaw convention must not change physical candidate values, while gravity, scale, target extent, camera direction, occlusion, and temporal order remain observable task variables. Exact global SE(3) invariance is therefore neither required nor claimed.

4.1.3 Local EFM3D evidence and the coverage gap

Implementation: partial **Evidence: pending**

EFM3D fields, their voxel pose, and finite extent are persisted. The current finite-horizon scorer consumes lossy global field moments; their sufficiency as the only long-horizon scene representation remains pending.

Literature: [Met25b, Str+24b]

Promotion gate: report target and candidate support against the persisted voxel pose and extent

Development source: docs/literature/tex-src/arXiv-EFM3D/method.tex, Sec. Egocentric Voxel Lifting, lines 2–44; docs/contents/literature/efm3d.qmd; docs/contents/theory/efm3d_scene_embeddings.qmd; aria_nbv/aria_nbv/data_handling/vin_store/writer.py

EFM3D encodes logged RGB frames before lifting selected evidence into a finite gravity-aligned voxel field anchored to a snippet pose. The stored voxel transform and extent define the support of lifted features and dense heads. They must not be interpreted as a global volume containing every region observed by the trajectory. A target or candidate outside this support can remain physically valid; it has missing local-EVL evidence rather than an ordinary zero feature or automatically invalid action.

The coverage limitation motivates a layered interface rather than repeated inference at hypothetical candidate poses. Local EVL features provide high-quality Aria-native evidence where supported. Broader semidense or fused point carriers can preserve observed surfaces beyond that volume. Sparse ray-aware state can distinguish occupied, free, and unknown regions and can be updated from selected geometry. Logged appearance descriptors can be attached to points only after visibility and source lineage are established. None of these carriers creates RGB, DINO, detector, or EVL evidence at an unvisited counterfactual pose.

4.1.4 Ranked representation design space

Implementation: exploratory **Evidence: pending**

This table ranks scene carriers only. The interaction architecture is specified separately in Section 4.4 so a carrier change does not silently redefine candidate or target semantics.

Literature: [Ave+24, Che+24, Fra+25, Jeo+26]

Promotion gate: promote only after matched support, leakage, runtime, and oracle-policy ablations

Development source: docs/literature/tex-src/arXiv-GenNBV/3-Method.tex, Secs. Formulation and Generalizable State Embedding, lines 10–49; docs/literature/tex-src/arXiv-Hestia/sec/3_method.tex, Sec. Methods, lines 14–58; docs/literature/tex-src/arXiv-scene-script/sections/structured_scene_language.tex; docs/contents/literature/efm3d.qmd

Carrier	Status	Inductive benefit	Cost and promotion gate
Root semidense moments	implemented S0 control	Cheap root-level context for contract and optimization tests.	No spatial support or causal update; never interpret as task-sufficient state.
Persisted EVL and snippet evidence	available; lossy moments consumed	Aria-native local fields, OBB hypotheses, calibrated cameras, and explicit extent.	The scorer consumes global moments rather than spatial EVL fields; limited support still requires coverage metadata.
Selected-surface point memory	implemented privileged identity-start control; unpromoted	Strictly causal surfaces follow selected views; fixed width and a zero-output residual preserve matched H0 initialization.	Confirmatory comparison requires an immutable receipt and untouched scene-disjoint test manifest.
Sparse ray-aware memory	planned primary extension	Separates surface, free, unknown, support, uncertainty, and causal updates.	Requires deterministic fusion and counterfactual-source masks.
Target-centred EVL re-lifting	adaptation ablation	Reuses logged frame features when the target lies outside the root field.	Domain shift in the 3D neck; compare against simple logged-feature pooling.
DINO-on-point	appearance ablation	Extends logged appearance to observed points beyond the EVL grid.	Needs visibility gating, compression, and missing-descriptor masks.
TSDF/SDF or sparse encoder	geometry/ encoder ablation	Compact metric geometry or learned coordinate-bearing tokens.	Must preserve observation weights and unknown-space semantics.
Object-aware 3DGS	renderable-memory ablation	Candidate rendering, soft target membership, and primitive uncertainty.	Per-scene optimization and mask supervision change the state contract.
SceneScript	global-context control	Broad ASE-aligned layout and object hypotheses from semidense input.	Not an ATEK drop-in and does not provide causal free/unknown updates.

Table 7: Ranked scene-carrier design space. Status describes the role in the thesis method, not empirical superiority.

Representation comparisons begin with the smallest planned scorer carrier and then add the smallest carrier that tests a diagnosed information loss. Learned point, sparse, equivariant, or renderable encoders are introduced only when a simpler carrier demonstrably limits target-RRI ranking or finite-horizon recovery. This ordering keeps representation capacity separate from target identity, candidate support, replay validity, and the candidate interaction architecture.

4.2 Descriptor and Encoding Protocol

4.2.1 Persisted replay interface

Implementation: implemented **Evidence: pending**

The factual replay schema, lazy reader, framework-neutral dataset seam, and dense `q_h/` view are implemented and unit-tested. Frozen scientific replay evidence remains pending. The schema is a storage contract; readability from the store does not make a field actor-visible.

Promotion gate: preserve row identity, masks, provenance, and source roles in every tensor reader

Development source: aria_nbv/aria_nbv/rollouts/zarr_store.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/aria_nbv/data_handling/qh_data/dataset.py; aria_nbv/tests/rollouts/test_qh_reader.py

The rollout store preserves source, target, rollout, step, candidate, diagnostic, and lineage tables. Its derived `q_h/` arrays provide a padded state–candidate view without changing factual identities or labels. Target pose and extents in the current oracle task remain privileged ground-truth-derived instructions. The implemented finite-horizon scorer consumes actor-side candidate geometry, selected-pose history, remaining budget, root evidence, and materialization support; target gains, ground-truth associations, mesh diagnostics, crops, current all-candidate renders, and Q labels remain supervision or audit fields. Previously selected depth becomes a later actor input only under a named counterfactual-observation protocol.

Carrier	Persisted content	Learning role
Target	identity, class, pose, extents, reference-relative pose, source and validity provenance	privileged oracle-task instruction; learned actors need observed or predicted equivalents
Candidate	stable row identities, world/root-relative pose, masks, reasons, sampler provenance, support fields	finite action row; privileged diagnostics remain source-gated
Selected chain	selected row, shell index, step order, policy, seed, successor link, terminal state	history and temporal-difference linkage
Selected observation	selected depth, valid mask, calibration, pose and source	later dynamic-state input only under a declared privileged, sensor-like, or actor-visible protocol; never an all-candidate student input
Oracle labels	target RRI, target root gain, errors, optional crops and candidate renders	supervision, evaluation, and audit only

Table 8: Implemented replay carriers and admissible learning roles.

4.2.2 Model-input and DTO contract

Implementation: partial **Evidence: pending**

The implemented DTO seam separates actor inputs, selected-transition supervision, and audit lineage for varying stored chain lengths. The scalar requested-horizon scorer returns a typed conditional-Q/feasibility result, with optional objective-specific CORAL training tensors that never participate in masking. Static scene context, target state, candidate rows, and causal history are implemented; privileged selected depth can now form the S1 point-set residual, while ray-aware dynamic state remains richer planned work.

Literature: [De +20, Bro+21, Sch+15, Zho+23]

Promotion gate: typed selected-observation state, positive-width actor-visible target path, source masks, leakage tests, and scientific policy evidence

Development source: aria_nbv/aria_nbv/data_handling/qh_data/views.py; aria_nbv/aria_nbv/lightning/qh_module.py; docs/contents/theory/candidate_view_dependence.qmd

The intended input for target e at step t is

$$\mathcal{I}_{t,e} = \left(\boldsymbol{h}_e^{\mathrm{tgt}}, \boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{H}_t, \boldsymbol{b}_t, t, H, \left\{\boldsymbol{x}_{t,i}, \boldsymbol{e}_{a|i}^{\mathrm{rel}}\right\}_{i=1}^{N_q}\right)$$

This equation is an information contract rather than one flat tensor. The corresponding DTO roles are:

DTO role	Candidate content	Visibility
`StaticSceneContext`	root semidense evidence, supported EVL tokens, root frame and EVL extent	actor input; immutable within one rollout
`DynamicSceneState`	selected geometry, free/unknown support, recency, source masks and ordered history	actor input; causal update only
`TargetState`	protocol-specific descriptor, target-local support and field-availability masks	oracle-task instruction or actor-visible target
`CandidateTable`	row identity, local pose, target relation, actor validity and padding mask	actor input; row-aligned
`ValueQuery`	scalar requested residual horizon h per state; omission means $h = b_t$	`bounded_scalar_v1`: $1 \le h \le b_t$; bundle support gates deployment
`CandidateSupervision`	one-step root gain, diagnostic target RRI and `q_train_mask`	supervision only
`SelectedTransition`	factual action index and row id, reward, discount, terminal and successor identity	training linkage only
`AuditLineage`	source/store/config hashes, policy, seed and reason vocabulary	CPU audit data; not a learned feature

Table 9: Finite-horizon scorer DTO roles. Model inputs, scalar horizon queries, supervision, transition linkage, and provenance remain distinct even when collated in one training batch.

The maximum supported horizon $H_{\max}$ is a scorer, data, and checkpoint contract. Remaining budget b_t is a factual rollout-state field, whereas requested horizon h selects one member of the scalar family $1 \leq h \leq b_t \leq H_{\max}$. Implemented `bounded_scalar_v1` validates this full syntactic domain: `None` means $h = b_t$, realized off-diagonal calls may request a shorter supported return, and padding alone uses $h = 0$. This admission is not an empirical capability claim. Lightning records the horizons that actually receive targets, and a verified inference bundle rejects any syntactically valid horizon absent from its manifest-bound promoted support. The public output remains $\left[B, S, N_q\right]$; multiple horizons use separate scalar calls, while a public vectorized horizon axis remains evidence-gated. The step index t remains lineage by default and becomes a learned feature only in a named non-stationarity ablation.

The executable boundary is `score(actor, requested_horizon=None)`. It hides scene, target, candidate, history, and time encoding behind one deep module and returns conditional Q plus feasibility logits in stored candidate order. The continuous `conditional_q` field retains one meaning for regression and CORAL: it alone enters Bellman backup and online ranking. CORAL additionally carries cumulative logits and its fixed support as training metadata. Static encodings may be reused privately, but the interface exposes no cache lifecycle, encoder handles, candidate sorting, or public horizon axis.

Padding masks, modality-presence masks, source-role masks, action masks, and training masks remain distinct. Out-of-range requests and bundle-unsupported in-range horizons fail closed and are never clamped to the available budget. This does not break recursion from h to $h-1$: the target scorer receives the explicit supported scalar query $h-1$ at the factual successor, whose remaining budget is $b_t - 1$. A missing modality must never be encoded as an ordinary zero observation or confused with padding.

The target descriptor separates identity, geometry, observed support, confidence, and source:

$$\boldsymbol{\varphi}_e = \mathrm{Enc}_{\mathrm{tgt}}\left(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{\pi}_e, A_e^{\mathrm{proj}}, n_e^{\mathrm{semi}}, n_e^{\mathrm{EVL}}, \omega_e^{\mathrm{EVL}}, \ell_e^{\mathrm{src}}, \boldsymbol{T}_{r_t,e}, \boldsymbol{T}_{c_t,e}\right)$$

The learned token combines that descriptor with target-local scene support:

$$\boldsymbol{h}_e^{\mathrm{tgt}} = \mathrm{MLP}_{\mathrm{tgt}}(\mathrm{concat}(\boldsymbol{\varphi}_e, \boldsymbol{g}_e^{\mathrm{tgt}}))$$

In the current oracle-task data, target geometry is ground-truth-derived and several generic descriptor fields are unmeasured placeholders. The model contract should therefore use protocol-specific target-state variants, or carry an explicit availability mask per optional field. The same numerical zero cannot simultaneously mean absent support and measured zero support.

Target-independent static scene encodings may be reused across several target tasks. The candidate table, however, is not generally shared: target-bearing, lateral-bypass, and target-looking candidates depend on the selected target. Multi-target evaluation must therefore either carry one candidate table per target or construct a union table with a target–candidate availability mask; only physically identical rows may share candidate encodings.

4.2.3 Relative candidate geometry

Implementation: partial **Evidence: pending**

Root-relative candidate geometry and a minimal candidate-local target relation are implemented. The complete candidate-frustum and dynamic-memory relation descriptor remains planned.

Literature: [Li+21, Zho+19, Zho+23]

Promotion gate: frame-transform, row-shuffle, and held-out descriptor ablations

Development source: aria_nbv/aria_nbv/data_handling/qh_data/views.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/vin/modules/pooling.py

Canonical rigid transforms remain in the store. The reader or tensor adapter derives a candidate pose relative to the current decision reference rather than learning the arbitrary world origin:

$$\boldsymbol{T}_{r_t,i}^{\mathrm{rel}} = \boldsymbol{T}_{w,r_t}^{-1}\boldsymbol{T}_{w,c_{t,i}}, \quad \boldsymbol{\delta}_{r_t,i}^{p} = \boldsymbol{R}_{r_t}^{\top}\left(\boldsymbol{c}_{t,i} - \boldsymbol{c}_{r_t}\right), \quad \boldsymbol{R}_{r_t,i} = \boldsymbol{R}_{r_t}^{\top}\boldsymbol{R}_{t,i}$$

$$\boldsymbol{h}_{t,i}^{\mathrm{pose}}(q_{t,i}; r_t) = \mathrm{concat}\left(\boldsymbol{\delta}_{r_t,i}^{p}, \mathrm{R6D}\left(\boldsymbol{R}_{r_t,i}\right), \left\|\boldsymbol{\delta}_{r_t,i}^{p}\right\|_2, \mathrm{atan2}\left(\delta_{r_t,i}^{y}, \delta_{r_t,i}^{x}\right), \Delta h_{t,i}, \boldsymbol{u}_{t,i}^{\mathrm{up/frustum}}\right)$$

The target relation is encoded separately so candidate self-motion cannot be confused with target conditioning:

$$\boldsymbol{h}^{\text{rel}}_{t,e|i}(q_{t,i}, e) = \operatorname{concat}\left(\boldsymbol{\delta}^{p}_{e|i}, \left\|\boldsymbol{\delta}^{p}_{e|i}\right\|_2, \cos\theta^{\text{opt}}_{t,e,i}, \beta^{\text{elev}}_{t,e,i}, \lambda^{\text{obb}}_{t,e,i}\right)$$

Continuous rotation features, metric range, bearing, elevation, height, and frustum variables preserve physical task structure. Query-local relative positional encoding is a later ablation for candidate–target, candidate–history, or candidate–candidate interactions:

$$\boldsymbol{\delta}^{\text{p}}_{a|i} = \boldsymbol{R}^{\top}_{t,i}(\boldsymbol{p}_a - \boldsymbol{c}_{t,i}), \quad \boldsymbol{\delta}^{\text{R}}_{a|i} = \boldsymbol{R}^{\top}_{t,i}\boldsymbol{R}_a$$

$$\boldsymbol{\eta}_{a|i} = \operatorname{concat}\left(\boldsymbol{\delta}^{\text{p}}_{a|i}, \left\|\boldsymbol{\delta}^{\text{p}}_{a|i}\right\|_2, \operatorname{enc}_R\left(\boldsymbol{\delta}^{\text{R}}_{a|i}\right)\right), \quad \boldsymbol{e}^{\text{rel}}_{a|i} = \psi_{\text{rel}}\left(\mathcal{F}\left(\boldsymbol{\eta}_{a|i}\right)\right), \quad a \in \{j, e, k\}$$

This adopts QCNet's query-centric geometric discipline, not its trajectory decoder or streaming claim [Zho+23]. World-frame copies remain audit facts; the learned interface should expose one authoritative local transform rather than recomputing the same relation through multiple fields.

4.2.4 Candidate rows and directional history

Implementation: partial **Evidence: pending**

The scorer exposes a versioned selected-pose history seam. H0 is the checkpoint-compatible masked mean; H1 is an implemented exploratory causal Transformer over current-camera-relative pose tokens and relative age. Directional scene memory and scientific longer-horizon evidence remain pending.

Literature: [e3n25, Lu+26]

Promotion gate: matched repeated-seed H0/H1 measurement, directional-novelty fixture, and held-out policy evidence

Development source: aria_nbv/aria_nbv/data_handling/qh_data/views.py; aria_nbv/aria_nbv/vin/modules/qh_history_encoders.py; aria_nbv/tests/vin/test_qh_history_encoders.py; docs/contents/theory/candidate_view_dependence.qmd

The candidate row is a local query, not a duplicate of the full state. The implemented trunk encodes root-relative and current-relative candidate pose plus global root-scene moments, predicts feasibility before target or horizon conditioning, and then adds a candidate-local target transform for conditional value prediction. Shared target, scene, causal-history, remaining-budget, and requested-horizon context are supplied as state tokens. Hard action validity is not a scorer feature;

candidate family or sampler provenance is audit-only by default and may enter the model only as a named ablation.

For each realized state, materialization must provide the complete strictly causal selected-pose prefix. Each selected pose is expressed from the current camera before the shared R6D–LFF pose encoder. H0 reproduces the original masked arithmetic mean exactly. H1 attaches relative age—zero for the immediate predecessor, increasing backward through the prefix—then uses causal self-attention and last-valid-token readout:

$$\boldsymbol{p}_{t,j}^{\text{hist}} = \text{PoseEnc}\Big(T_{c_t \leftarrow c_j}\Big), \quad j < t$$

$$a_{t,j}^{\text{hist}} = \frac{t-1-j}{H_{\max}}$$

$$\boldsymbol{h}_t^{\text{hist}^{\text{H0}}} = \text{HistProj}\left(\frac{1}{\max(1,t)} \sum_{j<t} \boldsymbol{p}_{t,j}^{\text{hist}}\right)$$

$$\boldsymbol{h}_t^{\text{hist}^{\text{H1}}} = \text{HistProj}\Big(\text{LastValid}\Big(\text{CausalTransformer}\Big(\Big[\boldsymbol{e}_{\text{empty}}, \Big\{\boldsymbol{p}_{t,j}^{\text{hist}} + g\Big(a_{t,j}^{\text{hist}}\Big)\Big\}_{j<t}\Big]\Big)\Big)\Big)$$

The learned empty token represents the realized root state with no selected predecessor; padded states remain zero and cannot enter a candidate output. Relative age makes chronology observable without adding absolute step index as a separate feature. Prefix cardinality is also factual causal state, but neither chronology nor cardinality substitutes for the separately encoded remaining budget b_t or requested value horizon h. H1 still observes poses only: it carries no selected depth, appearance, free/unknown update, target-local evidence, or new counterfactual render. It is therefore an `S0-pose` ablation rather than a sufficient dynamic reconstruction state.

Viewing history is not reducible to camera distance. For a target-local point or cell, selected camera directions define a signed first moment together with the second-moment memory

$$\boldsymbol{\mu}_t^{\text{dir}}(\boldsymbol{v}) = \frac{\sum_{k<t} w_k(\boldsymbol{v}) \boldsymbol{d}_k(\boldsymbol{v})}{\sum_{k<t} w_k(\boldsymbol{v}) + \varepsilon}, \quad \boldsymbol{M}^{\text{dir}}(\boldsymbol{v}) = \frac{\sum_{k<t} w_k(\boldsymbol{v}) \boldsymbol{d}_k(\boldsymbol{v}) (\boldsymbol{d}_k(\boldsymbol{v}))^\top}{\sum_{k<t} w_k(\boldsymbol{v}) + \varepsilon}$$

The signed first moment distinguishes opposite approach directions, whereas the second moment records concentration along viewing axes. Their query-relative projections form directional features without committing to a full spherical-harmonic field. Low-order spherical harmonics are promoted only if these moments improve over ordered pose history and still leave systematic directional errors.

4.2.5 Target-frame spherical diagnostics and calibrated frustum support

Implementation: partial **Evidence: pending**

The stored-rollout inspector implements target-frame movement, camera-forward, and calibrated proxy-surface frustum histograms. They diagnose factual selected paths; they are not yet scorer inputs and do not establish an S2 policy improvement.

Promotion gate: held-out coverage calibration, resolution-convergence study, and depth- or mesh-based occlusion validation before promotion into S2 memory

Development source: aria_nbv/aria_nbv/rollouts/inspection.py; aria_nbv/aria_nbv/app/panels/_stored_rollouts/s2_directions.py

The three spherical views answer different geometric questions and must not be conflated. The selected displacement direction $\hat{\boldsymbol{\delta}}^e_{j,t}$ describes **how the camera moved**. The optical-axis direction $\hat{\boldsymbol{v}}^e_{j,t}$ describes **where the camera pointed**. Neither point says which target-centred surface directions fell inside the camera image. For rollout chain j and persisted decision step t, both directions are expressed in the target object's orientation:

$$r_e = \left(a_x a_y a_z\right)^{\frac{1}{3}}$$

$$\hat{\boldsymbol{\delta}}^e_{j,t} = \operatorname{normalize}\left((\boldsymbol{R}^e_W)^\top \frac{\boldsymbol{c}^W_{j,t} - \boldsymbol{c}^W_{j,t-1}}{r_e} \right)$$

$$\hat{\boldsymbol{v}}^e_{j,t} = \operatorname{normalize}\Big((\boldsymbol{R}^e_W)^\top \boldsymbol{R}^W_{j,t} \boldsymbol{e}_z \Big)$$

The persisted OBB extents are full axis lengths, so a_x, a_y, a_z are their half-lengths. The scalar $r_e = \left(a_x a_y a_z\right)^{\frac{1}{3}}$ is their geometric mean: it scales linearly under uniform object scaling and supplies one permutation-invariant characteristic length. It is not the radius of a sphere with the same volume as the OBB; that radius would include the factor $\left(\frac{6}{\pi}\right)^{\frac{1}{3}}$. Dividing a displacement by r_e does not alter its eventual unit direction, but it retains a dimensionless movement magnitude for a future target-relative S2 descriptor.

The calibrated frustum diagnostic instead treats $\boldsymbol{d}^e \in \mathcal{S}^2$ as an outward surface direction and places the proxy point $\boldsymbol{x}^e = r_e \boldsymbol{d}^e$ on the target-centred sphere. The selected camera centre is subtracted **before** the target-to-camera rotation; omitting this translation would project only the frustum's orientation and would not describe target-centred surface support. A cell belongs to $\mathcal{F}^e_{j,t}$ only if its proxy-surface normal faces the camera, its camera-frame depth is positive, and the ARIA left–up–forward pinhole projection lies within the half-pixel image rectangle $\left[-\frac{1}{2}, W - \frac{1}{2}\right] \times \left[-\frac{1}{2}, H - \frac{1}{2}\right]$:

$$\boldsymbol{x}^e(\boldsymbol{d}) = r_e \boldsymbol{d}, \quad \boldsymbol{x}^c(\boldsymbol{d}) = (\boldsymbol{R}^c_e)^\top \left(\boldsymbol{x}^e(\boldsymbol{d}) - \boldsymbol{c}^e_{j,t}\right)$$

$$u(\boldsymbol{d}) = c_x - f_x \frac{x^c_x}{x^c_z}, \quad v(\boldsymbol{d}) = c_y - f_y \frac{x^c_y}{x^c_z}$$

$$\mathcal{F}^e_{j,t} = \left\{ \boldsymbol{d} \in \mathcal{S}^2 : \begin{cases} \boldsymbol{d}^\top \left(\boldsymbol{c}^e_{j,t} - \boldsymbol{x}^e(\boldsymbol{d})\right) > 0 & \text{front-facing} \\ x^c_z > 0 & \text{in front} \\ -\frac{1}{2} \le u(\boldsymbol{d}) \le W - \frac{1}{2} & \text{horizontal image support} \\ -\frac{1}{2} \le v(\boldsymbol{d}) \le H - \frac{1}{2} & \text{vertical image support} \end{cases} \right\}$$

$$\kappa^e_{j,t} = \mathrm{area}_{\mathcal{S}^2} \frac{\mathcal{F}^e_{j,t}}{4\pi}$$

The implementation partitions $\mathcal{S}^2$ uniformly in azimuth φ and target-frame height z. Because the sphere's area element is $\mathrm{d}\Omega = \mathrm{d}\varphi\, \mathrm{d}z$, all cells have equal solid angle. Complete per-cell counts therefore estimate the per-view fraction $\kappa^e_{j,t}$ and the union across factual selected views without the polar bias of uniform-elevation bins. The plotted incidence overlay is only a deterministic bounded reservoir; rollout colour preserves common chain heritage and marker shape preserves common step index. It never replaces the complete count grid.

Camera intrinsics also define an orientation-only field-of-view solid angle. If $\boldsymbol{q}_0, \ldots, \boldsymbol{q}_3$ are the normalized corner rays obtained from focal length, principal point, and image size, the spherical quadrilateral is split into two spherical triangles:

$$\Omega_{\triangle}(\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}) = 2\,\mathrm{atan2}\Big(\big|\boldsymbol{a}^{\top}(\boldsymbol{b}\times\boldsymbol{c})\big|, 1+\boldsymbol{a}^{\top}\boldsymbol{b}+\boldsymbol{b}^{\top}\boldsymbol{c}+\boldsymbol{c}^{\top}\boldsymbol{a}\Big)$$

$$\Omega_{j,t}^{\mathrm{FOV}} = \Omega_{\triangle}(\boldsymbol{q}_0,\boldsymbol{q}_1,\boldsymbol{q}_2) + \Omega_{\triangle}(\boldsymbol{q}_0,\boldsymbol{q}_2,\boldsymbol{q}_3)$$

This exact intrinsic $\Omega_{j,t}^{\mathrm{FOV}}$ and the numerically integrated target-proxy fraction answer different questions: the former depends only on calibration, whereas the latter additionally depends on target scale and camera pose. Neither is the fraction of the **true target mesh** observed. The sphere replaces the OBB shape, the front-facing test models only proxy self-occlusion, and scene occluders are ignored. A visibility claim requires intersecting calibrated rays with selected depth or the target and scene meshes, resolving the nearest surface along each ray, and reporting the corresponding target-surface measure. Until that evidence exists, the dashboard labels this quantity geometric potential visibility and the thesis treats it as an admission diagnostic for designing S2 memory rather than as a model result.

Optional appearance, ray, and directional blocks require three independent indicators where applicable: modality presence, batch padding, and evidence source. Removing one optional source must alter only its masked branch, and counterfactual-only geometry must not receive fabricated RGB, DINO, detector, or EVL descriptors.

4.3 Finite Candidate and Replay Contract

Implementation: implemented **Evidence: pending**

Finite candidate tables, hard masks, lineage, selected transitions, selected-depth persistence, the derived `q_h/` view, and a fail-closed dense-valid supervision profile are implemented and schema-tested. Frozen scientific policy evidence remains pending.

Promotion gate: preserve deterministic shell identity, source roles, and selected-transition validation

Development source: aria_nbv/aria_nbv/rollouts/replay/types.py; aria_nbv/aria_nbv/rollouts/replay/engine.py; aria_nbv/aria_nbv/rollouts/zarr_store.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/tests/rollouts/test_qh_reader.py

At step t, candidate generation returns a finite full-shell table $\mathcal{Q}_t$ with a hard-valid mask $\boldsymbol{m}_t$ and versioned invalid-reason bitsets. Scores are stored compactly only for hard-valid rows and are bound back to stable shell indices before selection. The admissible action set is

$$\mathcal{Q}_t = \left\{q_{t,i}\right\}_{i=1}^{N_q}, \quad \mathcal{A}_t = \left\{i \in \left\{1, ..., N_q\right\} : m_{t,i}^{\text{act}} = 1\right\}$$

Candidate orientations factor a component-specific base gaze from a local yaw–pitch perturbation. Production rollout mixtures resolve every component to the final seminar support: symmetric yaw and pitch caps of 60° and 30°, respectively, with zero roll jitter. A zero resolved yaw or pitch cap is rejected as a configuration error. The sampled residuals and their resolved bounds and a per-row bounded-support flag remain attached to the full shell for visual and statistical audit. Legacy zero-cap spherical samplers are explicitly unbounded rather than being misrepresented by a zero-area box; view jitter changes proposal support but does not bypass hard validity rules.

A paired proposal component may reuse one sampled world-space camera center for two distinct base-gaze families before applying their independent view jitter. Both rows remain first-class actions and retain a shared `position_pair_id` plus a `gaze_variant_id`; they are not averaged or collapsed before scoring. This controlled intervention separates the value of camera translation from the value of viewing direction at identical acquisition positions.

Invalid rows remain available for diagnostics and dense replay, but cannot be selected. A row enters the training mask only when it is actor-selectable and has a finite oracle target. Invalid rows have false masks and undefined labels; scene RRI is never substituted for a missing target-specific label. `valid_action_mask`, `q_train_mask`, padding, and any deployable feasibility estimate are distinct fields with distinct owners.

The persisted supervision profile makes label density an auditable contract rather than an inference from array shape. Historical stores declare `legacy_unspecified` together with `subset_of_action_v1`: their Q-label support may be any subset of hard-valid materialized rows and cannot establish dense one-step supervision. A fitted-Q store may instead declare `dense_valid` together with `equals_action_on_realized_steps_v1`. The writer then proves, and the reader revalidates, exact equality between `q_train_mask` and `valid_action_mask` on every realized state. Padding is excluded from both masks. Hard-invalid materialized rows remain useful binary examples for the physical feasibility head, but receive neither a fabricated Q target nor bootstrap support. Unknown or crossed profile pairs fail closed.

For benchmark reporting, **proposal support** is the persisted set of candidate rows that a declared generator attempted, with applicability and validity masks retained per family. **Proposal regret** is a descriptive comparison between the selected row and the best admissible row under the same persisted oracle contract; it is unavailable when no compatible oracle labels exist. Scene-paired aggregation first computes state-level quantities and then gives each scene equal weight, preventing scenes with more states from dominating a macro estimate. An immutable evidence bundle is the hash-bound JSON manifest and Parquet projection of these facts. Readers reject incomplete, stale, schema-mismatched, fixture, or hash-mismatched bundles; presentation layers consume the reader and do not recompute scientific quantities.

4.3.1 Implemented replay transition

Rollout expansion records the full candidate table, selected valid and shell indices, policy scores and probabilities, selection policy, and random seed. The implemented transition is

$$(x_{t+1}, \boldsymbol{H}_{t+1}, b_{t+1}, \mathcal{Q}_{t+1}) = \mathrm{Step}\big(x_t, \boldsymbol{H}_t, b_t, q_{t,a_t}, \xi_t\big)$$

where x_t is the current reference pose, $\boldsymbol{H}_t$ the selected-pose history, b_t the remaining budget, and ξ_t the deterministic generation context. The next candidate table is regenerated around the selected pose under the same target task, history constraints, and versioned generator configuration.

Proposal and action-selection randomness use separate deterministic streams. The proposal stream is keyed by the campaign proposal root, the ordered full-shell indices of previously selected actions, and an explicit proposal replica. It excludes transient frontier and beam indices. Consequently, reordering retained trajectories does not change the candidate table for the same replay state, while changing the replica yields an independent shell without perturbing action-selection randomness. Selection uses a distinct recipe seed keyed by the same selected-action history.

This transition is deliberately a replay-control transition, not yet a complete reconstruction-state update. It changes pose, selected-pose history, budget, lineage, and action support. It does not imply that the actor has received a new RGB observation, recomputed EFM3D field, or fused the selected depth into a spatial memory. Any implementation that consumes only these fields must be labelled `S0-pose`.

The persisted factual tables retain source and target identity, lineage hashes, step order, selected rows, masks, reasons, sampler provenance, rewards, selected-depth calibration, and support diagnostics. The derived `q_h/` view right-pads states, stores one-step and target-root-gain labels, and exposes only the factual selected transition needed for the temporal-difference backup defined in Section 4.5.3. It does not create counterfactual labels for unselected actions or make privileged selected depth actor-visible.

Training, validation, and test stores must share the same learning semantics: reward definition, mask meanings, target and actor-state protocols, horizon support, and label-support profile. They need not share a candidate-generator hash, rollout-recipe hash, or behavior-policy label. Those fields identify the sampled population rather than the value function, so each stage binds its own population provenance into the experiment bundle while the training contract remains the model identity. This permits a genuinely held-out distribution without weakening semantic checks. Stage acquisition is deterministic and scene-disjoint: complete eligible units are hash-ranked within each requested split, input-feasibility rejections are recorded, and no unit is replaced or selected using its oracle labels or model error.

4.3.2 Selected observation and planned scene-state transition

Implementation: planned **Evidence: pending**

A task-sufficient successor state must update only evidence produced by the selected observation. Current GT-mesh selected depth is privileged counterfactual evidence and requires an explicit `CF-GT` state protocol.

Literature: [Che+24, Lu+26]

Promotion gate: typed selected-observation reader, deterministic fusion, source masks, and no-future-observation tests

Development source: docs/contents/theory/efm3d_scene_embeddings.qmd; aria_nbv/aria_nbv/oracle/evidence.py; aria_nbv/aria_nbv/rollouts/zarr_store.py

A selected observation is a typed tuple containing depth, validity mask, calibration, root-relative camera pose, and an explicit source role. The geometry successor state consumes this selected evidence:

$$s_t^{\text{cf}+} = \left(s_t^{\text{cf}0}, \left(\boldsymbol{D}, \boldsymbol{V}, \mathcal{P}^{\text{cf}}, \boldsymbol{n}\right)_{1:t}\right)$$

The source role distinguishes privileged mesh depth, declared sensor-like simulation, and an actor-visible sensor observation. Unselected candidate renders at step t are never elements of the student state.

The existing geometry-level counterfactual uses a set union of retained points,

$$\mathcal{P}_{t+1} = \mathcal{P}_t \cup \mathcal{P}_{q_t}$$

but a planning memory must additionally preserve whether space is observed occupied, observed free, or still unknown. The proposed sparse update is

$$\boldsymbol{M}_{t+1}^{\text{ray}} = \text{Fuse}\left(\boldsymbol{M}_t^{\text{ray}}, \boldsymbol{P}_{t+1}^{\text{selected-obs}}(a_t), \mathcal{R}_{t+1}^{\text{selected-obs}}(a_t)\right), \quad a_t \in \mathcal{A}_t$$

Here both evidence terms are indexed by the selected action a_t; no unselected candidate render enters the successor state. The update may add selected surface evidence, carve observed free space, update support and uncertainty, and refresh target-local directional memory. It must not attach RGB, DINO, detector, or EVL descriptors to counterfactual geometry unless a corresponding actor-visible observation exists. Counterfactual-only cells instead carry explicit source and missing-modality masks.

Raw selected depth is an input to the state builder, not necessarily a permanent model token. The reader may emit either the typed selected-observation prefix or a deterministically derived `DynamicSceneState`, but those two representations must not be mixed silently. This split prevents two common errors: treating pose-only replay as complete visual simulation, and treating privileged selected-depth geometry as a deployable sensor stream.

4.4 Geometric and Mask Acceptance Tests

Remove or rewrite before submission: Retain acceptance properties only for the architecture that is actually implemented and evaluated. Proposed architecture alternatives belong in development notes until a measured failure motivates them.

Source: this section; scorer implementation and tests

Gate: one production scorer passes the stated permutation, masking, frame, and horizon tests

Implementation: partial **Evidence: pending**

The implemented A0/A1 scorer uses independent candidate rows, candidate-relative target/current-pose geometry, scalar requested horizons, and adapter-owned hard masks. Permutation, duplicate, invalid-row isolation, mask independence, and horizon bounds are unit-tested for the shared public contract; scientific policy evidence and richer state protocols remain pending.

Literature: [De +20, Bro+21, Lee+19, Zah+17]

Promotion gate: retain permutation, subset, duplicate, padding, mask-independence, frame, source, and horizon tests for every admitted state protocol

Development source: aria_nbv/aria_nbv/vin/models/target_finite_horizon.py; aria_nbv/aria_nbv/vin/modules/qh_history_encoders.py; aria_nbv/aria_nbv/vin/modules/qh_state_fusion.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/tests/vin/test_qh_history_encoders.py; aria_nbv/tests/vin/test_qh_state_fusion.py; aria_nbv/tests/vin/test_target_finite_horizon.py; aria_nbv/tests/lightning/test_qh_module.py

Candidate order carries no task meaning. For a per-candidate scorer f_θ, jointly permuting row-aligned inputs by Π must permute outputs by the same amount:

$$f_\theta(\Pi \boldsymbol{X}_t, \Pi \boldsymbol{m}_t^{\mathrm{cand}}) = \Pi f_\theta(\boldsymbol{X}_t, \boldsymbol{m}_t^{\mathrm{cand}})$$

Equivariance alone does not guarantee invalid-row isolation. Holding valid rows and the mask fixed while changing only invalid-row contents must satisfy

$$\mathrm{Mask}\big(\boldsymbol{X}_t, \boldsymbol{m}_t^{\mathrm{cand}}\big) = \mathrm{Mask}\big(\boldsymbol{X}_t', \boldsymbol{m}_t^{\mathrm{cand}}\big) \quad \Rightarrow \quad \mathrm{Mask}\big(f_\theta\big(\boldsymbol{X}_t, \boldsymbol{m}_t^{\mathrm{cand}}\big), \boldsymbol{m}_t^{\mathrm{cand}}\big) = \mathrm{Mask}\big(f_\theta\big(\boldsymbol{X}_t', \boldsymbol{m}_t^{\mathrm{cand}}\big), \boldsymbol{m}_t^{\mathrm{cand}}\big)$$

Mask ownership depends on the interaction. In implemented A0, each materialized candidate row is concatenated with the same five named state tokens and processed independently. In A1, materialized candidates are queries while scene, target, history, budget, and requested-horizon context are keys and values. `candidate_mask` sanitizes padding before either fusion. Neither scorer reads `action_mask`; Lightning uses it for Q-loss and bootstrap support, and online inference uses it for the final selectable set. Candidate masks become attention-key masks only in candidate-as-key architectures such as DeepSets context or a masked Set Transformer. Padding, action validity, Q-label support, feasibility-label support, and modality presence remain separate.

Duplicate-row and valid-count tests are required because candidate-set pooling or per-set normalization can otherwise change the absolute value of an unchanged physical candidate. A duplicate row may duplicate an output, but it must not silently change another row's value unless the tested architecture explicitly models candidate-set context.

Coordinate handling is deliberately weaker than exact SE(3) equivariance. World poses remain available for reproducibility, while model inputs use root-, target-, or candidate-relative geometry. Gravity, scale, height, yaw, camera direction, target orientation, motion limits, and frustum geometry remain physical variables. A global origin convention must not become a shortcut [Bro+21].

Actor/oracle provenance is the final acceptance condition. Target gains, GT associations, mesh distances, current candidate renders, and target crops may supervise or audit a model but may not enter its actor graph. Previously selected GT depth may enter only a `CF-GT` state branch. Source-dropout tests must show that removing an unavailable optional carrier changes only its masked branch.

The scalar requested-horizon interface adds its own acceptance contract: the mathematical boundary $Q_0 = 0$ is represented only by padded rows in the executable call; an $h = 1$ target contains no bootstrap; $h > b_t$ is rejected rather than clamped; and a target for $h > 1$ may depend only on a successor value requested at $h-1$ [De +20]. Public vectorization remains gated on real atomic callers, measured inadequacy of private batching, and scalar/vector parity tests. No monotonicity such as $Q_{h+1} \geq Q_h$ is assumed because valid actions may have negative immediate gain.

4.4.1 Orthogonal architecture ladder

Implementation: partial **Evidence: pending**

A0 independent-row MLP and A1 candidate-to-state cross-attention are implemented over the same `S0-pose` root-moments inputs and decoder seam. A1 remains the default; comparative evidence, richer scene carriers, and candidate interaction remain orthogonal measurements.

Literature: [Bre+23, Fuc+20, Lee+19, SHW21, Sch+15, Zah+17, Zho+23]

Promotion gate: measure identical-feature A0 versus A1 under the same fit/evaluation contract, then promote a level only after lower interaction controls pass on the same scene carrier and target/source protocol

Development source: aria_nbv/aria_nbv/data_handling/qh_data/views.py; docs/contents/theory/candidate_view_dependence.qmd

The shared A0/A1 query contains root- and current-relative candidate pose, an explicit candidate–target transform, and global root-scene moments. Target, scene, causal pose-history summary, remaining budget, and requested horizon are supplied as the same ordered state-token tuple. A0 maps `[query; vec(tokens)]` to one context with a row-shared MLP; A1 maps the query and tuple to the same context width with cross-attention. Both then expose `[query; context; query times context]` to the configured value decoder. This is feature matching, not parameter matching; every comparison reports parameters, runtime, and the frozen decoder/training identity. The physical trunk and feasibility head precede target/horizon conditioning. Candidate provenance and generator-family identity remain audit-only by default.

History representation is orthogonal to A0/A1 interaction. The default `None` alias preserves the original H0 masked mean and its old state-dictionary/config identity. Explicit `mean_pool_v1` names the same control; `causal_transformer_v1` is an implemented exploratory H1 carrier with relative-age encoding, a learned empty-prefix token, causal self-attention, and last-valid readout. Both emit one equal-width history token. H1 must retain exact-prefix, padding, future-perturbation, and candidate-equivariance tests, and it is not promoted from executability alone. Spatial scene memory remains a later internal replacement behind the same scorer interface [Sch+15].

Level	Interaction model	Scientific role
A0	Independent per-row MLP	Locks target/candidate descriptors, time context, and label learnability for a fixed scene carrier; adapters retain hard masks.
A1	Candidate-to-state cross-attention	Each candidate query independently reads shared target, scene, history, and budget context.
A2	DeepSets candidate context	Tests whether an unordered summary of valid candidates adds information without pairwise attention.
A3	Masked Set Transformer	Tests candidate–candidate interaction while preserving row equivariance and mask isolation.
A4	Query-local relation bias	Tests QCNet-style target, history, and candidate relations without importing its forecasting decoder.
A5	Temporal/recurrent state read	Tests whether ordered long-horizon history remains informative after explicit dynamic scene memory.
A6	Residual value head	Tests finite-horizon recovery over a calibrated continuous one-step root-gain control.
A7+	Exact-equivariant or graph interaction	Escalates only after local-frame controls reveal a symmetry-related failure.

Table 10: Interaction-architecture ladder. A0 and A1 are implemented; scene carrier, target/source protocol, time-query contract, decoder, and learning target stay fixed for their comparison.

Candidate-to-candidate attention is not required by the core task. It may improve relative policy context or diversity, but unrelated sampled rows must not silently redefine the absolute value of candidate $q_{t,i}$. Exact equivariant layers are similarly scoped to diagnosed support encoders or candidate graphs after local-frame scalar controls. This ordering favors reusable context encodings and keeps scalar requested-horizon conditioning separate from candidate-set interaction.

4.5 Target-Conditioned Finite-Horizon Value Model

Remove or rewrite before submission: Retain scalar requested-horizon conditional Q with direct continuous Huber regression as the core. Keep exact horizon two, CORAL, residual, behavior-return, and alternative estimator variants explicitly labelled as diagnostics or ablations.

Source: this section; aria_nbv/aria_nbv/lightning/qh_module.py

Gate: a frozen scorer interface, training objective, checkpoint, and matched policy evaluation

4.5.1 Implemented scorer and training interface

Implementation: partial **Evidence: pending**

The modular A0/A1 finite-horizon scorer, H0/H1 pose-history seam, S0 root-moments and privileged S1 selected-surface scene carriers, typed output, scalar horizon query, feasibility auxiliary, regression/CORAL value decoders, fitted-Q learner, bounded exact-Q2 certifier, hard-masked online adapter, and dense-valid data path are implemented. Bounded one-epoch CUDA contract fixtures cover both decoders for deployable CF0 bundle reconstruction and source-matched CF+ H0/S1 optimization. These tests establish executable updates, not population-level evidence. An initial five-chain CF+ comparison was inspected during S1 architecture development and therefore belongs to development evidence, not the confirmatory test split. Its numerical outputs have no immutable run receipt in this repository and support no thesis result. Neither CF+ profile is publishable as an inference bundle. A1 remains the default interaction and H0 the default history control; H1 and S1 are exploratory. The learned exact-Q2 and scientific policy gates remain unmet.

Literature: [CMR19, Fra+25, HGS15]

Promotion gate: retain the one-step scorer as a matched control, populate a held-out exact-Q2 receipt, and require independent oracle-rescored policy evidence

Development source: aria_nbv/aria_nbv/data_handling/qh_contracts.py; aria_nbv/aria_nbv/vin/models/target_finite_horizon.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/lightning/qh_q2_certification.py; aria_nbv/aria_nbv/oracle/pipelines/online_qh.py; aria_nbv/tests/lightning/test_qh_gpu_smoke.py; aria_nbv/tests/lightning/test_qh_q2_certification.py

The actor DTO carries root semidense evidence, target pose and extent admitted by the declared target-input protocol, root-relative candidate geometry, selected-pose history, remaining budget, and candidate materialization support. The non-deployable `v0_gt_input` protocol provides privileged GT-derived target geometry; deployable `v1_observed` requires an actor-visible descriptor and binds its source, construction provenance, and hash in the training and held-out receipts. The DTO deliberately excludes supervision and audit lineage from scorer inputs. In `root_moments_v1`, semidense support is the number of finite persisted points divided by that chain's persisted point length; the collated batch width is padding only and cannot change the scene token or candidate scores. The current model makes this an executable `S0-pose` baseline, but neither a trained checkpoint nor task-sufficient reconstruction state follows from interface tests alone.

The H0 scorer has two source roles. Deployable `qh_cf0_v1` requires the selected-observation carrier to be absent. Privileged `qh_cfplus_gt_depth_v1` requires the complete CF-GT prefix $\boldsymbol{o}_t = (D, V, K, T_{r\leftarrow c})_{j<t}^{\mathrm{sel}}$ with the same source tag, raster and pose shapes, dtypes, device, and exact causal support as the factual selected-pose history. In the source-matched H0 control these facts are admission conditions only: depth values, depth-valid entries, calibration values, and selected-camera poses do not enter either prediction branch. For any two structurally admissible carriers,

$$\mathrm{Struct}(\boldsymbol{o}_t) = \mathrm{Struct}(\boldsymbol{o}_t') \quad \Rightarrow \quad f_\theta^{\mathrm{CF+\text{-}H0}\left(s_t^{\mathrm{S0\text{-}pose}}, \boldsymbol{o}_t, \boldsymbol{\varphi}_e, \{q_{t,i}\}_{i=1}^{N_q}, h\right)} = f_\theta^{\mathrm{CF+\text{-}H0}\left(s_t^{\mathrm{S0\text{-}pose}}, \boldsymbol{o}_t', \boldsymbol{\varphi}_e, \{q_{t,i}\}_{i=1}^{N_q}, h\right)}$$

This invariance makes CF+ H0 the correct source-matched null model for a later S1 encoder: H0 and S1 can share the CF+ population, seed, optimizer, and carrier contract. Their difference nevertheless estimates the complete S1 package, including geometry consumption, added point-network capacity, initialization, and its stochastic path; it does not identify geometry alone. CF+ H0 is not a selected-depth value estimator, and CF0-versus-CF+ H0 differences would primarily confound source population rather than identify a representation gain. Dataset provenance reports admitted CF+ source rows and scenes relative to the immutable actor split. It also reports distinct admitted source–target tasks, but leaves their coverage fraction undefined because the actor store does not enumerate the complete target-task denominator. H0/S1 results therefore estimate performance conditional on CF-GT carrier availability and cannot silently become a claim about the complete root–target population. Lightning must explicitly mark the run privileged; the deployable bundle validator rejects the CF+ profile even if a manifest falsely clears that boolean marker.

The implemented S1 control consumes that carrier without widening the downstream scorer. For selected view $j < t$ and sampled valid pixel u, canonical float32 backprojection uses the persisted metric depth, linear pinhole, and root-from-camera pose before expressing the point from the factual current camera.

A shared point MLP and masked mean/maximum pooling form a fixed-width residual:

$$
\begin{aligned}
\boldsymbol{p}^{c_t}_{t,j,u} &= T_{c_t\leftarrow r}T_{r\leftarrow c_j}\pi^{-1}\Big(u, D^{\mathrm{sel}}_{j,u}\Big), \quad j<t \\
\boldsymbol{z}_{t,j,u} &= \varphi_{\mathrm{pt}\left(\frac{\boldsymbol{p}^{c_t}_{t,j,u}}{\sigma_{\mathrm{xyz}}}\right)} \\
\boldsymbol{g}^{\mathrm{S1}}_t &= \Big[\mathrm{Mean}\,\boldsymbol{z}, \mathrm{Max}\,\boldsymbol{z}, \rho^{\mathrm{present}}_t, \rho^{\mathrm{pixel}}_t, \rho^{\mathrm{view}}_t\Big] \\
\boldsymbol{\Phi}^{\mathrm{scene}\,\mathrm{S1}}_t &= \boldsymbol{\Phi}^{\mathrm{scene}\,\mathrm{root}}_t + W_{\mathrm{pt}}\boldsymbol{g}^{\mathrm{S1}}_t, \quad W^{0}_{\mathrm{pt}} = 0
\end{aligned}
$$

The zero-output condition $W^0_{\mathrm{pt}} = 0$ is an experimental-control constraint. Under a matched seed, fresh S1 and H0 models have identical common weights and exactly identical conditional-Q and feasibility predictions; the mere presence of a higher-capacity carrier cannot perturb the comparison before training. Only the bias-free final projection is zeroed. Its first gradient is proportional to the already nonzero pooled statistic $\boldsymbol{g}^{\mathrm{S1}}_t$, so ordinary backpropagation can open the residual immediately. The point MLP itself receives no task gradient through a zero projection on that first update and begins adapting after W_{pt} moves away from zero. This short delay is the intended cost of an exact nested-model comparison, and requires no learned gate, warm-up schedule, or checkpoint variant.

The metric divisor σ_{xyz}, pixel stride, view-chunk bound, point width, raster geometry, source profile, and identity-start rule are experiment identity. The executable carrier records this contract as `root_moments_plus_selected_surface_points_identity_start_v1`. The older `root_moments_plus_selected_surface_points_v1` discriminator did not distinguish random from zero-residual initialization; readers retain it as `unknown_legacy_v1` evidence, but new training, warm start, inference, and bundle publication reject it. PyTorch3D is pinned to the exact resolved VCS commit, and deployable manifests verify the installed direct-URL commit rather than recording only the package version. ρ^{pixel} is valid sampled pixels divided by causal sampled-pixel capacity; ρ^{view} is selected observations with any valid point divided by causal observation count. An empty or all-invalid prefix yields exactly the root-moments control throughout training. Mean pooling makes the carrier density weighted rather than duplicate invariant: globally replicating point support together with sampled capacity preserves the statistic, whereas duplicating only a surface subset changes its empirical weight. It retains neither source-view identity nor ray/free/unknown semantics, and it is target- and candidate-independent. Candidate-relative point attention and Déjà-View-style recurrent refinement remain later, separately promoted interactions.

The first five-chain CF+ comparison cannot serve as confirmatory evidence because its outcomes informed the next S1 initialization choice. It is reclassified as development evidence and its unreceipted numerical summaries are omitted. A valid S1 comparison requires an immutable research-run artifact binding the split role, store and scorer identities, checkpoints, seeds, selected fitted-Q row count, exact-Q2 row count, chain-level aggregation, and uncertainty. Model selection must then stop before evaluation on a new untouched scene-disjoint test manifest. Until that protocol is executed, S1 remains a package-level optimization probe: neither geometry benefit, candidate-local attention, nor recursive-value improvement is established.

The scorer emits two policy-facing tensors before masking:

$$\left(Q^{\text{cond}}_{h,\theta,e,i}, \ell^{\text{feas}}_{t,i}\right) = f_\theta\left(s_t, e, q_{t,i}, h\right), \quad h = b_t \text{ in implemented V1;} \quad 1 \le b_t \le H_{\max}$$

`conditional_q` is finite for every materialized row and is trained only where hard validity and Q-label support admit a target. `feasibility_logits` is produced by a target-, horizon-, and action-mask-independent physical trunk and may receive binary supervision on labelled materialized rows. A configured value decoder may attach training-only tensors, but neither policy masking nor feasibility reads them. Lightning owns hard-mask admission, decoder-specific value loss, optional feasibility BCE, exact $h \leftarrow h-1$ recursion checks, Double-Q targets, and target synchronization. Online inference compacts conditional values only through the authoritative hard mask.

4.5.2 Scalar requested-horizon scorer

Implementation: implemented **Evidence: pending**

One scalar h per state is implemented. Omitting the query uses factual remaining budget; public horizon vectorization remains evidence-gated.

Literature: [De +20, EGW05, Lee+19, Sch+15, Zah+17, Zho+23]

Promotion gate: exact-Q2 certification, A0/A1 controls, per-horizon support tests, and held-out oracle policy comparison

Development source: docs/typst/thesis/sections/04-method/04-01-scene-representation-requirements.typ; docs/typst/thesis/sections/04-method/04-02-descriptor-and-encoding-plan.typ; aria_nbv/aria_nbv/lightning/qh_module.py

Using the finite-horizon return and conditional value defined in Section 2.4, the return for target e over exactly the requested residual horizon h is

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \le h \le b_t \le H_{\max}$$

and the learned quantity is

$$Q_{h,e}^{\star}(s_t, i) = \sup_{\pi \in \Pi^{\text{act}}} \mathbb{E}_\pi \left[G_{t,e}^{(h)} \mid s_t, a_t = i \right], \quad i \in \mathcal{A}_t, \quad 1 \le h \le b_t \le H_{\max}, \quad Q_{0,e}^{\star}(s, i) = 0$$

The notation Q_H names a bounded finite-horizon family. One model query is

$$f_{\theta\left(s_t^{\text{S0-pose}}, \boldsymbol{\varphi}_e, \{q_{t,i}\}_{i=1}^{N_q}, h\right)} \to \left(\left\{ Q_{h,\theta,e,i}^{\text{cond}} \right\}_{i=1}^{N_q}, \left\{ \ell_{t,i}^{\text{feas}} \right\}_{i=1}^{N_q} \right)$$

rather than a value from a separately configured fixed-H model. The maximum $H_{\max}$ is a dataset, model, and checkpoint contract. Requested residual horizon h is a model input and remaining budget b_t determines whether the query is admissible. The admissible pairs form the triangular domain $\{(b_t, h) : 1 \le h \le b_t \le H_{\max}\}$: $h = b_t$ requests the full factual residual budget, while $h < b_t$ requests a truncated return from the same factual state. `None` selects the diagonal; it does not mean that the horizon is unknown. The boundary $Q_{0,e} = 0$ closes the recursion mathematically, while zero in executable tensors is reserved for padding and not a realized learned query. Syntactically in-range requests still fail closed when their horizon lacks manifest-bound training and promotion support. The step index t stays lineage unless a named non-stationarity ablation uses it.

One candidate row is one geometric query. Candidate pose, candidate–target relation, and root-scene support are encoded once; remaining budget and requested horizon stay separate named state tokens. The implemented A0 and A1 controls receive identical query and state-token values and return the same-width context:

$$
\begin{aligned}
\boldsymbol{Z}_t &= \left(\boldsymbol{\Phi}_t^{\text{scene}}, \boldsymbol{h}_e^{\text{tgt}}, \boldsymbol{h}_t^{\text{hist}}, \text{Emb}\left(\frac{b_t}{H_{\max}}\right), \text{Emb}\left(\frac{h}{H_{\max}}\right)\right) \\
\boldsymbol{c}_{t,i}^{\text{A0}} &= \text{MLP}_{\text{A0}}\big(\text{concat}\big(\boldsymbol{x}_{t,i}, \text{vec}(\boldsymbol{Z}_t)\big)\big) \\
\boldsymbol{c}_{t,i}^{\text{A1}} &= \text{CrossAttn}_{\text{A1}}\big(\boldsymbol{x}_{t,i}, \boldsymbol{Z}_t, \boldsymbol{Z}_t\big) \\
\boldsymbol{u}_{t,i}^{\text{Ak}} &= \text{concat}\big(\boldsymbol{x}_{t,i}, \boldsymbol{c}_{t,i}^{\text{Ak}}, \boldsymbol{x}_{t,i} \cdot \boldsymbol{c}_{t,i}^{\text{Ak}}\big), \quad \text{Ak} \in \{\text{A0}, \text{A1}\}
\end{aligned}
$$

followed by the same `[query, context, query times context]` decoder input and shared per-row value head. A0 flattens the five tokens in the fixed semantic order scene, target, causal history, budget, horizon and uses an independent-row MLP. A1 uses candidate-to-state cross-attention; candidates never become keys or values. The controls are feature-matched but not parameter-matched, so comparisons report trainable parameters and runtime rather than attributing a difference solely to attention. This conditioning follows the general principle of a shared value approximator queried by an explicit task variable [Sch+15]. Fixed-H models and separate $Q_1, ..., Q_H$ heads remain ablations rather than competing public interfaces.

The history token has its own versioned one-factor seam. H0 performs the original masked mean over selected poses expressed from the current camera. H1 keeps those geometric inputs fixed, adds normalized relative age $a_{t,j} = \frac{t-1-j}{H_{\max}}$, and applies a causal Transformer with a learned empty-prefix token and last-valid readout. The immediate predecessor has age zero; no absolute step embedding is added. Materialized support must equal the complete prefix $j < t$, while padded states remain zero. H1 exposes causal pose order and prefix length, but it does not add selected observations or change the `S0-pose` state estimand. It remains unpromoted until a matched repeated-seed comparison and held-out endpoint evidence justify its capacity.

The public interface scores one h per state and returns $\left[B, S, N_q\right]$. Multiple horizons use separate calls; private batching may reuse encodings. A public L axis is introduced only after two real atomic callers, measured inadequacy of private batching, and scalar/vector parity tests exist. Every query receives only the causal state available at step t; the learning target, not future scorer input, determines how many rewards the value represents.

The candidate materialization mask sanitizes padding. Changing only the valid-action mask cannot change raw conditional Q or feasibility logits:

$$\left(Q^{\text{cond}}_{h,\theta,e,i}, \ell^{\text{feas}}_{t,i}\right)\left(s_t, e, q_{t,i}, h, \boldsymbol{m}_t\right) = \left(Q^{\text{cond}}_{h,\theta,e,i}, \ell^{\text{feas}}_{t,i}\right)\left(s_t, e, q_{t,i}, h, \boldsymbol{m}'_t\right)$$

The valid-action mask instead gates the policy argmax, Q supervision, and every bootstrap maximum. The mask defines $\mathcal{A}^{\text{valid}}_t = \left\{i : m_{t,i} = 1\right\}$ for the thesis-core policy. A hard-invalid but materialized row may retain a finite network output for regular tensor operations, but that number has no supervised or deployable Q interpretation and can never enter selection or backup. The feasibility logit is a separate prediction of admissibility; multiplying it by conditional Q would mix units and changing invalid Q values to zero is unsafe when valid values can be negative. Target-independent root encodings may be reused across targets, but target-dependent candidate generators require per-target tables or a union table with an explicit target–candidate availability mask.

The value family is defined relative to a frozen state and source protocol. An `S0-pose` value, a privileged `CF-GT` selected-depth value, and a deployable observed-state value are different functions even when their tensors have similar shapes. The implemented CF+ H0 control belongs to the CF+ source population but, by construction, still estimates an S0-pose-conditioned function; the privileged selected-depth value begins only when S1 actually consumes the carrier. Likewise, "optimal" means optimal continuation only within the generated finite candidate support, hard-validity contract, represented state, and transition distribution. A pose-only state cannot be promoted to a task-sufficient reconstruction value merely by requesting a longer horizon.

4.5.3 Horizon-recursive offline learning

Batch fitted Q iteration learns a greedy action-value function from a fixed transition collection through successive supervised regression problems; it does not require online interaction [EGW05]. The frozen lower-horizon recursion is

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

and, for $h > 1$,

$$y_t^{(h,e)} = r_t^e + \gamma B_t^{(h,e)}$$

where the lower-horizon prediction is detached, frozen, or supplied by a delayed target copy. The essential structural rule for this candidate is $Q_h \leftarrow Q_{h-1}$: no horizon value bootstraps from itself. Fixed-horizon TD motivates this recursion and shows that horizon-indexed values can share parameters and be updated in parallel, although a staged $h = 1$ to H schedule remains the clearest initial control [De +20].

The stored evidence gives a particularly strong base case. The executable objective queries $h = 1$ for every realized state and supervises every finite hard-valid candidate with continuous one-step root-normalized gain. Candidate losses are averaged within state before state means are averaged within horizon; non-empty horizons then receive equal weight. Selected-transition recursion is disjoint and begins at $h > 1$. This prevents large candidate tables and abundant Q1 labels from silently dominating longer horizons. The bundle records the realized state/candidate support by horizon and online inference rejects a requested horizon absent from that support. If a selected first action has a successor table with dense one-step labels, then the exact finite-support H=2 target is

$$y_t^{(2,\text{exact})} = r_t^e + \gamma_t \max_{j:m_{t+1,j}^{\text{train}}=1} r_{t+1,j}^e$$

The implemented certification surface distinguishes two claims. A unit-level implementation control injects exact one-step values and proves that the recursive tensor path reproduces this target. A frozen-bundle population run instead measures the learned recursion error

$$\varepsilon_t^{(2)} = \left|y_t^{(2,\text{recursive})} - y_t^{(2,\text{exact})}\right|, \quad \varepsilon_t^{(2)} \le \tau_{\text{abs}} + \tau_{\text{rel}}\left|y_t^{(2,\text{exact})}\right|$$

on held-out supported rows. That second quantity includes the learned Q_1 approximation and is therefore model evidence, not another implementation parity test. The population is censused without actor-tensor materialization, stratified by scene and target identity, configured horizon, candidate-width bin, candidate generator, rollout recipe, and behavior policy, then selected by a deterministic balanced hash under explicit global and per-stratum bounds. The receipt reports chain coverage, exact-row support, per-stratum error, numeric tolerances, and —when CORAL is used—outer-class occupancy and values outside the fixed representative support. Longer-horizon interpretation remains gated until this learned Q_2 error passes its frozen support and tolerance contract and independent held-out endpoint evaluation establishes positive oracle-lookahead headroom. The existing persisted terminal-step contrast is a diagnostic proxy and cannot satisfy that endpoint gate.

The census denominator is the complete eligible held-out chain population, not only chains that happen to contain an exact horizon-two row. A chain that terminates or becomes unsupported before factual $h = 2$ contributes zero exact rows and therefore lowers support coverage. It must not disappear through post-hoc filtering. Consequently, an executable one-epoch fit, a valid bundle, or even low error on a few supported rows cannot promote $h > 2$ when the frozen minimum-row, coverage, or tolerance predicate fails.

Double Q is an optional estimator for the learned successor maximum. The nonterminal bootstrap first intersects hard action support with factual support at horizon $h - 1$:

$$\mathcal{A}_{t+1}^{(Q,h-1)} = \left\{i : m_{t+1,i}^{\text{act}} = 1 \wedge m_{t+1,i}^{Q,h-1} = 1\right\}$$

The online scorer then selects

$$B_t^{(h,e)} = \begin{cases} Q_{h-1,\theta^-,e}\Big(s_{t+1}, \operatorname{argmax}_{i\in\mathcal{A}_{t+1}^{(Q,h-1)}} Q_{h-1,\theta,e}(s_{t+1}, i)\Big) & \text{if } \ h > 1 \wedge d_t = 0 \wedge \mathcal{A}_{t+1}^{(Q,h-1)} \neq \emptyset \\ 0 & \text{otherwise} \end{cases}$$

and a delayed scorer to evaluate that row. This may reduce overestimation from maximizing noisy learned values [HGS15]. It is neither a requirement for offline learning nor the definition of the frozen scalar requested-horizon interface. It cannot repair unsupported long-horizon actions, an aliased actor state, or missing selected-observation evidence.

Complete stored chains also permit regression to truncated Monte-Carlo returns. Without off-policy correction, those returns estimate the continuation of the behavior policy that generated each chain, Q^{μ}, rather than the greedy finite-support value $Q^{\star}$ unless that behavior policy is itself the specified target policy [SB18, Secs. 5.2 and 5.5, pp. 96–97 and 103–109]. They are therefore useful controls and diagnostics, but they must not be mixed with optimal Bellman targets without naming the estimand.

The objective-design comparison is therefore:

1. dense continuous Q_1 supervision on every candidate admitted by `q_train_mask`;
2. exact selected-action Q_2 supervision as a base-case certification;
3. direct continuous regression versus provenance-bound fixed-support CORAL decoding over the same fitted-Q targets;
4. the shared scalar-horizon model versus fixed-H or separate-head ablations for $h = 1, ..., H_{\max}$;
5. Double-Q selector/evaluator backups as a max-bias ablation;
6. behavior-policy Monte-Carlo return regression as a separate estimand;
7. an uncentred one-step-plus-residual decomposition.

Because dense one-step rows vastly outnumber selected transitions at longer horizons, training and evaluation report state-normalized loss and continuous-unit mean absolute calibration error separately by horizon. Dense $h = 1$ additionally reports within-state pairwise ranking accuracy over unequal-target candidate pairs and greedy selected-action regret. The factual selected-transition labels at $h > 1$ do not identify either counterfactual quantity, so their per-horizon ranking-pair and regret-state support remains explicitly zero instead of fabricating a metric. Exact or independently oracle-rescored longer-horizon candidate tables are required before those fields can become nonzero. If requested horizons share one learner, their sampling or weighting must be explicit; an aggregate loss must not allow Q_1 to hide longer-horizon failure.

4.5.4 Modular continuous-value decoding

Implementation: implemented **Evidence: pending**

Direct regression and provenance-bound fixed-support CORAL are executable alternatives over the same candidate-state features and the same continuous fitted-Q targets. Regression remains canonical; CORAL is an ordinal ablation, not a new action-value estimand.

Literature: [CMR19, Dab+17]

Promotion gate: fit-data-only support selection, held-out per-horizon ranking and calibration, support-saturation reports, and matched policy evaluation

Development source: aria_nbv/aria_nbv/vin/modules/qh_value_decoders.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/tests/vin/test_qh_value_decoders.py

The representation trunk produces one feature vector per materialized candidate row. A terminal decoder maps that vector to the scalar `conditional_q` consumed by the unchanged Double-Q backup and hard-masked selector. The regression decoder applies a per-row MLP and linear output, trained with Huber loss in continuous root-gain return units. The CORAL decoder applies a per-row MLP and $K-1$ cumulative thresholds. For fitted-Q target y_n, fixed edges e_k^Q create the ordinal class c_n^Q standard cumulative binary cross-entropy supervises the thresholds [CMR19]:

$$
\begin{aligned}
c_n^Q &= \sum_{k=0}^{K-2} \mathbb{1}\left[y_n > e_k^Q\right], \quad l_{n,k} = \mathbb{1}\left[c_n^Q > k\right] \\
\mathcal{L}_Q^{\mathrm{CORAL}} &= -\sum_n \sum_{k=0}^{K-2} \left(l_{n,k} \log p_{n,k} + \left(1 - l_{n,k}\right) \log\left(1 - p_{n,k}\right)\right), \quad p_{n,k} = \sigma\left(o_{n,k}\right) \\
\pi_{n,k}^{\mathrm{raw}} &= p_{n,k-1} - p_{n,k}, \quad p_{n,-1} = 1, \quad p_{n,K-1} = 0 \\
\tilde{\pi}_{n,k} &= \frac{\max\left(\pi_{n,k}^{\mathrm{raw}}, 0\right)}{\sum_{j=0}^{K-1} \max\left(\pi_{n,j}^{\mathrm{raw}}, 0\right) + \varepsilon} \\
Q_n^{\mathrm{cond}} &= \sum_{k=0}^{K-1} \tilde{\pi}_{n,k} u_k^Q
\end{aligned}
$$

CORAL provides order, not metric distance. The continuous interpretation used by backup and ranking is an additional experiment contract: repaired class mass is averaged through fixed, strictly increasing representatives u_k^Q. Support provenance is a closed value object: `train_fitted_v1` may use training targets only, whereas `predeclared_physical_v1` names a target-independent physical rule and units. Both bind the construction method, source population, split role, ordered input digest, edges, representatives, and self-verifying artifact digest into scorer and bundle identity. Historical fixed-edge configurations remain inspection-only. Changing any of these facts defines another model even if K is unchanged. Invalid candidates are not mapped to the lowest class; hard-invalid and unsupported selected rows are excluded before either Huber or CORAL loss.

The outer CORAL classes are open-ended while decoded values saturate at the outer representatives. Every CORAL run consequently reports the fraction of fitted-Q targets below and above the representative support, the outer-class fraction, and pre-repair cumulative-probability order violations. Train-fitted support must be frozen before validation or test evaluation; physically predeclared support must be frozen before any empirical target outcome is inspected. Targets are never clipped into the outer classes. A one-epoch GPU smoke proves the executable optimization transaction and, only for profiles admitted by the publisher, bundle reconstruction. It does not prove scientific superiority or make Huber and cumulative-BCE loss magnitudes comparable.

If a third decoder is promoted, quantile regression is the most coherent next study: it estimates a return distribution whose expectation can preserve the scalar ranking seam, whereas CORAL only discretizes one scalar target [Dab+17]. Such a head is justified only after stochastic returns or actor-state aliasing are measured, and it requires a separately frozen distributional Bellman projection, quantile support, risk-neutral or risk-sensitive decoding rule, and calibration evaluation. It must not be introduced as a stylistic replacement for direct Q.

4.5.5 One-step base and finite-horizon residual

Implementation: exploratory **Evidence: pending**

The residual decomposition is a testable hypothesis. Direct continuous finite-horizon value prediction remains the required control under whichever time-query interface is selected.

Literature: [De +20, CMR19]

Promotion gate: target-specific root-gain calibration and direct-versus-residual ablation across horizons

Development source: docs/literature/tex-src/arXiv-VIN-NBV/sec/3_methods.tex, ordinal RRI paragraph, line 125; docs/contents/theory/candidate_view_dependence.qmd

The hypothesis separates calibrated immediate utility from downstream effects:

$$b_{\psi,i} = f_{\psi}^{\text{1-step}}\big(s_t^{\text{cf0}}, \boldsymbol{\varphi}_e, q_{t,i}\big), \quad \delta_{\theta,t,e,i}^{h} = g_{\theta}\big(\mathcal{I}_{t,e}, q_{t,i}, h\big), \quad Q_{h,\theta,e,i} = b_{\psi,i} + \delta_{\theta,t,e,i}^{h}$$

The base $b_{\psi,i}$ must predict continuous one-step target root gain

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

in the same additive units as the finite-horizon return. Historical state-relative RRI or an ordinal CORAL score may remain auxiliary ranking and calibration outputs, but they are not interchangeable with this additive base unless an explicit calibrated conversion is learned and validated. The residual captures candidate regeneration, selected-observation state changes, occlusion, support, and the requested residual horizon.

It is not exactly mean-centred within each candidate table because duplicate or unrelated rows would then change absolute value targets. Magnitude regularization may be tested without redefining the value field:

$$Q_{h,\theta,e,i} = \hat{r}^{e}_{\psi}\big(s^{\mathrm{cf0}}_{t}, \boldsymbol{\varphi}_{e}, q_{t,i}\big) + \delta^{h}_{\theta,t,e,i}\big(\mathcal{I}_{t,e}, q_{t,i}\big), \quad \mathcal{L}_{\delta} = \lambda_{\delta}\frac{1}{|\mathcal{A}_{t}|}\sum_{j\in\mathcal{A}_{t}}\big(\delta^{h}_{\theta,t,e,j}\big)^{2}$$

The historical one-step CORAL score remains a motivated myopic ranking and calibration interface, but it must not be confused with the implemented finite-horizon CORAL decoder above. The historical head discretizes one-step RRI; the new decoder discretizes the continuous fitted-Q target and decodes back to return units. Fixed-H and requested-horizon models, the exact H=2 control, Double-Q ablation, Monte-Carlo control, and residual learner form separate comparisons rather than one implicit architecture.

Research TODO: Compare staged and joint shared-parameter variable-horizon fitted Q against separate per-horizon heads; include dense Q1, exact Q2, Double Q, behavior-return regression, and the uncentred residual after positive oracle-lookahead headroom is established.

Source: variable-horizon learning-target contract

Gate: A0/A1 learning, per-horizon support report, and headroom report

Open decision: Freeze the maximum horizon, horizon-sampling or loss-weighting rule, lower-horizon target-update schedule, and target-network rule in the resolved training manifest. For the residual ablation, also record whether the one-step base is frozen, slow-fine-tuned, or jointly trained.

Source: variable-horizon fitted-value hypothesis

Gate: model-selection protocol freeze

The scientific endpoint is not training loss. For every claimed horizon, selected trajectories are regenerated under the documented candidate and state protocol and re-scored by the same target-specific oracle used for the baselines. After positive oracle-lookahead headroom is established against oracle-greedy, the model succeeds only if it closes a prespecified fraction of the endpoint gap from the actor-visible learned-myopic control to oracle lookahead on held-out scenes without violating horizon, mask, provenance, source, or support constraints.

5 Experimental Design

The experiments separate feasibility evidence from policy evidence. Feasibility requires a frozen scene-level population, reproducible target and candidate support, valid replay linkage, and measured oracle throughput. Policy inference additionally requires a held-out scene split, matched acquisition budgets, independent endpoint oracle evaluation, and an implemented actor-visible decision rule. The current train-only bandwidth pilots address only the first category: incomplete runs and resource failures characterize the generation system, but they cannot support comparisons between policies.

5.1 Study Population and Evidence Gates

Validation TODO: Preregister the eligible population, exclusions, primary estimands, aggregation unit, number of independent runs, uncertainty interval, minimum meaningful effect, and multiplicity policy before inspecting confirmatory results.

Source: experiment manifest and analysis specification

Gate: immutable analysis plan plus matched held-out policy table

5.1.1 Candidate-Generation Realism Analysis Contract

RQ4 treats the scene as the independent unit. Candidate-support quality-control observations are computed per state, averaged across states within scene, and finally macro-averaged across the scene cohort; no candidate row is treated as an independent scene replicate. These within-profile diagnostics remain descriptive. Any comparative claim instead uses a paired per-scene difference over eligible scene/target/root tasks observed under both profiles, equal candidate budgets, and the frozen candidate-generation manifest. Its report identity is `paired_scene_mean_difference`, with the paired scene count as its denominator and an interval computed over the paired scene differences.

Missing or failed states are retained in the audit tables. Support counts and fractions use the preregistered failure values stated in Table 3; an undefined target-relative direction prevents that paired scene from entering a geometric contrast rather than creating an angle by imputation. The excluded-scene count and reason remain reported. Thus the descriptive state–scene macro summaries and the paired comparative estimand have distinct populations and cannot be substituted for one another.

Minimum operational support is reported as a worst-case diagnostic; the fifth percentile of per-state valid support is the more stable lower-tail estimand, accompanied by actor-valid fraction, configured-family zero rate, target-side balance, and circular target-relative orbit span. Failed roots and zero-valid configured family/state pairs remain part of the denominator. The projection fraction, oracle opportunity, and jitter compliance are secondary diagnostics: projection is calibrated framing rather than visibility, oracle opportunity is finite-support headroom rather than policy performance, and jitter compliance must preserve the nonzero production seminar-jitter invariant.

The source population comprises ASE/ATEK snippet windows from scenes for which the configured GT mesh and object-box table resolve. A frozen manifest assigns entire scenes to train, validation, or test before model selection; no scene may cross these boundaries through another snippet. Each reported run records the manifest hash and the exact counts of scenes, snippets, admitted target tasks, rollout chains, transitions, and retained candidate rows. A capped train-only pilot is therefore a throughput and support probe, not a sample from which held-out policy performance can be estimated.

The present data generator defines oracle target tasks by seeded sampling from geometry-valid GT OBB rows. Its task-coverage report therefore describes the available GT pool, sampled tasks, classes, scenes, and later oracle-evaluation failures. It does not measure proposal matching, IoU ambiguity, projected visibility, or actor-observation support. Those quantities belong to a future observed-target selector required for deployable-input claims. Until that selector exists, GT target geometry may define labels and bounded oracle references but cannot be presented as actor-visible input. The privileged-supervision boundary in Figure 1 applies equally to render-derived evidence: dense GT candidate depth may produce labels, returns, or explicitly named privileged ablations, but it is not a legal actor input. A separate teacher policy, distillation path, or current-belief renderer remains a hypothesis until implemented and evaluated, so none is promoted to a main-text process figure.

The first policy gate is an actor-visible myopic scorer over the same finite candidate table intended for Q_H. The scorer must expose one value per candidate, respect the hard action mask, and be assessed by candidate ranking, calibration, and oracle-rescored selected actions. The existing scene-level VIN scorer is historical substrate; it is not a target-conditioned control until the observed target descriptor is wired into the model and evaluated on a frozen held-out split.

The second policy gate estimates whether bounded oracle lookahead has headroom over one-step oracle greedy:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

Only if the preregistered analysis classifies this headroom as meaningful is Q_H evaluated for closure of the separate actor-visible learned-myopic-to-oracle-lookahead endpoint gap:

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Success is measured by matched endpoint oracle evaluation, not predicted values or training loss. If lookahead has no meaningful headroom, the result is scoped to the frozen split, target protocol, candidate generator, horizon, branch factor, and validity regime. If headroom exists but the learned model does not close the prescribed actor-visible-myopic-to-oracle-lookahead gap, target observability, action support, replay coverage, reward construction, and model capacity remain separate candidate explanations.

Claim	Primary evidence	Decision rule
Population	scene-split manifest and coverage bundle	Inference is restricted to the frozen held-out scene population.
Task protocol	GT pool, sampled tasks, classes, and oracle failures	Current evidence is oracle-task coverage, not observed-target matching.
Myopic control	ranking, calibration, and oracle-rescored selections	Actor-visible target conditioning must be implemented before comparison.
Planning headroom	Δ_{look} and learned-control gap-closure ratio η_Q	Q_H gap closure is evaluated only after a meaningful oracle-headroom gate.

Table 11: Objective-to-evidence matrix.

5.2 Replay Eligibility and Finite-Horizon Learning Gate

The factual rollout tables determine which records may supervise each learning problem. The action-validity mask defines actions selectable under the admitted oracle geometry contract. The stricter Q-training eligibility additionally requires a valid target/ground-truth label state and finite target-root-gain and diagnostic target-RRI labels. Padding, actor validity, one-step training eligibility, transition eligibility, modality presence, source role, and horizon availability remain separate masks. Masked rows remain available for support and failure analysis but cannot enter action selection, supervised loss, or bootstrap maximization.

All eligible candidate rows can support dense one-step supervision. Exact H=2 supervision is narrower: the factual first action must have a stored reward and either an explicit terminal outcome, whose continuation is exactly zero, or a valid successor step with finite one-step root-gain labels for every hard-valid successor candidate. This completeness condition makes the masked successor maximum exact over the stored action set; one finite label is insufficient because an unlabeled hard-valid action could have the true maximum. General recursive supervision is narrower again: it requires a factual selected action, reward, terminal flag, discount, a defined training horizon, and—when nonterminal—a reproducible successor state, hard mask, and lower-horizon factual support. The derived `q_h/` arrays align these fields on a padded state–candidate view; they do not create labels for unobserved transitions, make selected GT depth actor-visible, or turn sparse long-horizon action support into dense support.

Learning surface	Purpose	Minimum factual evidence
dense $h = 1$	immediate candidate value	actor-selectable row with finite one-step root-gain label
exact $h = 2$	base-case finite-support value	selected reward plus either terminal outcome or complete finite one-step labels over the hard-valid successor action set
recursive $h > 1$	variable-horizon fitted value	selected transition, terminal and discount; nonterminal rows also require successor actor state, hard mask, and lower-horizon target support
behavior return	policy-conditioned Monte-Carlo control	complete retained reward prefix and behavior-policy identity

Table 12: Required replay evidence by learning target. Dense immediate labels, exact H=2 targets, recursive optimal-continuation targets, and behavior-policy returns are distinct supervision surfaces.

Gate	Available evidence	Required before inference
Oracle data	target labels, masks, lineage, replay validation, and selected-depth persistence	held-out coverage, source-role audit, and matched endpoint evaluation
Myopic control	scene-level VIN substrate	actor-visible target-conditioned Q_1 scorer and frozen checkpoint
Finite-horizon scorer	feature-matched A0/A1–S0-pose/root-moments scalar-horizon controls and fitted-Q seam	exact-Q2 certification, parameter/runtime report, compatible checkpoint, frozen state protocol, and oracle-rescored policy
Requested horizons	fail-closed scalar query and exact $h \leftarrow h-1$ recursion tests	supported targets through $H_{\max}$, positive headroom, and per-horizon validation
Dynamic Q_H	selected-observation persistence and planned state update	typed dynamic-state reader, deterministic fusion, source masks, and held-out policy evaluation
Policy claim	train-only feasibility pilots	completed held-out paired comparison under equal budget

Table 13: Learning-readiness gates. A runnable scorer establishes executable readiness, not a scientific policy result.

Implementation: partial **Evidence: pending**

The scalar requested-horizon A0/A1 controls, dense-Q1 plus selected-recursion Double-Q learner, feasibility auxiliary, manifest-bound trained-horizon support, and bounded stratified exact-Q2 certification surface are implemented. A1 remains the default; a qualifying population receipt, task-sufficient dynamic state, and comparative policy evidence remain unavailable.

Literature: [De +20, EGW05, FMP19, HGS15, Kum+20]

Promotion gate: populated held-out exact-Q2 receipt, supported H>2 targets, compatible checkpoint, frozen state protocol, and independent held-out oracle re-evaluation

Development source: aria_nbv/aria_nbv/data_handling/qh_data/batching.py; aria_nbv/aria_nbv/lightning/qh_datamodule.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/lightning/qh_q2_certification.py; aria_nbv/aria_nbv/rollouts/qh_reader.py

The finite-candidate value model decodes actions only over valid candidate rows:

$$\boldsymbol{u}_{t,i} = \mathrm{Enc}_\theta\big(\mathcal{I}_{t,e}, q_{t,i}\big)$$

$$a_t^\theta = \operatorname*{argmax}_{i:m^{\mathrm{act}}_{t,i}=1} Q_{h,\theta,e}(s_t, i)$$

The masked argmax is already the discrete decision rule. A separate actor network and online data collection are not required to train or execute this finite-candidate policy. Batch fitted Q iteration explicitly learns a greedy Q function from a fixed collection of transitions by repeatedly solving supervised regression problems [EGW05].

5.2.1 Training unit, causal batching, and prediction support

A retained rollout chain supplies several realized decision states rather than one monolithic recurrent sample. The root is one state; after each selected action, its factual successor is another state with an updated current pose, selected prefix, remaining budget, and candidate table. The `q_h/` reader collates these states on padded batch and state axes. At state t, every actor carrier is restricted to the causal prefix available before action a_t: a later state may be present elsewhere in the same tensor batch, but its selected poses, observations, and candidate table are not inputs to the prediction at t.

The scorer evaluates the realized states and their candidate rows in parallel. For one scalar horizon query per state it emits one conditional value for every materialized candidate row. This parallel tensor layout is not recurrent inference, autoregressive rollout generation, or temporal attention across the rollout. The current scorer does not advance the environment, feed one prediction back into its next state, or iteratively refine one candidate value. During policy execution it is invoked again only after the selected action has produced a genuinely new causal state and candidate table. Causal masking across the state axis would become relevant only for a future joint temporal state encoder; it is not required when each state already carries its independently materialized causal prefix.

Prediction support and loss support are therefore different. The scorer may emit finite values for every materialized candidate. Dense one-step loss uses every hard-valid candidate with a finite immediate label. For $h > 1$, recursive loss is limited to the factual selected transition. Nonterminal targets additionally require a supported factual successor; terminal targets have zero continuation. The padded candidate dimension must not be read as dense multi-step supervision, and an unselected candidate receives no invented successor or return.

5.2.2 Scalar requested-horizon targets

The scorer represents $Q_{\theta(s_t,e,i,h)}$ for each residual horizon h admitted by $1 \leq h \leq b_t \leq H_{\max}$. Remaining budget b_t is a factual field of the represented state; requested horizon h selects how many rewards the current value is meant to summarize. Consequently, a shorter query $h < b_t$ changes the value estimand without changing the state or pretending that less acquisition budget was available. Truncating the return of an eight-step suffix to five steps is conceptually valid only when that five-step target is explicitly constructed and factually supported; changing b_t itself is not ordinary data augmentation. The boundary target is

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

and the recursive target is

$$y_t^{(h,e)} = r_t^e + \gamma B_t^{(h,e)}$$

Here **recursion** means Bellman target recursion across factual successor states. For a nonterminal $h > 1$ target, the current reward is combined with a shorter-horizon successor value queried at $h - 1$; a terminal selected transition has zero continuation and requires no successor state. This construction combines the Bellman decomposition of action value into immediate reward and successor value with temporal-difference bootstrapping from an estimate of that successor [SB18, Secs. 3.5–3.6 and 6.1, pp. 58–67 and 119–124]. The lower-horizon prediction is treated as a fixed regression target by stop-gradient, a frozen stage checkpoint, or a delayed target copy. The defining relation is $Q_h \leftarrow Q_{h-1}$ rather than $Q_h \leftarrow Q_h$, and the executable learner rejects a nonterminal successor whose factual horizon is not exactly $h - 1$. This terminology does not denote an RNN hidden-state loop, repeated invocation of a refinement block, or backpropagation through an unrolled future rollout. Fixed-horizon TD was introduced precisely for predictions over a bounded number of future rewards and avoids same-horizon self-bootstrapping; its horizon functions may use shared parameters and parallel updates [De +20].

The executable joint objective uses one shared horizon-conditioned network. Its public scorer interface still admits one scalar h per state. Privately, Lightning duplicates the complete actor batch along the leading batch axis and concatenates two query domains into one scorer transaction:

1. query $h = 1$ for every realized state and fit every finite hard-valid candidate reward;
2. query the factual residual horizon $h = b_t$ for the same realized states, retaining recursive loss only for selected transitions with $h > 1$;
3. select the successor action with the online scorer at factual $h - 1$ and evaluate it with the delayed target scorer;
4. average candidate losses within each Q1 state, state losses within each horizon, and non-empty horizon means uniformly;
5. persist the realized per-horizon state and candidate counts, report state-normalized loss and absolute calibration error for every supported horizon,

report pairwise ranking and greedy selected-action regret only where a dense counterfactual candidate table exists, and reject online queries absent from manifest-bound training support.

This private batching avoids a second model transaction while preserving the scalar-horizon estimand. It does not add a public vectorized horizon axis and it does not enumerate every off-diagonal query $1 < h < b_t$. Sampling or enumerating additional supported state–horizon pairs would be a separate, versioned learning-contract change with its own weighting and support rules. The present objective preserves one inference interface and shared encoders while making target lineage and weighting explicit. Staged backward induction, fixed-H networks, separate per-horizon heads, and broader off-diagonal horizon enumeration remain matched alternatives rather than implicit runtime behavior.

For remaining horizon two, the store supplies an exact target whenever the successor table has dense one-step labels:

$$y_t^{(2,\text{exact})} = r_t^e + \gamma_t \max_{j: m_{t+1,j}^{\text{train}}=1} r_{t+1,j}^e$$

This target uses no learned successor value or target network. The executable base-case evaluation has two layers: exact-table injection tests the recursion implementation, while the frozen-bundle certification compares the learned recursive target against the factual control using

$$\varepsilon_t^{(2)} = \left| y_t^{(2,\text{recursive})} - y_t^{(2,\text{exact})} \right|, \quad \varepsilon_t^{(2)} \leq \tau_{\text{abs}} + \tau_{\text{rel}} \left| y_t^{(2,\text{exact})} \right|$$

The latter is evaluated on a deterministic bounded sample from a metadata-only census. Its versioned stratum contains scene, target row, configured horizon, candidate-width bin, candidate- and rollout-configuration hashes, and behavior policy. Within each stratum wave, the selector prefers previously unseen scenes before taking another chain from an already represented scene. This rule improves scene coverage without pretending that several rows from one scene are independent replications.

The `qh-exact-q2-certification-receipt-v2` receipt binds the scorer and module configuration, actor-state and learning contracts, ordered test-store manifests and paths, test provenance, selection seed and bounds, and absolute and relative tolerances. Its independent unit is the pair of ordered-store-manifest digest and scene identity. Thesis-core promotion requires at least five selected independent units, at least one factual selected-action exact Q2 row in every selected unit, and the same error gate to pass in every unit. Pooled row-level error remains diagnostic and cannot compensate for a failing scene. The receipt exposes a denominator ladder from eligible census chains through materialized successors and complete hard-valid successor labels to factual selected-action exact Q2 rows. It also reports per-stratum support and provenance-bound fixed-support CORAL saturation separately from recursion error. Missing support is a failed gate rather than zero error. Development-schema `qh-exact-q2-certification-receipt-v1` receipts remain readable historical evidence but cannot promote a claim. Agreement with the exact control remains necessary but insufficient for interpreting longer horizons: positive oracle-lookahead headroom must come from independent held-out endpoint evaluation. A persisted terminal-step role contrast is retained as a diagnostic proxy and is manifest-bound, but it cannot promote an $h > 2$ claim.

5.2.3 Double-Q and behavior-return controls

Double Q changes how a noisy learned successor maximum is estimated; it does not define the scalar horizon interface. The online path selects

$$B_t^{(h,e)} = \begin{cases} Q_{h-1,\theta^-,e}\Big(s_{t+1}, \operatorname{argmax}_{i \in \mathcal{A}_{t+1}^{(Q,h-1)}} Q_{h-1,\theta,e}(s_{t+1}, i)\Big) & \text{if } \; h > 1 \wedge d_t = 0 \wedge \mathcal{A}_{t+1}^{(Q,h-1)} \neq \emptyset \\ 0 & \text{otherwise} \end{cases}$$

and the delayed path evaluates $Q_{|(\theta)}(s_{t+1}, e, j^\star, h-1)$. This selector/evaluator split can reduce overestimation caused by maximizing noisy action values [HGS15]. It remains an ablation against the simpler frozen lower-horizon maximum. It is relevant in an offline setting only because a learned maximum is present, not because online learning is planned.

As distinguished from the greedy value objective in Section 4.5.3, a retained chain also yields the truncated Monte-Carlo target

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \le h \le b_t \le H_{\max}$$

for its behavior policy μ. Regression to this fixed target is a useful policy-conditioned control, but it estimates Q^μ, not the greedy finite-support value $Q^\star$, unless μ is explicitly the target continuation policy. Behavior returns from random-valid, greedy, softmax, and oracle-lookahead chains must therefore remain identified rather than pooled as if they represented one optimal value function.

Double Q addresses one maximization bias. It does not create missing successor transitions, make unsupported actions reliable, or repair state aliasing. CQL and BCQ motivate explicit offline-support diagnostics because a greedy learned policy can select actions whose multi-step consequences are weakly represented in the behavior data [FMP19, Kum+20]. Conservative regularization is introduced only if those diagnostics reveal systematic unsupported-action overestimation.

5.2.4 Horizon-balanced replay and evaluation

Dense one-step rows vastly outnumber selected-action targets at longer horizons. An unweighted row mean can therefore optimize the myopic task while obscuring longer-horizon failure. The implemented learning contract freezes candidate-within-state, state-within-horizon, equal-horizon aggregation as `uniform-horizon-state-balanced-dense-q1-selected-recursion-v2`. It binds this rule with the maximum horizon, replay candidate/rollout configuration identities, behavior-policy vocabulary, reward/return semantics, and discount, and rejects cross-stage mismatches. The bundle additionally records realized state and candidate counts for every trained horizon; syntactically valid horizons without positive training support fail closed at online inference. Checkpoint admission binds the actor-state identity separately. Alternative sampling or weighting schemes require new versioned learning identities. Candidate alternatives are:

- a horizon-stratified sampler;

- per-horizon loss weights w_h;
- or a fixed number of admitted targets per horizon and scene.

Every run reports, separately for each h:

- admitted states, selected actions, behavior policies, candidate families, scenes, and targets;
- value loss and signed target error;
- candidate ranking and top-action regret where oracle comparison is available;
- bootstrap, terminal, and no-valid-successor fractions;
- online/target disagreement for Double-Q runs;
- marginal selected target RRI and the selected-chain cumulative target RRI diagnostic;
- cumulative target root gain, kept distinct from cumulative RRI;
- endpoint performance of the masked policy requested at the remaining budget.

A single scalar validation loss is insufficient for model selection unless its horizon aggregation is frozen in advance. Cross-stage corpus admission enforces the same maximum horizon, reward and return semantics, discount, state/source protocol, candidate/reason vocabulary, replay-support identity, and horizon-weighting rule.

The optimality claim remains bounded under either interface. A learned finite-horizon value can approximate the best continuation only within the sampled finite candidate generator, hard-validity regime, represented actor state, and offline transition support. The same checkpoint cannot silently mix a pose-only state with privileged selected-depth, sensor-like, or actor-visible dynamic states. Longer horizons increase—not decrease—the need for selected-observation geometry and a sufficiently Markov scene state.

The current scorer-independent learner establishes infrastructure readiness only. Its horizon metrics are clustered by factual state: candidate errors are first averaged within state, pairwise ranking accuracies are first formed within state, and greedy regret contributes one value per supported state. Only dense Q1 currently has ranking/regret support; selected-transition recursion at longer horizons reports zero counterfactual support. Confirmatory interpretation requires the source-owner scorer decision, a canonical oracle-task corpus, dense-Q1 and exact-Q2 controls, supported targets for every claimed horizon, cross-stage learning-contract equality, checkpoint compatibility, a frozen actor-state protocol, horizon-specific support diagnostics, and matched held-out endpoint oracle re-evaluation. A dynamic-state claim additionally requires selected-observation fusion and no-future-observation leakage tests.

5.3 Policy Comparison and Statistical Protocol

Implementation: partial **Evidence: pending**

The aggregation and artifact-driven reporting seam exists, but confirmatory paired policy evidence does not.

Promotion gate: frozen held-out manifest, completed paired endpoints, and validated report bundle

Development source: aria_nbv/aria_nbv/rollouts/reporting.py; docs/typst/thesis/experiment_data.typ

The experimental unit is the scene. Every comparison is paired by scene, target task, root state, candidate-generation distribution, hard validity regime, and acquisition horizon. Planner depth may differ because it defines the decision rule, but every policy receives the same acquisition budget. Repeated snippets, targets, and seeds are aggregated within scene before the primary comparison so that densely sampled scenes do not dominate the estimand.

The primary estimand is the paired per-scene difference in fixed-budget endpoint target-quality gain, using $J_e^{(H)}$. Its $\Delta_0^e + \varepsilon$ denominator is distinct from the $\max(\Delta_0^e, \varepsilon)$ denominator of cumulative target-root gain, while state-relative RRI has a state-varying denominator; both remain secondary diagnostics rather than alternate names for the endpoint. The centralized analysis artifact freezes the scene aggregation function, uncertainty procedure, interval level, comparison set, and minimum meaningful effect before final aggregation. Invalid-action rate, runtime, path length, and task/candidate coverage likewise diagnose mechanism and feasibility but do not replace the endpoint estimand. The current replay store alone cannot estimate this endpoint comparison because it does not persist an independent post-budget reconstruction for every policy; confirmatory analysis therefore requires matched endpoint oracle re-evaluation or an equivalent frozen endpoint record.

Support and failure strata are fixed before policy inspection. The report distinguishes source rows without an admitted target task, admitted tasks whose oracle evaluation fails, roots without a valid action, trajectories that terminate before the shared budget, and completed paired trajectories. A policy that cannot continue remains in the comparison with no further gain; it is not silently dropped. These strata delimit the analysis population and expose whether a policy difference is instead a support or systems failure.

Policy	Decision input	Acquisition budget	Scientific role
π_{rand}	actor-visible	H	valid-action lower reference
$\pi_{\text{learned-1}}$	actor-visible	H	learned myopic control
$\pi_{\text{oracle-1}}$	oracle immediate reward	H	one-step oracle-greedy comparator
$\pi_{\text{oracle-look}}$	bounded oracle lookahead	H	conditional finite-support upper reference
π_{Q}	actor-visible	H	planned learned finite-horizon gap closure

Table 14: Matched policy comparison. All rows acquire the same number of views; oracle access and planner depth define the decision rule rather than the acquisition budget.

The first inferential comparison is bounded oracle lookahead against one-step oracle greedy. Its evaluation contract freezes one target mesh and crop, ASE-depth source, renderer and backprojection, fusion and point cap, and point–mesh metric configuration for every policy. Repeated evaluation must reproduce the same scores within the declared numerical tolerance; it verifies deterministic execution under this contract, not invariance to a different mesh, sampling process, or metric. The analysis artifact also freezes the meaningful-headroom rule before learned-policy inspection. Learned-control gap-closure ratios are reported only after oracle headroom passes that gate and only when their distinct actor-visible-myopic-to-oracle-lookahead denominator is admissible. The learned comparison then contrasts Q_H with the actor-visible myopic control under the same endpoint evaluation and acquisition budget; because neither learned target-conditioned control is presently complete, this comparison remains prospective.

Validation TODO: Populate the evidence chain in order: evaluation-contract identity and deterministic repeatability, candidate and target support, oracle-lookahead headroom, myopic-control calibration, finite-horizon gap closure, then representation and architecture ablations. Missing upstream evidence blocks downstream claims rather than becoming a zero result.

Source: thesis objective-to-evidence contract

Gate: artifact-backed Results bundle

Validation TODO: Run row-shuffle, mask-isolation, duplicate-row, valid-count, frame-transform, target-source-dropout, and horizon-boundary tests for every model admitted to the policy comparison.

Source: geometric acceptance contract

Gate: architecture validity report

Open decision: Freeze the scene aggregation, interval procedure, interval level, comparison family, and meaningful-headroom threshold in the resolved analysis manifest rather than in thesis prose.

Source: artifact-driven reporting plan

Gate: confirmatory analysis freeze

5.4 Outcome Logic

Meaningful, stable lookahead headroom establishes only that the frozen finite-candidate setup contains exploitable non-myopic structure. No meaningful headroom is a setup-specific negative result, not evidence that target-aware planning is universally myopic. A non-reproducible or mismatched evaluation contract, or insufficient paired support, blocks downstream policy claims. Conclusions remain conditional on the frozen metric contract and are not claims of representation independence. Likewise, train-only bandwidth pilots and failed generation attempts, including memory-exhaustion failures, inform compute configuration and throughput planning only; incomplete artifacts cannot establish headroom or policy superiority.

6 Results

Validation TODO: Replace fixture- and readiness-oriented output with confirmatory results that answer each research question using matched endpoint evaluation, denominators, exclusions, aggregation units, independent-run uncertainty, and immutable artifact provenance.

Source: confirmatory report bundle and analysis manifest

Gate: submission evidence bundle passes all report assertions and every stated result resolves to raw and derived artifacts

The thesis report bundle is loaded through the strict schema checked in `experiment_data.typ`. Its declared evidence class is *pilot*. Schema validity establishes only that provenance, missingness, and table contracts are readable; it does not turn a development fixture or an incomplete pilot into scientific evidence.

Result	Required evidence	Status
Population and targets	population facts with denominators in every validated store	not available
Candidate-support QC	five descriptive diagnostics with the frozen state–scene macro identities and exact scene denominators	not available
Paired policy effect	paired per-scene effect, interval, exact paired-scene count, and headroom decision in every validated store	not available
Resource feasibility	wall time, peak GPU memory, and storage in every validated store	not available

Table 15: Evidence availability in the loaded report bundle. A status of “not available” means that no thesis result is inferred from fixture values, train-only attempts, or an incomplete store.

6.1 Candidate and Store Feasibility

The available artifacts show that the finite-candidate rollout path reaches mesh rendering, target-specific oracle scoring, and selected-action replay on training sources. They therefore support an implementation-readiness claim only. A CUDA out-of-memory failure in an unbatched candidate render and later memory-bounded attempts identify rendering as a scale gate; neither establishes rollout throughput, storage cost, candidate-family support, or policy quality for the intended study population.

6.2 Target-Task Coverage

The implemented pipeline samples geometry-valid GT target tasks and records target and validity fields in the rollout schema. The loaded bundle does not contain a validated population and exclusion table for the study. Scene coverage, eligible-target coverage, invalid-reason frequencies, and their denominators are consequently unavailable as results.

6.3 Policy Comparison

The primary estimand is not estimable from the current evidence because no validated held-out bundle contains the matched policy outcomes and aggregation inputs. Oracle repeatability, oracle-lookahead headroom, learned one-step performance, finite-horizon recovery, uncertainty intervals, and sensitivity analyses are therefore not reported.

6.4 Runtime and Storage Gate

The runtime and storage claim requires completed validated stores whose resolved manifests, failure records, and resource summaries refer to the same run. Until that evidence exists for both rollout profiles, the large-scale data-generation gate remains pending. The current attempts justify memory-bounded rendering and retained failure provenance, but no extrapolation to cluster throughput or dataset volume.

7 Discussion

Remove or rewrite before submission: Rewrite the discussion after confirmatory results exist. The present implementation-readiness narrative and architecture bridge registry are development material; the final chapter must interpret measured answers, alternative explanations, scope, limitations, and failure modes.

Source: Chapter 6 and the frozen analysis plan

Gate: every interpretation points to a reported result or is explicitly bounded as a limitation

The current evidence supports a narrow implementation conclusion. ARIA-NBV has an executable finite-candidate path that separates actor-visible policy inputs from oracle-only target geometry, applies invalidity as a hard decision constraint, evaluates target-specific reconstruction change offline, and exposes the resulting store through one typed reporting seam. This establishes that the proposed experiment can be represented and audited; it does not establish that one candidate family, rollout policy, representation, or learned model performs better than another.

The rollout attempts also expose a systems limitation. Counterfactual scoring repeatedly renders a large mesh for a candidate set, so renderer memory and latency constrain feasible branch factor and rollout volume before statistical efficiency becomes relevant. An out-of-memory observation is evidence for batching and resource measurement, not for or against the candidate distribution or planning objective. Completed validated stores are required before throughput, storage, or failure rates are generalized.

The scientific limitations are more restrictive. Current attempts use training sources, no held-out paired policy table supports the endpoint estimand, metric repeatability and meaningful oracle-lookahead headroom remain unestablished, and the target-conditioned finite-horizon comparison has not yet been demonstrated. The present evidence therefore cannot distinguish negligible non-myopic structure from inadequate candidate support, target observability, replay coverage, representation support, or model failure.

7.1 Evidence-conditioned architecture bridges

Implementation: exploratory **Evidence: pending**

Alternative representations and policies remain part of the thesis design registry, but they are promoted only as responses to diagnosed bottlenecks.

Promotion gate: a measured limitation that the proposed bridge directly tests

Development source: Table 7; Table 10

If the local EFM3D field is the limiting variable, broad semidense memory, sparse occupied/free/unknown state, target-centred re-lifting, or logged appearance features provide progressively stronger representation tests. If compact descriptors lose relevant spatial structure, point, sparse, or equivariant encoders become justified. If independent candidate queries leave errors correlated with the valid candidate set, DeepSets or masked candidate interaction becomes relevant. If selected-view history fails specifically by approach direction, signed first- plus second-moment memory, followed only if needed by spherical harmonics, becomes the targeted ablation. A renderable 3DGS state is appropriate only when candidate rendering or soft instance membership is itself the measured missing capability.

Continuous and hierarchical policies change the action contract rather than merely increasing model capacity. A target-then-pose policy could first select or mask a target token and then condition pose queries on target, horizon, and step; multiple targets can share scene encodings and be evaluated in parallel. This remains post-core work because continuous pose generation requires separate feasibility, safety, support, and simulator-realism evidence. The finite-candidate model is retained as the calibrated control even if a continuous policy is later introduced.

Sequence or recurrent models are similarly conditional. They become scientifically motivated when errors persist after explicit history, time, horizon, and scene-memory conditioning, not simply because the dataset contains trajectories. Looping or recurrent refinement must preserve causal masks and candidate-row equivariance and must be compared against the same fixed-budget endpoint oracle evaluation.

Research TODO: After the primary evidence chain is complete, promote only bridges tied to a measured failure: EVL support, target observability, directional history, valid-set interaction, long-horizon credit, or action-support mismatch.

Source: method design-space registry

Gate: artifact-backed failure analysis

The next evidential step remains procedural: complete validated stores under resolved manifests, freeze the eligible population and exclusions, establish oracle stability and headroom, train the matched controls, and only then interpret architectural differences. The registry above preserves concrete follow-up hypotheses without presenting them as validated conclusions.

8 Conclusion

Validation TODO: Rewrite the conclusion as direct, evidence-calibrated answers to RQ1–RQ4. The current conditional outcome tree is useful development guidance but is not a final scientific conclusion.

Source: Chapter 6; Chapter 7

Gate: confirmatory results and discussion support one concise answer per research question

This thesis defines a leakage-auditable experiment for target-conditioned finite-candidate next-best-view planning. Its present contribution is the separation of actor-visible state from oracle supervision, the target-specific reconstruction objective, hard validity and replay contracts, and an artifact-driven reporting seam that keeps provenance and missingness attached to later results.

The available evidence does not answer whether bounded oracle lookahead improves fixed-budget target reconstruction over oracle-greedy or whether a learned finite-horizon policy closes the separate endpoint gap from an actor-visible learned-myopic control to oracle lookahead. The current training-source rollout attempts establish pipeline reachability and reveal a renderer resource gate; the development report fixture establishes the data contract only. Neither supports a held-out policy claim, a population-level effect, or a scale estimate.

The final scientific conclusion is therefore conditional on evidence that is not yet available. A stable oracle metric and positive paired lookahead effect would define measurable headroom for the evaluated finite support. Negligible headroom would be a setup-specific negative result. Unstable oracle evaluation would block planning claims, and stable headroom without the prescribed learned-control gap closure would remain a learned-control failure with several unresolved mechanisms. These outcomes delimit the thesis without extending it to continuous control, online reinforcement learning, or real-device deployment.

Development roadmap

This page is the compact strategic view of thesis development. It is omitted from submission output. Scientific claims remain owned by the active thesis; executable behavior and measurements remain owned by source, tests, configuration, and immutable evidence artifacts.

Implementation: partial **Evidence: pending**

Reviewed 31 Aug 2026 refresh due 14 Sep 2026. Target: reproducible full-draft freeze by 30 November 2026, followed by a safety buffer until submission on 7 January 2027.

Promotion gate: Close P1 before confirmatory policy claims.

Development source: `docs/typst/thesis/development/roadmap.toml`; active thesis, package, test, configuration, and evidence owners

TL;DR

Done **Pilot infrastructure**

Candidate-benchmark smoke evidence, a bounded two-scene orbit MVP, and a pilot rollout report are committed. These prove the pipeline can produce reviewable artifacts, not that population-scale support or policy gains are established.

Now **Evidence substrate**

Freeze the multi-scene store, scene-level split, camera/depth and target-RRI fidelity, candidate-support admission, and throughput receipts while the thesis narrative and conceptual figures remain under review.

Blocked **Confirmatory claims**

Lookahead-headroom, learned-value, and population-coverage claims remain blocked until paired multi-scene evidence and an immutable experiment registry exist. The current rollout bundle is explicitly pilot evidence.

Next **Paired evidence to freeze**

Run equal-budget baselines, train and evaluate the finite-horizon value model under actor-visible inputs, synthesize results and limitations, and reach the reproducible full-draft freeze recorded below.

Critical path

The strategic dependency is evidence-first: close the substrate, establish paired baseline headroom, evaluate the learned value model, synthesize the claims, and freeze the full draft. The December-to-January interval is a safety buffer, not planned scientific work.

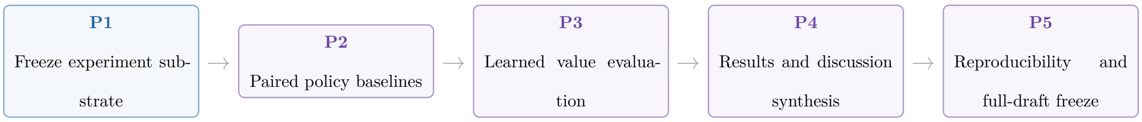


Figure 7: Critical development gates from experiment substrate to the 30 November 2026 full-draft freeze.

Milestones

ID	Phase	Dates	State	Exit gate
P0	Pilot infrastructure	7 Jul–30 Aug 2026	**done**	Committed smoke and MVP artifacts compile and remain explicitly bounded as pilot evidence.
P1	Freeze experiment substrate	31 Aug–30 Sep 2026	**in progress**	Multi-scene immutable store, scene-level split, camera/depth and target-RRI fidelity, candidate-support admission, and throughput receipts.
P2	Paired policy baselines	1 Oct–20 Oct 2026	**planned**	Equal-budget random-valid, greedy, and bounded-lookahead comparisons report uncertainty, support, and oracle receipts.
P3	Learned value evaluation	21 Oct–10 Nov 2026	**planned**	Held-out ranking, calibration, and paired policy value are reported under actor-visible inputs.
P4	Results and discussion synthesis	11 Nov–22 Nov 2026	**planned**	Final tables, failure cases, coverage limits, and claim dispositions resolve to the immutable evidence registry.
P5	Reproducibility and full-draft freeze	23 Nov–30 Nov 2026	**planned**	Full draft, reproducible configurations, final figures, and release checks pass with no unresolved placeholders.
P6	Safety buffer and submission	1 Dec–7 Jan 2027	safety buffer	Advisor feedback, formal checks, final upload, and the authenticated submission receipt are closed without reopening the scientific core.

Table 16: Evidence-gated schedule from the completed pilot infrastructure to submission. Dependencies are checked against the adjacent roadmap data.

Blockers

- **B1 — Multi-scene experiment substrate** (affects P1): The committed artifacts do not yet establish a fresh multi-scene store, scene-level split, camera/depth fidelity, target-RRI robustness, and representative throughput in one immutable receipt chain.
- **B2 — Candidate-support scale admission** (affects P1, P2): The positive two-scene orbit MVP is explicitly not scale-up admission; family survival, framing, diversity, and proposal regret still require broader evidence.
- **B3 — Confirmatory experiment registry** (affects P2, P3, P4): The available report bundle is a one-scene pilot. Final comparisons need immutable run/config identities, coverage accounting, uncertainty, and claim-to-artifact resolution.

The primary evidence pointers remain beside each record in `roadmap.toml`; the roadmap contract checks their existence, while scientific review decides whether they are sufficient for promotion.

Promotion queue

Promotion queue — candidate: Promote bounded-lookahead and learned-value results only after paired held-out evidence is immutable.
Source: P2-P3 experiment registry and rollout reports target: Results and Discussion gate: positive headroom, actor-visible evaluation, uncertainty, and support coverage

Promotion queue — blocked: Resolve candidate-support and substrate limits before claiming population coverage.
Source: P1 evidence receipts and candidate benchmark manifests target: Experimental Design and Results gate: multi-scene support, family survival, scene-level split, and representative throughput

Promotion queue — deferred: Keep online discrete and continuous target-then-pose extensions outside the core until the offline finite-candidate gates close.
Source: Discussion and roadmap P1-P3 target: Discussion gate: stable offline learned-value evidence and an explicit post-core scope decision

Maintenance contract

`roadmap.toml` is the single owner for the snapshot, dates, states, dependencies, blockers, release checks, and evidence pointers rendered above. Update it at least every 14 days and whenever a gate changes state. `make thesis-roadmap-contract` rejects stale review dates, broken dependencies, missing local evidence pointers, divergence from the thesis submission date, or references to the retired M1 snapshot. Internal trackers and hosted issue state may inform a refresh but are not imported as public thesis truth.

The native table, cards, and critical-path strip are deliberately sufficient: they show exact milestones and the consequential dependency chain without a second diagram data model. A date-dense Gantt can be reconsidered only if this compact projection no longer answers the planning question.

HM/FK07 release gate

These checks are human-owned and apply to the exact final candidate and the author's authenticated records; the repository cannot satisfy them by itself.

- **R1 — Applicable rules:** Confirm the individually assigned programme and SPO version in PRIMUSS; public reading versions establish only the current baseline.
- **R2 — Registration record:** Retain the PRIMUSS confirmation, official start date, individual deadline, assigned examiners, supervisor-specific form requirements, and requested attachments.
- **R3 — Exact upload set:** Before the individual deadline, verify the exact final PDF and every PRIMUSS-marked mandatory document or attachment. A repository build, digest, or upload alone is not submission evidence.
- **R4 — Conditional paper copy:** Provide an identical paper copy only if the examiner or authenticated PRIMUSS record requires one under the current FK07 process.
- **R5 — Final-PDF review:** Check the declaration and AI/tool disclosure in the exact PDF prepared for upload; confirm any mandatory signature or attachment against the authenticated record.
- **R6 — Submission closure:** Finalize with the PRIMUSS action "Abschlussarbeit abgeben", retain the authenticated timestamp, and archive the exact submitted PDF; only that receipt closes the gate.

Official baseline checked 31 August 2026: HM examination regulations, FK07 thesis process, PRIMUSS registration guide, PRIMUSS submission guide. At the full-draft freeze, recheck the official versions and the individually assigned instructions in authenticated PRIMUSS records.

Freeze and submission

By 30 November 2026, the full draft, experiment registry, configurations, figures, and release checks must be reproducible and free of unresolved placeholders. The interval from 1 December to 7 January 2027 is reserved for advisor feedback, formal HM/FK07 checks, final upload, and recovery from unexpected defects. Only the authenticated submission receipt closes the final gate.

9 Development Pilots

9.1 Candidate-Family Support

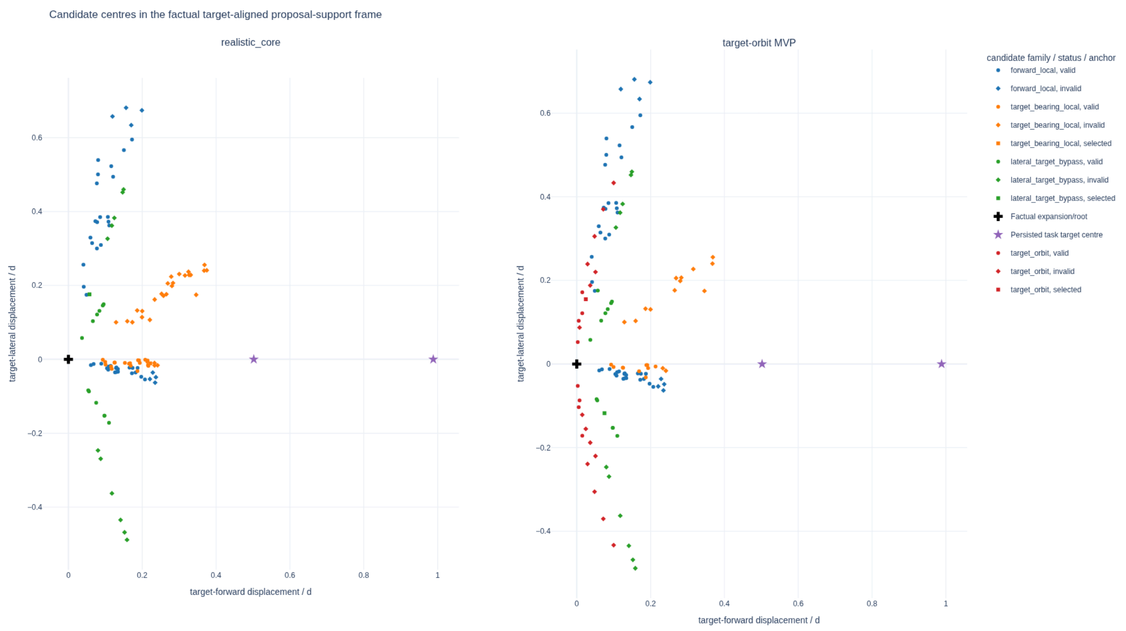


Figure 8: Attempted candidate centres from a two-scene, two-state real-data method audit. Each state is centred on its factual expansion pose, yaw-aligned to the oracle-evaluation target with world Z-up, and divided by its current three-dimensional target distance. The cross marks the expansion/root pose, the stars mark the two oracle task target centres, colour denotes proposal family, and marker shape denotes actor-valid or selected status. The figure diagnoses proposal geometry under the `v0_gt_input` target protocol; it is neither a population estimate nor evidence of actor-visible target discovery or policy quality.

The target-orbit challenger adds attempted support on both target-relative sides in this bounded pilot, while pruning still leaves invalid rows and one configured family/state pair without actor-valid support. The associated immutable evidence manifest records the common-frame reducer, store-manifest hashes, expected state index, committed reduced candidate rows, resolved challenger profile, reproduction commands, metric denominators, and plot hashes. No frame-mixed side-balance or orbit-span values are used. The interactive evidence and Streamlit view can optionally add short arrows for the valid candidates' camera optical axes; the static thesis figure omits them to preserve legibility.

9.2 Target-Frame $\mathcal{S}^2$ Diagnostics

The following frozen figures are generated by the same Python reporting and plotting owners as the stored-rollout application. For the selected real-data shard, the report records these support counts: source samples 1, snippets 1, scenes 1, targets 1, retained rollout chains 1, and selected steps 5. It remains pilot evidence: its purpose is to validate the representation and reporting contract, not to estimate policy quality. The two point-direction views contain 5 factual displacements and 5 factual optical axes. Colour denotes rollout-chain identity, marker shape denotes decision-step identity, and the sphere heat map is computed from the complete equal-solid-angle count grid rather than from the bounded incidence overlay.

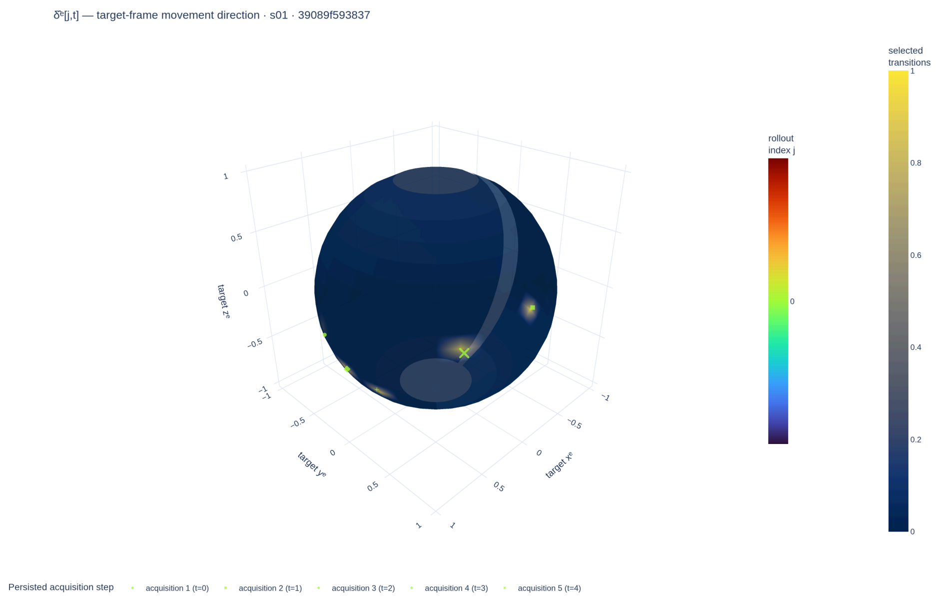


Figure 9: Target-frame $\mathcal{S}^2$ movement directions from the frozen pilot report. This view records how the camera moved; it is not a visibility measurement.

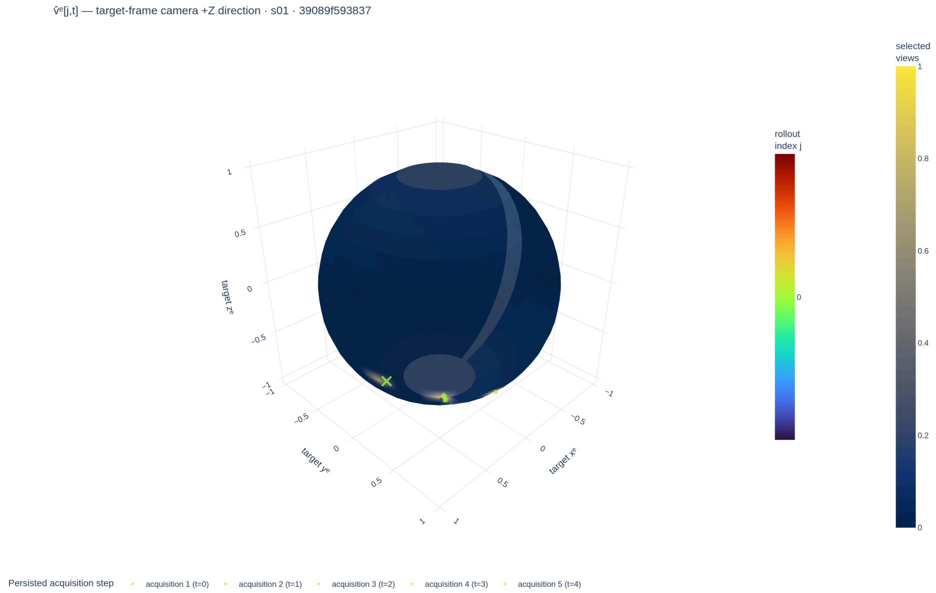


Figure 10: Target-frame $\mathcal{S}^2$ selected-camera optical-axis directions from the same frozen pilot report. This view records where the camera pointed; it is distinct from both camera motion and calibrated surface support.

The calibrated proxy diagnostic integrates 5 selected frusta. Their mean intrinsic field-of-view solid angle is 2.49 sr. On the geometric-mean-scale proxy sphere, one view covers a mean fraction of 0.168, while their union covers 0.312. These fractions express geometric potential support under the proxy and front-facing tests; they exclude true target shape, depth ordering, and scene occlusion.

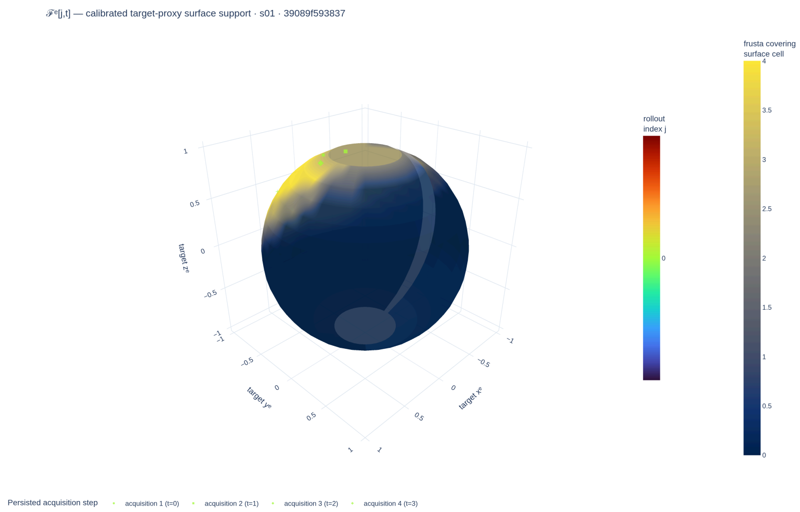


Figure 11: Calibrated selected-frustum support on the target-centred proxy sphere. The surface heat map is the complete equal-solid-angle coverage field; overlaid incidence samples preserve rollout and time heritage. This is not measured target-mesh visibility.

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A Reproducibility Record

The report bundle is the single numerical interface between validated rollout artifacts and this manuscript. Its schema version, evidence status, store manifests, typed parameter rows, failure records, and sidecar hashes preserve provenance without duplicating analysis logic in Typst. The document loader rejects schema or status mismatches before any table is rendered. Compact labels and digest prefixes are used below; complete store identifiers, paths, and hashes remain in the machine-readable bundle.

The implementation anchors behind that interface are the immutable `vin_offline/` source store, the task-specific `rollouts.zarr/` replay store, and the derived `q_h/` training cache. Their exact schemas, normalized joins, chunking, codecs, receipt versions, and execution commands remain manifest-owned reproducibility metadata rather than scientific entities in Chapter 3. A reported row is admissible only when the bundle resolves these identities and preserves source role, missingness, and failure records.

Evidence class	Store	Manifest digest	Validated
Development fixture	store S1	not-a-real-m...	yes

Table 17: Loaded development-fixture provenance. The fixture verifies the document interface only and is not scientific evidence.

The bundled development fixture contains synthetic values solely to test typed loading and missingness. Its numerical rows are deliberately not rendered or interpreted.

B Development Diary

Archived source note: This marked diary is a development-only record. It is excluded from submission mode; only evidence-backed definitions, protocols, and results graduate into the thesis body.

Source: thesis mode contract

B.1 Decisions Retained

The thesis keeps the privileged oracle and actor-visible policy as separate contracts, evaluates finite candidate sets under a shared validity mask, and treats target-specific reconstruction improvement as the common comparison signal. Rollout stores retain selected transitions and provenance rather than becoming a second implementation of the oracle or training pipeline.

Open decision: Freeze the scene-disjoint analysis population and the minimum evidence scale only when the resolved manifests and exclusion ledger are available.

Source: experimental-design contract

Gate: confirmatory bundle freeze

B.2 Failures That Changed the Plan

Early rollout attempts reached the mesh-rendering path but exceeded available accelerator memory when candidate views were rendered without a bounded batch. This failure narrowed the immediate claim to implementation readiness and made peak memory, throughput, failure provenance, and completed-store validation explicit scale gates.

Validation TODO: Require completed stores, matching manifests, resource summaries, and recorded failures before extrapolating rollout bandwidth or storage demand.

Source: rollout attempt artifacts

Gate: cluster-readiness evidence

B.3 Rejected Scope

Continuous actions, online simulators, additional scene representations, and alternative reinforcement-learning families are excluded from the core study while the finite-candidate target-conditioned comparison remains unvalidated. They would change both the action contract and the supervision regime, so adding them now would weaken rather than extend the primary experiment.

Remove or rewrite before submission: Delete bridge material that cannot be tied to a concrete limitation or a matched follow-up experiment after the confirmatory analysis is complete.

Source: scope review

Gate: discussion freeze

B.4 Directional-Memory Hypothesis

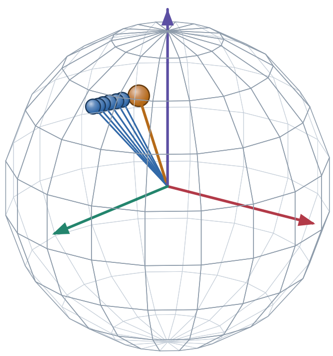
Research TODO: Treat target-centred directional memory as an ablation hypothesis, not an actor feature or contribution. The representation must be implemented, mask-audited, and compared against pose distance and overlap before any predictive claim enters the manuscript.

Source: RQ3 representation hypothesis

Gate: paired held-out ablation

A. Target-centred directions on $\mathbb{S}^2$

actual logged camera centres, expressed in the target-object frame

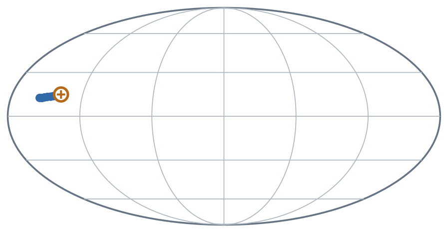


● logged history direction ⊕ example query

Scene 81286, sample ASE_81286_Atek_000024, target instance 128 frames 0, 2, 4, 6, 8, 10 and query frame 15.

B. Complete-domain view and prospective compression

Mollweide is equal-area; no smooth field or SH lobe is implied



$+x_e$ at map centre · $+z_e$ north · wrap seam at $\pm 180°$

$$\boldsymbol{M}_{\mathrm{dir}(v)} = \textstyle\sum_{k<t} w_{k(v)} \boldsymbol{d}_{k(v)} \boldsymbol{d}_{k(v)}^{\top}$$

$$\nu_{i(v)} = 1 - \frac{\boldsymbol{d}_i^{\top} \boldsymbol{M}_{\mathrm{dir}} \boldsymbol{d}_i}{\operatorname{tr} \boldsymbol{M}_{\mathrm{dir}} + \varepsilon}$$

This fixture uses uniform weights and a logged future pose only to make the second-moment query inspectable. It does not claim that the learner currently stores $\boldsymbol{M}_{\mathrm{dir}}$ or that this novelty predicts target RRI.

HYPOTHESIS / ABLATION MATERIAL. Keep outside submission claims until the representation exists and paired evaluation tests whether it adds information beyond pose distance and overlap.

Figure 12: Development-only directional-memory fixture. The sphere and Mollweide panels show the same logged ASE camera directions in a target-object frame; the orange query is another logged pose used to inspect one prospective second-moment novelty definition. No smooth field, spherical-harmonic representation, or learned benefit is claimed.

B.5 Next Evidence Gates

The next admissible evidence is a validated report bundle that fixes the study population, records target and candidate exclusions, supplies paired endpoint outcomes with uncertainty, and joins runtime and storage statistics to the same manifests. Only then can the manuscript replace the present availability statements with policy comparisons.

Implementation TODO: Export the confirmatory report bundle, compile both thesis modes against it, and inspect the resulting tables and failure cases before freezing claims.

Source: reporting contract

Gate: claim freeze

Ehrenwoertliche Erklaerung

Ich versichere, dass ich diese Masterarbeit selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen oder Hilfsmittel benutzt sowie wörtliche oder sinngemäße Zitate als solche gekennzeichnet habe.

Muenchen, 7. Jan 2027 Jan Duchscherer